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The equation $2x^4 + xy + y^3 + 2y - 1 = 0$ has no integer solutions

Size $h = 49$, length $l = 12$

Theorem 1. *The equation*

$$2x^4 + xy + y^3 + 2y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof combines quadratic Hilbert symbols over $\mathbb{Q}$ and cubic Hilbert symbols over $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of auxiliary polynomials H_i with rational coefficients, linked by a graph, and the values of auxiliary polynomials J_i with coefficients in $\mathbb{Z}[\omega]$, linked by a weighted graph. By the product formula and the reciprocity law, the products of the quadratic symbols over all places are 1, and the sums of the cubic symbols over all places are 0. We show that the local contributions vanish at the primes outside a finite set S , determine the sign contributions at the real place, compute the local contributions jointly at the primes of S , and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, finite computations with residues, and basic properties of the Legendre symbol and of the quadratic Hilbert symbol, the proof uses two standard facts from class field theory: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues $1, 3, 5, 7$ modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. Let a, b, c be real numbers with $a > 0$, $c > 0$ and $256ac^3 > 27b^4$. Then $at^4 + bt + c > 0$ for every real number t .

Proof. For $t = 0$ this is clear. For $t \neq 0$, the inequality between the arithmetic and geometric means of at^4 , $c/3$, $c/3$ and $c/3$ gives $at^4 + c \geq 4(at^4 c^3/27)^{1/4} = (256ac^3/27)^{1/4}|t| > |b||t| \geq -bt$. $\square$

2 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [3, Chapter V, §3]. We write $(a, b)_v = \omega^{\theta_v(a, b)}$ with $\theta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [3, Chapter V, §3 and Chapter VI, §8].

- (H1) $\theta_v(a, bc) = \theta_v(a, b) + \theta_v(a, c)$, $\theta_v(a, b) = -\theta_v(b, a)$, and $\theta_v(a, c^3) = 0$. In particular, $\theta_v(a, b)$ depends only on the classes of a and b in $K_v^\times/K_v^{\times 3}$.
- (H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\theta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\theta_v(a, b) = 0$ if a and b are both units.
- (H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\theta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \theta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\theta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values θ_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $\text{cl}_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), cl_v is additive, $\text{cl}_v(a)$ determines the class of a modulo cubes, and $\theta_v(a, b) = e\chi(w) - f\chi(u)$ for $\text{cl}_v(a) = (e, \chi(u))$ and $\text{cl}_v(b) = (f, \chi(w))$. We put $\text{cl}_p(a) = \text{cl}_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $\text{cl}_p(a) = (\text{cl}_{v_r}(a), \text{cl}_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, θ_p is the sum of the forms θ_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 8. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $\text{cl}_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then cl_3 is additive, and $\theta_v(A, B) = \text{cl}_3(A)^\top M_3 \text{cl}_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute θ_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), θ_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write θ_3 for the form of Lemma 8. Since $3 = -\omega^2 \pi^2$ and $4 = 2^2$, we have $\text{cl}_3(3) = (2, 2, 0, 0)$ and $\text{cl}_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 9. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $\text{cl}_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = \text{cl}_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $\text{cl}_{v_r}(z)$ are those of $\text{cl}_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = \text{cl}_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.

(iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = \text{cl}_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j \text{cl}_3(3)$ and D is spanned by $\text{cl}_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $\text{cl}_3(3)$ and $\text{cl}_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 8, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 10. Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\theta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\theta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then

$$\sum_{(i,j)} w_{ij} \theta_p(c_i, c_j) + \sum_i \theta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \theta_p(b_i, b_j) + \sum_i \theta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As θ_p is bilinear and $\theta_p(u, v) = -\theta_p(v, u)$, the left side equals the right side plus $\sum_i \theta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \theta_p(d_i, d_j)$, and both sums vanish. $\square$

3 The quadratic certificate

Put $F(x, y) = 2x^4 + xy + y^3 + 2y - 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y, \\ H_1 &= 3x^2 - 3xy + y^2 + y + 1, \\ H_2 &= x^2 + 3x + 3, \\ H_3 &= x^2 - xy + x + y^2 - 2y + 1, \\ H_4 &= x^2 + xy + 2x + y^2 + y + 1, \\ H_5 &= x^2 - xy - x + y^2 - y + 1, \\ H_6 &= x^2 + xy - x + y^2 - 2y + 1, \\ H_7 &= -y + 1, \\ H_8 &= 3x^2 + 3xy + y^2 + y + 1, \\ H_9 &= x^2 - x + 1, \\ H_{10} &= x^2 + xy - 2x + y^2 - y + 1, \\ H_{11} &= x^2 + 2xy - x + y^2 - y + 1, \\ H_{12} &= -2y + 1, \\ H_{13} &= x^2 + xy + y^2, \\ H_{14} &= x^2 - xy + y^2, \end{aligned}$$

$$\begin{aligned}
H_{15} &= 2xy + 4x, \\
H_{16} &= -x - 1, \\
H_{17} &= -x + 1, \\
H_{18} &= -x^2 - 2x + y + 1, \\
H_{19} &= -1, \\
H_{20} &= -x^2 - xy - x + y^2 + y, \\
H_{21} &= -x^2 + 2xy - 3x - y^2 + y, \\
H_{22} &= y - 1, \\
H_{23} &= -x^2 + 2xy - 2x - y^2 + y + 1, \\
H_{24} &= x^2 + xy - x + 2y, \\
H_{25} &= x^2 - xy - 2x - y^2 - y + 1, \\
H_{26} &= x - 1, \\
H_{27} &= y, \\
H_{28} &= x^2 + xy - 3x + y^2, \\
H_{29} &= x^2 + 2xy - 3x + y^2 - y, \\
H_{30} &= 2x - 1, \\
H_{31} &= x^2 + 3x + 2y, \\
H_{32} &= x^2 + xy + x + y^2 - y + 1, \\
H_{33} &= x^4 - x^2y + y^2, \\
H_{34} &= x^3 - x^2y + y^2 + y, \\
H_{35} &= x^2 - xy + x + y^2 + y + 1, \\
H_{36} &= -y - 1, \\
H_{37} &= x^3 + x^2 + y^2, \\
H_{38} &= x^3 - x^2 + y^2, \\
H_{39} &= x^3 - x^2y + x^2 + y^2 + y, \\
H_{40} &= x^2 - xy + 3x + y^2 + 3, \\
H_{41} &= -x^3 - x^2y + xy + y, \\
H_{42} &= -x^3 - x^2y + y, \\
H_{43} &= x^4 - x^3 + x^2y + x^2 + xy + y^2, \\
H_{44} &= -x^3 + x^2 - x - y, \\
H_{45} &= -x^3 + x^2 - xy + x - y^2 + y, \\
H_{46} &= -x^3 - x^2y + 2x^2 - x - y^2, \\
H_{47} &= -x^3 + x^2y + x^2 - x + y^2 - y, \\
H_{48} &= x^3 + x^2y + x + y, \\
H_{49} &= x, \\
H_{50} &= x^2 + x + 1, \\
H_{51} &= xy + x, \\
H_{52} &= -xy + x - y, \\
H_{53} &= -xy - y + 1, \\
H_{54} &= -2x - y,
\end{aligned}$$

$$\begin{aligned}
H_{55} &= x + 1, \\
H_{56} &= -xy + x + y + 1, \\
H_{57} &= -xy + 2x - 1, \\
H_{58} &= -x^2 - x - y + 1, \\
H_{59} &= -xy - x, \\
H_{60} &= xy - x - 1, \\
H_{61} &= -xy - 2x - 2, \\
H_{62} &= -x^2 - x + y - 1, \\
H_{63} &= -xy - x + y, \\
H_{64} &= -xy - 2x + y, \\
H_{65} &= -xy - 2y - 1, \\
H_{66} &= x + y + 1, \\
H_{67} &= -2xy - y, \\
H_{68} &= -2xy + y, \\
H_{69} &= -2xy - y^2, \\
H_{70} &= -x + y^2, \\
H_{71} &= 2xy - x - y^2 + y, \\
H_{72} &= 2xy + y, \\
H_{73} &= x + y^2 - y, \\
H_{74} &= x - y, \\
H_{75} &= x - y^2 + 1, \\
H_{76} &= -2xy + y^2 - 2y + 1, \\
H_{77} &= -x^2 + xy - 2x + y^2 - y, \\
H_{78} &= x^2 + xy + 3x + y^2 + 3y + 3, \\
H_{79} &= 2xy + y^2, \\
H_{80} &= -2xy + 2x + y - 1, \\
H_{81} &= -2xy + x + 2y - 1, \\
H_{82} &= -2x - y^2 + 2y - 1, \\
H_{83} &= -xy - 2x - y^2 - 1, \\
H_{84} &= y + 1, \\
H_{85} &= -x^2 - xy, \\
H_{86} &= 3, \\
H_{87} &= 6x - 4y - 2, \\
H_{88} &= 12, \\
H_{89} &= -x - y - 1, \\
H_{90} &= 2, \\
H_{91} &= -x^2 - y, \\
H_{92} &= x + 2, \\
H_{93} &= -4x - 5, \\
H_{94} &= 21, \\
H_{95} &= -4,
\end{aligned}$$

$$\begin{aligned}
H_{96} &= 2x + 1, \\
H_{97} &= 7, \\
H_{98} &= -x - 4, \\
H_{99} &= x - y + 1, \\
H_{100} &= 2x + 2, \\
H_{101} &= 2x + 4y - 2, \\
H_{102} &= x + y - 1, \\
H_{103} &= 2x + 4, \\
H_{104} &= x + y + 2, \\
H_{105} &= x - y + 2, \\
H_{106} &= 1.
\end{aligned}$$

The edges are $\{0, 1\}, \{0, 2\}, \{0, 3\}, \{0, 4\}, \{0, 5\}, \{0, 6\}, \{0, 13\}, \{1, 15\}, \{1, 16\}, \{1, 18\}, \{1, 20\}, \{1, 21\}, \{1, 23\}, \{1, 24\}, \{1, 87\}, \{2, 65\}, \{2, 66\}, \{2, 98\}, \{3, 21\}, \{3, 22\}, \{3, 26\}, \{3, 45\}, \{3, 71\}, \{3, 75\}, \{3, 76\}, \{3, 77\}, \{3, 99\}, \{4, 26\}, \{4, 27\}, \{4, 37\}, \{4, 55\}, \{4, 56\}, \{4, 60\}, \{4, 66\}, \{4, 67\}, \{4, 69\}, \{4, 72\}, \{4, 75\}, \{4, 79\}, \{5, 7\}, \{5, 22\}, \{5, 23\}, \{5, 26\}, \{5, 28\}, \{5, 30\}, \{5, 47\}, \{5, 57\}, \{5, 70\}, \{5, 80\}, \{5, 81\}, \{5, 90\}, \{6, 7\}, \{6, 12\}, \{6, 29\}, \{6, 73\}, \{6, 80\}, \{6, 82\}, \{6, 102\}, \{7, 8\}, \{7, 9\}, \{7, 10\}, \{7, 11\}, \{7, 86\}, \{8, 15\}, \{8, 17\}, \{8, 25\}, \{8, 26\}, \{8, 28\}, \{8, 29\}, \{8, 30\}, \{8, 31\}, \{8, 89\}, \{9, 18\}, \{9, 44\}, \{9, 49\}, \{9, 56\}, \{9, 57\}, \{9, 58\}, \{9, 63\}, \{9, 64\}, \{9, 96\}, \{10, 25\}, \{10, 30\}, \{10, 38\}, \{10, 46\}, \{10, 49\}, \{10, 68\}, \{10, 69\}, \{10, 79\}, \{10, 81\}, \{10, 90\}, \{10, 101\}, \{11, 19\}, \{11, 24\}, \{11, 42\}, \{11, 54\}, \{11, 83\}, \{11, 85\}, \{11, 90\}, \{12, 14\}, \{13, 30\}, \{13, 55\}, \{13, 66\}, \{13, 70\}, \{14, 20\}, \{14, 51\}, \{14, 52\}, \{14, 59\}, \{14, 63\}, \{14, 67\}, \{14, 68\}, \{14, 71\}, \{14, 72\}, \{14, 73\}, \{14, 74\}, \{19, 50\}, \{19, 97\}, \{26, 33\}, \{26, 40\}, \{26, 43\}, \{30, 97\}, \{31, 32\}, \{32, 41\}, \{32, 48\}, \{32, 55\}, \{32, 58\}, \{32, 62\}, \{32, 66\}, \{32, 77\}, \{32, 85\}, \{32, 105\}, \{33, 34\}, \{33, 37\}, \{33, 38\}, \{33, 39\}, \{33, 41\}, \{33, 42\}, \{33, 90\}, \{34, 35\}, \{35, 36\}, \{35, 49\}, \{35, 64\}, \{35, 65\}, \{35, 84\}, \{35, 100\}, \{36, 40\}, \{39, 40\}, \{40, 49\}, \{40, 53\}, \{40, 61\}, \{40, 76\}, \{40, 82\}, \{40, 83\}, \{40, 103\}, \{43, 44\}, \{43, 45\}, \{43, 46\}, \{43, 47\}, \{43, 48\}, \{43, 91\}, \{50, 51\}, \{50, 52\}, \{50, 53\}, \{50, 54\}, \{50, 59\}, \{50, 60\}, \{50, 61\}, \{50, 62\}, \{50, 92\}, \{50, 93\}, \{66, 78\}, \{74, 78\}, \{74, 86\}, \{78, 84\}, \{78, 104\}, \{86, 92\}, \{86, 105\}, \{87, 88\}, \{88, 90\}, \{88, 101\}, \{90, 97\}, \{93, 94\}, \{94, 95\}, \{96, 97\}, \{97, 98\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs, we use

$$\begin{aligned}
L_0 &= y, \\
Q_0 &= y^3 + 2y - 1, \\
\Delta_0 &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
&\quad + 3072y - 512, \\
L_7 &= y, \\
Q_7 &= y^3 + 2y - 1, \\
\Delta_7 &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2
\end{aligned}$$

$$\begin{aligned}
& + 3072y - 512, \\
L_{12} &= y, \\
Q_{12} &= y^3 + 2y - 1, \\
\Delta_{12} &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
& + 3072y - 512, \\
Q_{18} &= x^4 + 4x^3 + x^2y + 2x^2 + 2xy - 3x + y^2 - y + 3, \\
M_{18} &= x^6 + 6x^5 + 11x^4 - 3x^3 - 5x^2 + 9x - 4, \\
E_{18} &= 3x^4 + 12x^3 + 6x^2 - 8x + 11, \\
L_{22} &= y, \\
Q_{22} &= y^3 + 2y - 1, \\
\Delta_{22} &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
& + 3072y - 512, \\
L_{27} &= y, \\
Q_{27} &= y^3 + 2y - 1, \\
\Delta_{27} &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
& + 3072y - 512, \\
Q_{31} &= x^4 + 6x^3 - 2x^2y + 9x^2 - 6xy + 4x + 4y^2 + 8, \\
M_{31} &= x^6 + 9x^5 + 11x^4 + 31x^3 + 20x^2 + 24x + 8, \\
E_{31} &= 12x^4 + 72x^3 + 108x^2 + 64x + 128, \\
L_{36} &= y, \\
Q_{36} &= y^3 + 2y - 1, \\
\Delta_{36} &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
& + 3072y - 512, \\
Q_{54} &= 4x^2 - 2xy + x + y^2 + 2, \\
M_{54} &= 2x^4 - 8x^3 - 2x^2 - 4x - 1, \\
E_{54} &= 12x^2 + 4x + 8, \\
Q_{58} &= x^4 + 2x^3 - x^2y - x^2 - xy - x + y^2 + y + 3, \\
M_{58} &= x^6 + 3x^5 - 2x^4 - 4x^3 + 3x^2 + 4x - 2, \\
E_{58} &= 3x^4 + 6x^3 - 3x^2 - 2x + 11, \\
Q_{62} &= x^4 + 2x^3 + x^2y + 3x^2 + xy + 3x + y^2 + y + 3, \\
M_{62} &= x^6 + 3x^5 + 8x^4 + 8x^3 + 9x^2 + 6x + 2, \\
E_{62} &= 3x^4 + 6x^3 + 9x^2 + 10x + 11, \\
Q_{66} &= x^2 - xy + 3x + y^2 - y + 3, \\
M_{66} &= 2x^4 - x^3 - 4x^2 - 6x - 4, \\
E_{66} &= 3x^2 + 10x + 11, \\
Q_{74} &= x^2 + xy + x + y^2 + 2, \\
M_{74} &= 2x^4 + x^3 + x^2 + 2x - 1, \\
E_{74} &= 3x^2 + 4x + 8, \\
L_{84} &= y, \\
Q_{84} &= y^3 + 2y - 1,
\end{aligned}$$

$$\begin{aligned}
\Delta_{84} &= 512y^9 + 3072y^7 - 1536y^6 + 6144y^5 - 6171y^4 + 5632y^3 - 6144y^2 \\
&\quad + 3072y - 512, \\
Q_{87} &= 36x^2 + 24xy - 8x + 16y^2 - 8y + 36, \\
M_{87} &= 128x^4 + 216x^3 - 120x^2 + 232x - 136, \\
E_{87} &= 1728x^2 - 128x + 2240, \\
Q_{89} &= x^2 - xy + 3x + y^2 - y + 3, \\
M_{89} &= 2x^4 - x^3 - 4x^2 - 6x - 4, \\
E_{89} &= 3x^2 + 10x + 11, \\
Q_{91} &= x^4 - x^2y + x + y^2 + 2, \\
M_{91} &= x^6 - 2x^4 + x^3 + 2x^2 + 1, \\
E_{91} &= 3x^4 + 4x + 8, \\
Q_{99} &= x^2 + xy + 3x + y^2 + y + 3, \\
M_{99} &= 2x^4 + x^3 + 4x^2 + 6x + 2, \\
E_{99} &= 3x^2 + 10x + 11, \\
Q_{101} &= 4x^2 - 8xy + 8x + 16y^2 + 8y + 36, \\
M_{101} &= 128x^4 - 8x^3 - 8x^2 - 56x + 8, \\
E_{101} &= 192x^2 + 640x + 2240, \\
Q_{102} &= x^2 - xy - x + y^2 + y + 3, \\
M_{102} &= 2x^4 - x^3 + 2x^2 - 4x + 2, \\
E_{102} &= 3x^2 - 2x + 11, \\
Q_{104} &= x^2 - xy + 5x + y^2 - 2y + 6, \\
M_{104} &= 2x^4 - x^3 - 7x^2 - 16x - 13, \\
E_{104} &= 3x^2 + 16x + 20, \\
Q_{105} &= x^2 + xy + 5x + y^2 + 2y + 6, \\
M_{105} &= 2x^4 + x^3 + 7x^2 + 16x + 11, \\
E_{105} &= 3x^2 + 16x + 20,
\end{aligned}$$

and the identities

$$F = 2x^4 + L_0x + Q_0, \tag{4}$$

$$\Delta_0 = 512Q_0^3 - 27L_0^4, \tag{5}$$

$$4H_1 = (3x - 2y - 1)^2 + 3(x + 1)^2. \tag{6}$$

$$4H_2 = (2x + 3)^2 + 3. \tag{7}$$

$$4H_3 = (x - 2y + 2)^2 + 3x^2. \tag{8}$$

$$4H_4 = (x + 2y + 1)^2 + 3(x + 1)^2. \tag{9}$$

$$4H_5 = (x - 2y + 1)^2 + 3(x - 1)^2. \tag{10}$$

$$4H_6 = (x + 2y - 2)^2 + 3x^2. \tag{11}$$

$$F = 2x^4 + L_7x + Q_7, \tag{12}$$

$$\Delta_7 = 512Q_7^3 - 27L_7^4, \tag{13}$$

$$4H_8 = (3x + 2y + 1)^2 + 3(x - 1)^2. \tag{14}$$

$$4H_9 = (2x - 1)^2 + 3. \tag{15}$$

$$4H_{10} = (x + 2y - 1)^2 + 3(x - 1)^2. \quad (16)$$

$$4H_{11} = (2x + 2y - 1)^2 + 3. \quad (17)$$

$$F = 2x^4 + L_{12}x + Q_{12}, \quad (18)$$

$$\Delta_{12} = 512Q_{12}^3 - 27L_{12}^4, \quad (19)$$

$$4H_{13} = (x + 2y)^2 + 3x^2. \quad (20)$$

$$4H_{14} = (x - 2y)^2 + 3x^2. \quad (21)$$

$$-F = Q_{18}(-H_{18}) - M_{18}, \quad (22)$$

$$4Q_{18} = (x^2 + 2x + 2y - 1)^2 + E_{18}. \quad (23)$$

$$F = 2x^4 + L_{22}x + Q_{22}, \quad (24)$$

$$\Delta_{22} = 512Q_{22}^3 - 27L_{22}^4, \quad (25)$$

$$F = 2x^4 + L_{27}x + Q_{27}, \quad (26)$$

$$\Delta_{27} = 512Q_{27}^3 - 27L_{27}^4, \quad (27)$$

$$4H_{28} = (x + 2y)^2 + 3(x - 2)^2 - 12. \quad (28)$$

$$8F = Q_{31}H_{31} - M_{31}, \quad (29)$$

$$16Q_{31} = (-2x^2 - 6x + 8y)^2 + E_{31}. \quad (30)$$

$$4H_{32} = (x + 2y - 1)^2 + 3(x + 1)^2. \quad (31)$$

$$4H_{33} = (x^2 - 2y)^2 + 3(x^2)^2. \quad (32)$$

$$4H_{35} = (x - 2y - 1)^2 + 3(x + 1)^2. \quad (33)$$

$$F = 2x^4 + L_{36}x + Q_{36}, \quad (34)$$

$$\Delta_{36} = 512Q_{36}^3 - 27L_{36}^4, \quad (35)$$

$$4H_{40} = (x - 2y)^2 + 3(x + 2)^2. \quad (36)$$

$$4H_{43} = (x^2 + x + 2y)^2 + 3(x^2 - x)^2. \quad (37)$$

$$4H_{50} = (2x + 1)^2 + 3. \quad (38)$$

$$-F = Q_{54}H_{54} - M_{54}, \quad (39)$$

$$4Q_{54} = (-2x + 2y)^2 + E_{54}. \quad (40)$$

$$F = Q_{58}(-H_{58}) - M_{58}, \quad (41)$$

$$4Q_{58} = (-x^2 - x + 2y + 1)^2 + E_{58}. \quad (42)$$

$$-F = Q_{62}(-H_{62}) - M_{62}, \quad (43)$$

$$4Q_{62} = (x^2 + x + 2y + 1)^2 + E_{62}. \quad (44)$$

$$-F = Q_{66}(-H_{66}) - M_{66}, \quad (45)$$

$$4Q_{66} = (-x + 2y - 1)^2 + E_{66}. \quad (46)$$

$$-F = Q_{74}H_{74} - M_{74}, \quad (47)$$

$$4Q_{74} = (x + 2y)^2 + E_{74}. \quad (48)$$

$$H_{76} = (y)(-2x + y - 2) + 1, \quad (49)$$

$$4H_{78} = (x + 2y + 3)^2 + 3(x + 1)^2. \quad (50)$$

$$(-H_{83}) = (x + y - 2)(y + 2) + 5, \quad (51)$$

$$F = 2x^4 + L_{84}x + Q_{84}, \quad (52)$$

$$\Delta_{84} = 512Q_{84}^3 - 27L_{84}^4, \quad (53)$$

$$-64F = Q_{87}H_{87} - M_{87}, \quad (54)$$

$$64Q_{87} = (24x + 32y - 8)^2 + E_{87}. \quad (55)$$

$$-F = Q_{89}H_{89} - M_{89}, \quad (56)$$

$$4Q_{89} = (-x + 2y - 1)^2 + E_{89}. \quad (57)$$

$$F = Q_{91}(-H_{91}) - M_{91}, \quad (58)$$

$$4Q_{91} = (-x^2 + 2y)^2 + E_{91}. \quad (59)$$

$$-F = Q_{99}H_{99} - M_{99}, \quad (60)$$

$$4Q_{99} = (x + 2y + 1)^2 + E_{99}. \quad (61)$$

$$-64F = Q_{101}(-H_{101}) - M_{101}, \quad (62)$$

$$64Q_{101} = (-8x + 32y + 8)^2 + E_{101}. \quad (63)$$

$$-F = Q_{102}(-H_{102}) - M_{102}, \quad (64)$$

$$4Q_{102} = (-x + 2y + 1)^2 + E_{102}. \quad (65)$$

$$-F = Q_{104}(-H_{104}) - M_{104}, \quad (66)$$

$$4Q_{104} = (-x + 2y - 2)^2 + E_{104}. \quad (67)$$

$$-F = Q_{105}H_{105} - M_{105}, \quad (68)$$

$$4Q_{105} = (x + 2y + 2)^2 + E_{105}. \quad (69)$$

4 The cubic certificate

The certificate consists of 40 polynomials J_i with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i \in \mathbb{Z}[\omega]$, listed in Table 3, and of 163 *edges* (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$, listed below. For an edge (i, j) we put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij} \bmod 3$, and $\rho_{ij} = 0$ if i and j are not joined by an edge. The edges $(i, j; w_{ij})$ are $(0, 4; 1)$, $(0, 5; 2)$, $(0, 14; 1)$, $(0, 21; 1)$, $(0, 24; 1)$, $(0, 34; 1)$, $(1, 3; 1)$, $(1, 5; 1)$, $(1, 11; 2)$, $(1, 12; 1)$, $(1, 13; 2)$, $(1, 15; 1)$, $(1, 22; 1)$, $(1, 38; 2)$, $(1, 39; 2)$, $(2, 6; 2)$, $(2, 7; 2)$, $(2, 11; 1)$, $(2, 13; 1)$, $(2, 14; 1)$, $(2, 16; 2)$, $(2, 24; 2)$, $(2, 34; 2)$, $(2, 37; 2)$, $(2, 38; 2)$, $(2, 39; 1)$, $(3, 4; 2)$, $(3, 5; 1)$, $(3, 7; 2)$, $(3, 10; 2)$, $(3, 11; 1)$, $(3, 12; 2)$, $(3, 16; 2)$, $(3, 17; 2)$, $(3, 18; 2)$, $(3, 28; 2)$, $(3, 29; 2)$, $(3, 31; 1)$, $(3, 32; 2)$, $(3, 37; 2)$, $(3, 38; 1)$, $(3, 39; 2)$, $(4, 7; 1)$, $(4, 10; 2)$, $(4, 11; 2)$, $(4, 15; 2)$, $(4, 29; 2)$, $(4, 30; 2)$, $(5, 6; 1)$, $(5, 8; 1)$, $(5, 11; 1)$, $(5, 16; 2)$, $(5, 17; 2)$, $(5, 20; 2)$, $(5, 21; 1)$, $(5, 27; 2)$, $(5, 29; 2)$, $(5, 31; 2)$, $(6, 13; 1)$, $(6, 15; 2)$, $(6, 24; 2)$, $(6, 28; 1)$, $(6, 31; 1)$, $(6, 37; 2)$, $(7, 8; 1)$, $(7, 11; 2)$, $(7, 24; 1)$, $(7, 28; 2)$, $(7, 35; 2)$, $(7, 39; 2)$, $(8, 9; 1)$, $(8, 10; 1)$, $(8, 16; 2)$, $(8, 20; 1)$, $(8, 21; 2)$, $(8, 27; 1)$, $(8, 31; 2)$, $(8, 32; 1)$, $(8, 34; 1)$, $(9, 22; 2)$, $(9, 23; 2)$, $(9, 29; 1)$, $(9, 34; 2)$, $(9, 35; 2)$, $(9, 38; 1)$, $(9, 39; 1)$, $(10, 18; 1)$, $(10, 38; 2)$, $(10, 39; 1)$, $(11, 21; 1)$, $(11, 36; 2)$, $(12, 26; 2)$, $(12, 34; 1)$, $(13, 16; 1)$, $(13, 19; 2)$, $(13, 25; 1)$, $(13, 38; 1)$, $(14, 24; 1)$, $(14, 26; 2)$, $(14, 32; 2)$, $(14, 35; 2)$, $(15, 19; 1)$, $(15, 26; 1)$, $(15, 27; 1)$, $(15, 29; 1)$, $(15, 35; 2)$, $(15, 38; 1)$, $(15, 39; 1)$, $(16, 17; 2)$, $(16, 24; 1)$, $(16, 34; 2)$, $(16, 37; 1)$, $(17, 30; 1)$, $(17, 31; 2)$, $(17, 37; 1)$, $(17, 39; 1)$, $(18, 33; 2)$, $(18, 37; 2)$, $(19, 21; 2)$, $(19, 27; 2)$, $(19, 31; 2)$, $(19, 35; 2)$, $(19, 37; 1)$, $(19, 39; 2)$, $(20, 21; 2)$, $(20, 30; 1)$, $(20, 33; 2)$, $(20, 34; 2)$, $(20, 35; 2)$, $(20, 37; 2)$, $(21, 27; 2)$, $(21, 35; 2)$, $(21, 39; 2)$, $(22, 23; 2)$, $(22, 25; 1)$, $(22, 26; 2)$, $(22, 31; 1)$, $(22, 32; 1)$, $(23, 31; 2)$, $(23, 32; 1)$, $(23, 37; 1)$, $(23, 38; 2)$, $(25, 31; 1)$, $(26, 28; 2)$, $(26, 29; 2)$, $(26, 31; 2)$, $(26, 33; 1)$, $(26, 37; 2)$, $(27, 28; 1)$, $(27, 39; 1)$, $(28, 29; 1)$, $(28, 31; 1)$, $(28, 34; 2)$, $(28, 38; 1)$, $(29, 31; 2)$, $(29, 35; 1)$, $(29, 39; 2)$, $(30, 38; 2)$, $(31, 32; 1)$, $(31, 34; 2)$, $(31, 37; 2)$, $(31, 38; 1)$, $(34, 39; 2)$. Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $J_i = \varepsilon_i(\eta - r_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$ and a polynomial r_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 3 gives $\eta = r_i(\xi)$ for the linear vertices. For a linear vertex i , we put $f_i(t) = F$ and $g_{ij}(t) = J_j$ evaluated at $\xi = t$, $\eta = r_i(t)$; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 4 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (70)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For every other vertex i , Table 4 gives an integer $D_i \geq 1$ and polynomials T_i , U_i and V_i in x and y with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 c_i \prod_j J_j^{\rho_{ij}} - T_i^3 = U_i F + V_i J_i. \quad (71)$$

For every edge (i, j) with a linear endpoint, Table 5 gives the first linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{kl})$, where $\{k, l\} = \{i, j\}$; it is non-zero. For the other edges, Table 6 gives a non-zero constant $D_{ij} \in \mathbb{Z}[\omega]$ and polynomials λ_{ij} , μ_{ij} and ν_{ij} with coefficients in $\mathbb{Z}[\omega]$ such that

$$\lambda_{ij} F + \mu_{ij} J_i + \nu_{ij} J_j = D_{ij}. \quad (72)$$

These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$.

5 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. For $i \in \{16, 17, 23, 25, 26, 30, 33, 37, 38, 41, 42, 43, 44, 46, 48, 52, 55, 57, 60, 63, 64, 65, 67, 68, 69, 72, 79, 81, 93, 96\}$, checking the 2^2 residue pairs modulo 2, we find that F and H_i have no common zero modulo 2, so $H_i \neq 0$. For $i \in \{29, 49, 56, 71\}$, checking the 3^2 residue pairs modulo 3, we find that F and H_i have no common zero modulo 3, so $H_i \neq 0$. For $i \in \{21, 24, 47, 53, 77, 100\}$, checking the 4^2 residue pairs modulo 4, we find that F and H_i have no common zero modulo 4, so $H_i \neq 0$. For $i \in \{15, 34, 92, 103\}$, checking the 7^2 residue pairs modulo 7, we find that F and H_i have no common zero modulo 7, so $H_i \neq 0$. Checking the 8^2 residue pairs modulo 8, we find that F and H_{82} have no common zero modulo 8, so $H_{82} \neq 0$. For $i \in \{20, 45, 61, 80\}$, checking the 11^2 residue pairs modulo 11, we find that F and H_i have no common zero modulo 11, so $H_i \neq 0$. Checking the 13^2 residue pairs modulo 13, we find that F and H_{39} have no common zero modulo 13, so $H_{39} \neq 0$. For $i \in \{51, 59\}$, checking the 19^2 residue pairs modulo 19, we find that F and H_i have no common zero modulo 19, so $H_i \neq 0$. Checking the 23^2 residue pairs modulo 23, we find that F and H_{98} have no common zero modulo 23, so $H_{98} \neq 0$. We have $H_{85} = -x(x + y)$, and checking residue pairs, we find that F and x have no common zero modulo 3 and F and $x + y$ have no common zero modulo 2; hence $H_{85} \neq 0$. The other non-constant H_i are non-zero by Step 2. For $i \in \{0, 1, 2, 3, 4, 5, 7, 8, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 24, 25, 30, 31, 32, 33, 34, 35\}$, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{6, 9, 21, 23, 27, 37, 38\}$, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{28, 29, 36\}$, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{26, 39\}$, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. Also, all c_i are non-zero.

Step 2: signs. *The sign of H_0 .* Suppose that $H_0 \leq 0$. Since $H_0 = -y$, this means $y \geq 0$. By (4), F is $2x^4 + L_0x + Q_0$, where L_0 and Q_0 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_0 and Δ_0 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_0 > 0$ and $\Delta_0 > 0$. For $y = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every x , so (1) has no solution with this y . Hence $Q_0 > 0$ and $\Delta_0 > 0$, so Lemma 7 with $a = 2$, $b = L_0$, $c = Q_0$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_0 > 0$.

The sign of H_1 . If $H_1 \leq 0$, then (6) gives $(3x - 2y - 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|3x - 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -2)$, and it is not a solution of (1) with $H_1 \leq 0$. Hence $H_1 > 0$.

The sign of H_2 . By (7), $H_2 > 0$.

The sign of H_3 . If $H_3 \leq 0$, then (8) gives $(x - 2y + 2)^2 \leq 0$ and $3x^2 \leq 0$, so $|x - 2y + 2| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 1)$, and it is not a solution of (1) with $H_3 \leq 0$. Hence $H_3 > 0$.

The sign of H_4 . If $H_4 \leq 0$, then (9) gives $(x + 2y + 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x + 2y + 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, 0)$, and it is not a solution of (1) with $H_4 \leq 0$. Hence $H_4 > 0$.

The sign of H_5 . If $H_5 \leq 0$, then (10) gives $(x - 2y + 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x - 2y + 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, 1)$, and it is not a solution of (1) with $H_5 \leq 0$. Hence $H_5 > 0$.

The sign of H_6 . If $H_6 \leq 0$, then (11) gives $(x + 2y - 2)^2 \leq 0$ and $3x^2 \leq 0$, so $|x + 2y - 2| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 1)$, and it is not a solution of (1) with $H_6 \leq 0$. Hence $H_6 > 0$.

The sign of H_7 . Suppose that $H_7 \leq 0$. Since $H_7 = -y + 1$, this means $y \geq 1$. By (12), F is $2x^4 + L_7x + Q_7$, where L_7 and Q_7 depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_7 and Δ_7 as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_7 > 0$ and $\Delta_7 > 0$. Hence $Q_7 > 0$ and $\Delta_7 > 0$, so Lemma 7 with $a = 2$, $b = L_7$, $c = Q_7$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_7 > 0$.

The sign of H_8 . If $H_8 \leq 0$, then (14) gives $(3x + 2y + 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|3x + 2y + 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, -2)$, and it is not a solution of (1) with $H_8 \leq 0$. Hence $H_8 > 0$.

The sign of H_9 . By (15), $H_9 > 0$.

The sign of H_{10} . If $H_{10} \leq 0$, then (16) gives $(x + 2y - 1)^2 \leq 0$ and $3(x - 1)^2 \leq 0$, so $|x + 2y - 1| \leq 0$ and $|x - 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(1, 0)$, and it is not a solution of (1) with $H_{10} \leq 0$. Hence $H_{10} > 0$.

The sign of H_{11} . By (17), $H_{11} > 0$.

The sign of H_{12} . Suppose that $H_{12} \leq 0$. Since $H_{12} = -2y + 1$, this means $y \geq 1$. By (18), F is $2x^4 + L_{12}x + Q_{12}$, where L_{12} and Q_{12} depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_{12} and Δ_{12} as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_{12} > 0$ and $\Delta_{12} > 0$. Hence $Q_{12} > 0$ and $\Delta_{12} > 0$, so Lemma 7 with $a = 2$, $b = L_{12}$, $c = Q_{12}$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_{12} > 0$.

The sign of H_{13} . If $H_{13} \leq 0$, then (20) gives $(x + 2y)^2 \leq 0$ and $3x^2 \leq 0$, so $|x + 2y| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 0)$, and it is not a solution of (1) with $H_{13} \leq 0$. Hence $H_{13} > 0$.

The sign of H_{14} . If $H_{14} \leq 0$, then (21) gives $(x - 2y)^2 \leq 0$ and $3x^2 \leq 0$, so $|x - 2y| \leq 0$ and $|x| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(0, 0)$, and it is not a solution of (1) with $H_{14} \leq 0$. Hence $H_{14} > 0$.

The sign of H_{18} . Since $F(x, y) = 0$, (22) gives $Q_{18}(-H_{18}) = M_{18}$, and M_{18} and E_{18} depend only on x . We show that $M_{18} > 0$ and $E_{18} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -3$, write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{18} and E_{18} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{18} > 0$ and $E_{18} > 0$. It remains to consider $-2 \leq x \leq 0$. For $x = -2$ we have $M_{18} > 0$ and $E_{18} > 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has

no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . Hence $M_{18} > 0$ and $E_{18} > 0$, so (23) gives $4Q_{18} \geq E_{18} > 0$, and $Q_{18}(-H_{18}) = M_{18} > 0$ gives $(-H_{18}) > 0$. Hence $H_{18} < 0$.

The sign of H_{22} . Suppose that $(-H_{22}) \leq 0$. Since $(-H_{22}) = -y + 1$, this means $y \geq 1$. By (24), F is $2x^4 + L_{22}x + Q_{22}$, where L_{22} and Q_{22} depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_{22} and Δ_{22} as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_{22} > 0$ and $\Delta_{22} > 0$. Hence $Q_{22} > 0$ and $\Delta_{22} > 0$, so Lemma 7 with $a = 2$, $b = L_{22}$, $c = Q_{22}$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $(-H_{22}) > 0$. Hence $H_{22} < 0$.

The sign of H_{27} . Suppose that $(-H_{27}) \leq 0$. Since $(-H_{27}) = -y$, this means $y \geq 0$. By (26), F is $2x^4 + L_{27}x + Q_{27}$, where L_{27} and Q_{27} depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_{27} and Δ_{27} as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_{27} > 0$ and $\Delta_{27} > 0$. For $y = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every x , so (1) has no solution with this y . Hence $Q_{27} > 0$ and $\Delta_{27} > 0$, so Lemma 7 with $a = 2$, $b = L_{27}$, $c = Q_{27}$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $(-H_{27}) > 0$. Hence $H_{27} < 0$.

The sign of H_{28} . If $H_{28} \leq 0$, then (28) gives $(x + 2y)^2 \leq 12$ and $3(x - 2)^2 \leq 12$, so $|x + 2y| \leq 3$ and $|x - 2| \leq 2$. Solving these two linear forms for x and y , we find that the only integer points satisfying both inequalities are $(0, -1)$, $(0, 0)$, $(0, 1)$, $(1, -2)$, $(1, -1)$, $(1, 0)$, $(1, 1)$, $(2, -2)$, $(2, -1)$, $(2, 0)$, $(3, -3)$, $(3, -2)$, $(3, -1)$, $(3, 0)$, $(4, -3)$, $(4, -2)$, $(4, -1)$, and none of them is a solution of (1) with $H_{28} \leq 0$. Hence $H_{28} > 0$.

The sign of H_{31} . Since $F(x, y) = 0$, (29) gives $Q_{31}H_{31} = M_{31}$, and M_{31} and E_{31} depend only on x . We show that $M_{31} > 0$ and $E_{31} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -9$, write $x = -9 - z$ with $z \geq 0$. In both cases, expanding M_{31} and E_{31} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{31} > 0$ and $E_{31} > 0$. It remains to consider $-8 \leq x \leq 0$. For $x = 0$ we have $M_{31} > 0$ and $E_{31} > 0$. For $x \in \{-7, -5, -3, -1\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = -6$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -8$ and $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with such x . For $x = -4$, we have $F(x, y) \not\equiv 0 \pmod{23}$ for every y , as one checks for the 23 residues of y modulo 23, so (1) has no solution with this x . Hence $M_{31} > 0$ and $E_{31} > 0$, so (30) gives $16Q_{31} \geq E_{31} > 0$, and $Q_{31}H_{31} = M_{31} > 0$ gives $H_{31} > 0$.

The sign of H_{32} . If $H_{32} \leq 0$, then (31) gives $(x + 2y - 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x + 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, 1)$, and it is not a solution of (1) with $H_{32} \leq 0$. Hence $H_{32} > 0$.

The sign of H_{33} . By (32), $H_{33} \geq 0$, and $H_{33} \neq 0$ by Step 1, so $H_{33} > 0$.

The sign of H_{35} . If $H_{35} \leq 0$, then (33) gives $(x - 2y - 1)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x - 2y - 1| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -1)$, and it is not a solution of (1) with $H_{35} \leq 0$. Hence $H_{35} > 0$.

The sign of H_{36} . Suppose that $H_{36} \leq 0$. Since $H_{36} = -y - 1$, this means $y \geq -1$. By (34), F is $2x^4 + L_{36}x + Q_{36}$, where L_{36} and Q_{36} depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_{36} and Δ_{36} as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_{36} > 0$ and $\Delta_{36} > 0$. For $y = -1$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every x , so (1) has no solution with this y . For $y = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every x , so (1) has no solution with this y . Hence $Q_{36} > 0$ and $\Delta_{36} > 0$, so Lemma 7 with $a = 2$, $b = L_{36}$, $c = Q_{36}$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $H_{36} > 0$.

The sign of H_{40} . If $H_{40} \leq 0$, then (36) gives $(x - 2y)^2 \leq 0$ and $3(x + 2)^2 \leq 0$, so $|x - 2y| \leq 0$ and $|x + 2| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-2, -1)$, and it is not a solution of (1) with $H_{40} \leq 0$. Hence

$H_{40} > 0$.

The sign of H_{43} . By (37), $H_{43} \geq 0$, and $H_{43} \neq 0$ by Step 1, so $H_{43} > 0$.

The sign of H_{50} . By (38), $H_{50} > 0$.

The sign of H_{54} . Since $F(x, y) = 0$, (39) gives $Q_{54}H_{54} = M_{54}$, and M_{54} and E_{54} depend only on x . We show that $M_{54} > 0$ and $E_{54} > 0$. If $x \geq 5$, write $x = 5 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{54} and E_{54} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{54} > 0$ and $E_{54} > 0$. It remains to consider $0 \leq x \leq 4$. For $x = 1$ and $x = 3$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = 4$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , as one checks for the 5 residues of y modulo 5, so (1) has no solution with this x . For $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , as one checks for the 29 residues of y modulo 29, so (1) has no solution with this x . Hence $M_{54} > 0$ and $E_{54} > 0$, so (40) gives $4Q_{54} \geq E_{54} > 0$, and $Q_{54}H_{54} = M_{54} > 0$ gives $H_{54} > 0$.

The sign of H_{58} . Since $F(x, y) = 0$, (41) gives $Q_{58}(-H_{58}) = M_{58}$, and M_{58} and E_{58} depend only on x . We show that $M_{58} > 0$ and $E_{58} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{58} and E_{58} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{58} > 0$ and $E_{58} > 0$. It remains to consider $-3 \leq x \leq 0$. For $x = -3$ and $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_{58} > 0$ and $E_{58} > 0$, so (42) gives $4Q_{58} \geq E_{58} > 0$, and $Q_{58}(-H_{58}) = M_{58} > 0$ gives $(-H_{58}) > 0$. Hence $H_{58} < 0$.

The sign of H_{62} . Since $F(x, y) = 0$, (43) gives $Q_{62}(-H_{62}) = M_{62}$, and M_{62} and E_{62} depend only on x . We show that $M_{62} > 0$ and $E_{62} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{62} and E_{62} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{62} > 0$ and $E_{62} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{62} > 0$ and $E_{62} > 0$. Hence $M_{62} > 0$ and $E_{62} > 0$, so (44) gives $4Q_{62} \geq E_{62} > 0$, and $Q_{62}(-H_{62}) = M_{62} > 0$ gives $(-H_{62}) > 0$. Hence $H_{62} < 0$.

The sign of H_{66} . Since $F(x, y) = 0$, (45) gives $Q_{66}(-H_{66}) = M_{66}$, and M_{66} and E_{66} depend only on x . We show that $M_{66} > 0$ and $E_{66} > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{66} and E_{66} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{66} > 0$ and $E_{66} > 0$. It remains to consider $-1 \leq x \leq 2$. For $x = -1$ we have $M_{66} > 0$ and $E_{66} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , as one checks for the 29 residues of y modulo 29, so (1) has no solution with this x . Hence $M_{66} > 0$ and $E_{66} > 0$, so (46) gives $4Q_{66} \geq E_{66} > 0$, and $Q_{66}(-H_{66}) = M_{66} > 0$ gives $(-H_{66}) > 0$. Hence $H_{66} < 0$.

The sign of H_{70} . Suppose that $H_{70} \leq 0$, and put $Y = -H_{70}$, so that $Y \geq 0$. Since $H_{70} = -x + y^2$, we have $x = y^2 + Y$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{70,1}(X, Y)$ in the first case and $F(x, y) = G_{70,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{70,1}(X, Y) &= 2X^8 + 8X^6Y + 12X^4Y^2 + 2X^3 + 8X^2Y^3 + XY + 2X + 2Y^4 - 1, \\ G_{70,2}(X, Y) &= 2X^8 + 16X^7 + 8X^6Y + 56X^6 + 48X^5Y + 112X^5 + 12X^4Y^2 \\ &\quad + 120X^4Y + 140X^4 + 48X^3Y^2 + 160X^3Y + 110X^3 \\ &\quad + 8X^2Y^3 + 72X^2Y^2 + 120X^2Y + 50X^2 + 16XY^3 + 48XY^2 \end{aligned}$$

$$+ 47XY + 8X + 2Y^4 + 8Y^3 + 12Y^2 + 7Y - 3.$$

Expanding $G_{70,1}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{70,1}(i, 1 + Y)$ and $G_{70,1}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{70,1}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{70,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{70,2}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{70,2}(i, 1 + Y)$ and $G_{70,2}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{70,2}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{70,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{70} > 0$.

The sign of H_{73} . Suppose that $H_{73} \leq 0$, and put $Y = -H_{73}$, so that $Y \geq 0$. Since $H_{73} = x + y^2 - y$, we have $x = -y^2 + y - Y$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{73,1}(X, Y)$ in the first case and $F(x, y) = G_{73,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{73,1}(X, Y) &= 2X^8 - 8X^7 + 8X^6Y + 12X^6 - 24X^5Y - 8X^5 + 12X^4Y^2 \\ &\quad + 24X^4Y + 2X^4 - 24X^3Y^2 - 8X^3Y + 8X^2Y^3 + 12X^2Y^2 \\ &\quad + X^2 - 8XY^3 - XY + 2X + 2Y^4 - 1, \\ G_{73,2}(X, Y) &= 2X^8 + 24X^7 + 8X^6Y + 124X^6 + 72X^5Y + 360X^5 + 12X^4Y^2 \\ &\quad + 264X^4Y + 642X^4 + 72X^3Y^2 + 504X^3Y + 720X^3 \\ &\quad + 8X^2Y^3 + 156X^2Y^2 + 528X^2Y + 497X^2 + 24XY^3 \\ &\quad + 144XY^2 + 289XY + 192X + 2Y^4 + 16Y^3 + 48Y^2 + 65Y + 30. \end{aligned}$$

Expanding $G_{73,1}(1 + X, 1 + Y)$, and, for $i = 0$, $G_{73,1}(i, 1 + Y)$ and $G_{73,1}(1 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{73,1}(i, j) \neq 0$ for $i = j = 0$. Hence $G_{73,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. All coefficients of $G_{73,2}$ have the same sign and its constant term is non-zero, so $G_{73,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{73} > 0$.

The sign of H_{74} . Since $F(x, y) = 0$, (47) gives $Q_{74}H_{74} = M_{74}$, and M_{74} and E_{74} depend only on x . We show that $M_{74} > 0$ and $E_{74} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{74} and E_{74} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{74} > 0$ and $E_{74} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . Hence $M_{74} > 0$ and $E_{74} > 0$, so (48) gives $4Q_{74} \geq E_{74} > 0$, and $Q_{74}H_{74} = M_{74} > 0$ gives $H_{74} > 0$.

The sign of H_{75} . Suppose that $(-H_{75}) \leq 0$, and put $Y = H_{75}$, so that $Y \geq 0$. Since $(-H_{75}) = -x + y^2 - 1$, we have $x = y^2 + Y - 1$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{75,1}(X, Y)$ in the first case and $F(x, y) = G_{75,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{75,1}(X, Y) &= 2X^8 + 8X^6Y - 8X^6 + 12X^4Y^2 - 24X^4Y + 12X^4 + 2X^3 \\ &\quad + 8X^2Y^3 - 24X^2Y^2 + 24X^2Y - 8X^2 + XY + X + 2Y^4 - 8Y^3 \\ &\quad + 12Y^2 - 8Y + 1, \\ G_{75,2}(X, Y) &= 2X^8 + 16X^7 + 8X^6Y + 48X^6 + 48X^5Y + 64X^5 + 12X^4Y^2 \\ &\quad + 96X^4Y + 32X^4 + 48X^3Y^2 + 64X^3Y - 2X^3 + 8X^2Y^3 \\ &\quad + 48X^2Y^2 - 6X^2 + 16XY^3 - XY - 7X + 2Y^4 - Y - 4. \end{aligned}$$

Expanding $G_{75,1}(2 + X, 2 + Y)$, and, for $0 \leq i < 2$, $G_{75,1}(i, 2 + Y)$ and $G_{75,1}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{75,1}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{75,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{75,2}(2 + X, 2 + Y)$, and, for

$0 \leq i < 2$, $G_{75,2}(i, 2 + Y)$ and $G_{75,2}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{75,2}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{75,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{75}) > 0$. Hence $H_{75} < 0$.

The sign of H_{76} . Suppose that $H_{76} \leq 0$. Then (49) gives $(y)(-2x + y - 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $y \geq 1$ and $-2x + y \leq 1$, or $y \leq -1$ and $-2x + y \geq 3$. In the first case, write $y = X + 1$ and $-2x + y = -Y + 1$ with integers $X, Y \geq 0$, so that $x = \frac{X+Y}{2}$ and $y = X + 1$. Then $8F(x, y) = G_{76,1}(X, Y)$, where

$$G_{76,1}(X, Y) = X^4 + 4X^3Y + 8X^3 + 6X^2Y^2 + 28X^2 + 4XY^3 + 4XY + 44X + Y^4 + 4Y + 16.$$

All coefficients of $G_{76,1}$ have the same sign and its constant term is non-zero, so $G_{76,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. In the second case, write $y = -X - 1$ and $-2x + y = Y + 3$ with integers $X, Y \geq 0$, so that $x = \frac{-X-Y-4}{2}$ and $y = -X - 1$. Then $8F(x, y) = G_{76,2}(X, Y)$, where

$$G_{76,2}(X, Y) = X^4 + 4X^3Y + 8X^3 + 6X^2Y^2 + 48X^2Y + 76X^2 + 4XY^3 + 48XY^2 + 196XY + 236X + Y^4 + 16Y^3 + 96Y^2 + 260Y + 240.$$

All coefficients of $G_{76,2}$ have the same sign and its constant term is non-zero, so $G_{76,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{76} > 0$.

The sign of H_{78} . If $H_{78} \leq 0$, then (50) gives $(x + 2y + 3)^2 \leq 0$ and $3(x + 1)^2 \leq 0$, so $|x + 2y + 3| \leq 0$ and $|x + 1| \leq 0$. Solving these two linear forms for x and y , we find that the only integer point satisfying both inequalities is $(-1, -1)$, and it is not a solution of (1) with $H_{78} \leq 0$. Hence $H_{78} > 0$.

The sign of H_{83} . Suppose that $(-H_{83}) \leq 0$. Then (51) gives $(x + y - 2)(y + 2) < 0$, so the two factors have opposite signs. As these factors are integers, this means that $x + y \geq 3$ and $y \leq -3$, or $x + y \leq 1$ and $y \geq -1$. In the first case, write $x + y = 3 + X$ and $y = -3 - Y$ with $X, Y \geq 0$, so that $x = X + Y + 6$ and $y = -Y - 3$. Then $F(x, y) = G_{83,1}(X, Y)$, where

$$G_{83,1}(X, Y) = 2X^4 + 8X^3Y + 48X^3 + 12X^2Y^2 + 144X^2Y + 432X^2 + 8XY^3 + 144XY^2 + 863XY + 1725X + 2Y^4 + 47Y^3 + 422Y^2 + 1690Y + 2540.$$

All coefficients of $G_{83,1}$ have the same sign and its constant term is non-zero, so $G_{83,1}(X, Y) \neq 0$. In the second case, write $x + y = 1 - X$ and $y = -1 + Y$ with $X, Y \geq 0$, so that $x = -X - Y + 2$ and $y = Y - 1$. Then $F(x, y) = G_{83,2}(X, Y)$, where

$$G_{83,2}(X, Y) = 2X^4 + 8X^3Y - 16X^3 + 12X^2Y^2 - 48X^2Y + 48X^2 + 8XY^3 - 48XY^2 + 95XY - 63X + 2Y^4 - 15Y^3 + 44Y^2 - 56Y + 26.$$

Expanding $G_{83,2}(4 + X, 4 + Y)$, and, for $0 \leq i < 4$, $G_{83,2}(i, 4 + Y)$ and $G_{83,2}(4 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{83,2}(i, j) \neq 0$ for $0 \leq i, j < 4$. Hence $G_{83,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $(-H_{83}) > 0$. Hence $H_{83} < 0$.

The sign of H_{84} . Suppose that $(-H_{84}) \leq 0$. Since $(-H_{84}) = -y - 1$, this means $y \geq -1$. By (52), F is $2x^4 + L_{84}x + Q_{84}$, where L_{84} and Q_{84} depend only on y . If $y \geq 1$, write $y = 1 + z$ with $z \geq 0$; expanding Q_{84} and Δ_{84} as polynomials in z gives non-negative coefficients and positive constant terms, so $Q_{84} > 0$ and $\Delta_{84} > 0$. For $y = -1$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every x , so (1) has no solution with this y . For $y = 0$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every x , so (1) has no solution with this y . Hence $Q_{84} > 0$ and $\Delta_{84} > 0$, so Lemma 7 with $a = 2$, $b = L_{84}$, $c = Q_{84}$ and $t = x$ gives $F(x, y) > 0$, a contradiction. Hence $(-H_{84}) > 0$. Hence $H_{84} < 0$.

The sign of H_{87} . Since $F(x, y) = 0$, (54) gives $Q_{87}H_{87} = M_{87}$, and M_{87} and E_{87} depend only on x . We show that $M_{87} > 0$ and $E_{87} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -3$,

write $x = -3 - z$ with $z \geq 0$. In both cases, expanding M_{87} and E_{87} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{87} > 0$ and $E_{87} > 0$. It remains to consider $-2 \leq x \leq 0$. For $x = -1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_{87} > 0$ and $E_{87} > 0$, so (55) gives $64Q_{87} \geq E_{87} > 0$, and $Q_{87}H_{87} = M_{87} > 0$ gives $H_{87} > 0$.

The sign of H_{89} . Since $F(x, y) = 0$, (56) gives $Q_{89}H_{89} = M_{89}$, and M_{89} and E_{89} depend only on x . We show that $M_{89} > 0$ and $E_{89} > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{89} and E_{89} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{89} > 0$ and $E_{89} > 0$. It remains to consider $-1 \leq x \leq 2$. For $x = -1$ we have $M_{89} > 0$ and $E_{89} > 0$. For $x = 1$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = 0$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , as one checks for the 29 residues of y modulo 29, so (1) has no solution with this x . Hence $M_{89} > 0$ and $E_{89} > 0$, so (57) gives $4Q_{89} \geq E_{89} > 0$, and $Q_{89}H_{89} = M_{89} > 0$ gives $H_{89} > 0$.

The sign of H_{91} . Since $F(x, y) = 0$, (58) gives $Q_{91}(-H_{91}) = M_{91}$, and M_{91} and E_{91} depend only on x . We show that $M_{91} > 0$ and $E_{91} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{91} and E_{91} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{91} > 0$ and $E_{91} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{91} > 0$ and $E_{91} > 0$. Hence $M_{91} > 0$ and $E_{91} > 0$, so (59) gives $4Q_{91} \geq E_{91} > 0$, and $Q_{91}(-H_{91}) = M_{91} > 0$ gives $(-H_{91}) > 0$. Hence $H_{91} < 0$.

The sign of H_{99} . Since $F(x, y) = 0$, (60) gives $Q_{99}H_{99} = M_{99}$, and M_{99} and E_{99} depend only on x . We show that $M_{99} > 0$ and $E_{99} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{99} and E_{99} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{99} > 0$ and $E_{99} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{99} > 0$ and $E_{99} > 0$. Hence $M_{99} > 0$ and $E_{99} > 0$, so (61) gives $4Q_{99} \geq E_{99} > 0$, and $Q_{99}H_{99} = M_{99} > 0$ gives $H_{99} > 0$.

The sign of H_{101} . Since $F(x, y) = 0$, (62) gives $Q_{101}(-H_{101}) = M_{101}$, and M_{101} and E_{101} depend only on x . We show that $M_{101} > 0$ and $E_{101} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -2$, write $x = -2 - z$ with $z \geq 0$. In both cases, expanding M_{101} and E_{101} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{101} > 0$ and $E_{101} > 0$. It remains to consider $-1 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{101} > 0$ and $E_{101} > 0$. Hence $M_{101} > 0$ and $E_{101} > 0$, so (63) gives $64Q_{101} \geq E_{101} > 0$, and $Q_{101}(-H_{101}) = M_{101} > 0$ gives $(-H_{101}) > 0$. Hence $H_{101} < 0$.

The sign of H_{102} . Since $F(x, y) = 0$, (64) gives $Q_{102}(-H_{102}) = M_{102}$, and M_{102} and E_{102} depend only on x . We show that $M_{102} > 0$ and $E_{102} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -1$, write $x = -1 - z$ with $z \geq 0$. In both cases, expanding M_{102} and E_{102} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{102} > 0$ and $E_{102} > 0$. It remains to consider $x = 0$. For $x = 0$ we have $M_{102} > 0$ and $E_{102} > 0$. Hence $M_{102} > 0$ and $E_{102} > 0$, so (65) gives $4Q_{102} \geq E_{102} > 0$, and $Q_{102}(-H_{102}) = M_{102} > 0$ gives $(-H_{102}) > 0$. Hence $H_{102} < 0$.

The sign of H_{104} . Since $F(x, y) = 0$, (66) gives $Q_{104}(-H_{104}) = M_{104}$, and M_{104} and E_{104} depend only on x . We show that $M_{104} > 0$ and $E_{104} > 0$. If $x \geq 3$, write $x = 3 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{104} and E_{104} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{104} > 0$ and $E_{104} > 0$. It remains to consider $-3 \leq x \leq 2$. For $x \in \{-3, -1, 1\}$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with such x . For $x = 0$,

we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , as one checks for the 3 residues of y modulo 3, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . For $x = 2$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , as one checks for the 29 residues of y modulo 29, so (1) has no solution with this x . Hence $M_{104} > 0$ and $E_{104} > 0$, so (67) gives $4Q_{104} \geq E_{104} > 0$, and $Q_{104}(-H_{104}) = M_{104} > 0$ gives $(-H_{104}) > 0$. Hence $H_{104} < 0$.

The sign of H_{105} . Since $F(x, y) = 0$, (68) gives $Q_{105}H_{105} = M_{105}$, and M_{105} and E_{105} depend only on x . We show that $M_{105} > 0$ and $E_{105} > 0$. If $x \geq 1$, write $x = 1 + z$ with $z \geq 0$; if $x \leq -4$, write $x = -4 - z$ with $z \geq 0$. In both cases, expanding M_{105} and E_{105} as polynomials in z gives non-negative coefficients and positive constant terms, so $M_{105} > 0$ and $E_{105} > 0$. It remains to consider $-3 \leq x \leq 0$. For $x = -1$ and $x = 0$ we have $M_{105} > 0$ and $E_{105} > 0$. For $x = -3$, we have $F(x, y) \not\equiv 0 \pmod{2}$ for every y , as one checks for the 2 residues of y modulo 2, so (1) has no solution with this x . For $x = -2$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , as one checks for the 7 residues of y modulo 7, so (1) has no solution with this x . Hence $M_{105} > 0$ and $E_{105} > 0$, so (69) gives $4Q_{105} \geq E_{105} > 0$, and $Q_{105}H_{105} = M_{105} > 0$ gives $H_{105} > 0$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 28, 31, 32, 33, 35, 36, 40, 43, 50, 54, 70, 73, 74, 76, 78, 86, 87, 88, 89, 90, 94, 97, 99, 105, 106\}$ and $H_i < 0$ for $i \in \{18, 19, 22, 27, 58, 62, 66, 75, 83, 84, 91, 95, 101, 102, 104\}$. For $i \in \{15, 16, 17, 20, 21, 23, 24, 25, 26, 29, 30, 34, 37, 38, 39, 41, 42, 44, 45, 46, 47, 48, 49, 51, 52, 53, 55, 56, 57, 59, 60, 61, 63, 64, 65, 67, 68, 69, 71, 72, 77, 79, 80, 81, 82, 85, 92, 93, 96, 98, 100, 103\}$, the sign of H_i is not needed: $H_j > 0$ for every edge $\{i, j\}$, so $(H_i, H_j)_\infty = 1$. Counting the edges whose two values are negative, we find $B_\infty = 1$.

Step 3: the quadratic graphs at the odd primes outside S . Let $S = \{2, 3, 5, 7, 13\}$. It contains 2 and the primes dividing the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: cubic reciprocity. For every place v of K , put

$$\mathcal{B}_v = \sum_{(i,j)} w_{ij} \theta_v(J_i, J_j) + \sum_i \theta_v(J_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (73)$$

the first sum over the edges, and, for a rational prime p , put $\mathcal{B}_p = \sum_{v|p} \mathcal{B}_v$. Applying (H3) to every pair (J_i, J_j) with an edge, multiplied by w_{ij} , and to every pair (J_i, c_i) , and adding, we find that $\mathcal{B}_v = 0$ for all but finitely many v and $\sum_p \mathcal{B}_p = 0$, as the complex place contributes 0. By (H1) and the definition of cl_p , $\mathcal{B}_p = \sum_{(i,j)} w_{ij} \theta_p(\text{cl}_p(J_i), \text{cl}_p(J_j)) + \sum_i \theta_p(\text{cl}_p(J_i), \text{cl}_p(c_i))$, where θ_p is the form of Lemma 8 if $p = 3$.

Step 5: the cubic certificates at the primes outside S . The set S also contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} and of the D_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides J_i and J_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $J_k = \varepsilon_k(\eta - r_k(\xi))$, we have $\eta \equiv r_k(t)$ modulo v , hence $f_k(t) \equiv F(x, y) = 0$ and $g_{kl}(t) \equiv J_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with

coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. Otherwise, (72) shows that v divides D_{ij} , again a contradiction. By (H2), a term of (73) in which all arguments are units vanishes. Now let v divide J_i . Collecting the terms of (73) which contain J_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\theta_v(J_i, \Pi_i)$ with $\Pi_i = c_i \prod_j J_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in \mathbb{Z}[\omega]$. If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv J_j$ and $f_i(t) \equiv 0$ modulo v as above, and (70) gives the claim with $W = w_i(t)$. Otherwise, (71) gives the claim with $W = T_i(x, y)$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\theta_v(J_i, \Pi_i) = v(J_i)\chi(\Pi_i) = 0$ by (H2). Hence $\mathcal{B}_v = 0$, and $\mathcal{B}_p = 0$.

Step 6: the primes in S . Let $p \in S$, and suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Lemma 5 gives, for every vertex H_i of the quadratic graph, a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$, and Lemma 9 gives the analogous data for the values J_i of the cubic certificate. If the hypotheses of Lemma 6 hold for every quadratic graph and those of Lemma 10 hold for every cubic certificate, then B_p and $\mathcal{B}_p$ are determined by a , b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a$, $y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until all these hypotheses hold, or until no solution remains. Table 7 lists the resulting terminal classes together with the joint value $(\mathcal{B}_p; B_p)$; every integer solution lies in one of them. The possible joint values are: for $p = 2$, $(0; -1)$; for $p = 3$, $(1; 1)$, $(2; -1)$; for $p = 5$, $(0; 1)$; for $p = 7$, $(2; 1)$; for $p = 13$, $(0; 1)$.

Step 7: contradiction. By Steps 4 and 5, $\sum_{p \in S} \mathcal{B}_p = 0$. By Theorem 2(iv) and Step 3, $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 6, the pair $(\sum_{p \in S} \mathcal{B}_p; \prod_{p \in S} B_p)$ lies in the set consisting of the pairs $(0; -1)$ and $(1; 1)$, which does not contain $(0; 1)$. This contradiction proves the theorem. $\square$

References

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Table 1: The data of the identities (2).

$i = 0, m = 4$, neighbours 1, 2, 3, 4, 5, 6, 13	
W	$14x^3 + 8x^2 + 14x + 12$
A	$24x^{10} + 96x^9 + 176x^8 + 224x^7 + 332x^6 + 448x^5 + 488x^4 + 464x^3 + 340x^2$ $+ 336x + 144$
B	$-112x^{12}y + 224x^{12} + 32x^{11}y^2 - 368x^{11}y + 904x^{11} - 192x^{10}y^3$ $+ 616x^{10}y^2 - 1104x^{10}y + 1824x^{10} + 64x^9y^4 - 784x^9y^3 + 2064x^9y^2$ $- 2592x^9y + 2608x^9 - 176x^8y^5 + 768x^8y^4 - 2416x^8y^3 + 4208x^8y^2$ $- 4288x^8y + 3408x^8 + 96x^7y^6 - 816x^7y^5 + 2576x^7y^4 - 4976x^7y^3$ $+ 6448x^7y^2 - 4784x^7y + 3452x^7 - 128x^6y^7 + 688x^6y^6 - 2224x^6y^5$ $+ 4784x^6y^4 - 6864x^6y^3 + 7900x^6y^2 - 5472x^6y + 3448x^6 + 64x^5y^8$ $- 688x^5y^7 + 2320x^5y^6 - 4528x^5y^5 + 5968x^5y^4 - 5616x^5y^3$ $+ 5360x^5y^2 - 3984x^5y + 3384x^5 - 48x^4y^9 + 320x^4y^8 - 1520x^4y^7$ $+ 4208x^4y^6 - 6992x^4y^5 + 6544x^4y^4 - 4384x^4y^3 + 2952x^4y^2$ $- 1712x^4y + 2384x^4 + 32x^3y^{10} - 272x^3y^9 + 848x^3y^8 - 2000x^3y^7$ $+ 4464x^3y^6 - 6832x^3y^5 + 5776x^3y^4 - 2352x^3y^3 + 928x^3y^2$ $- 352x^3y + 1588x^3 - 16x^2y^{11} + 144x^2y^{10} - 592x^2y^9 + 1280x^2y^8$ $- 2032x^2y^7 + 2960x^2y^6 - 4144x^2y^5 + 4352x^2y^4 - 2416x^2y^3$ $+ 676x^2y^2 + 128x^2y + 1064x^2 - 48xy^{11} + 240xy^{10} - 576xy^9 + 1056xy^8$ $- 1584xy^7 + 1680xy^6 - 1392xy^5 + 1296xy^4 - 1056xy^3 + 720xy^2 + 48xy + 768x$ $- 48y^{11} + 144y^{10} - 192y^9 + 240y^8 - 336y^7 + 384y^6 - 336y^5 + 240y^4$ $- 192y^3 + 288y^2 - 48y + 288$
$i = 1, m = 16$, neighbours 0, 15, 16, 18, 20, 21, 23, 24, 87	
W	$x^4 - 5x^3y + 176x^3 - 11x^2y^2 + 197x^2y - 330x^2 - 22xy^3 + 70xy^2 - 433xy$ $+ 432x - 34y^4 - 382y^3 - 311y^2 - 1009y + 159$
A	$-18432x^{11} + 3072x^{10}y - 69120x^{10} - 1536x^9y - 45312x^9 - 74496x^8y$ $+ 107136x^8 - 205440x^7y + 251328x^7 - 262080x^6y + 384672x^6 - 245664x^5y$ $+ 722160x^5 - 306288x^4y + 1471288x^4 - 585328x^3y + 1391236x^3 - 466456x^2y$ $+ 557236x^2 - 85500xy + 651644x - 264040y + 68288$
B	$12288x^{13} + 10240x^{12}y + 46080x^{12} + 7168x^{11}y^2 + 45056x^{11}y + 26112x^{11}$ $+ 1024x^{10}y^3 + 28672x^{10}y^2 + 78848x^{10}y - 86784x^{10} - 2048x^9y^4$ $+ 3072x^9y^3 + 43520x^9y^2 + 97280x^9y - 182400x^9 + 6144x^8y^5$ $- 6144x^8y^4 - 3072x^8y^3 + 84736x^8y^2 + 59904x^8y - 250560x^8$ $- 1024x^7y^6 + 20480x^7y^5 - 10240x^7y^4 - 34816x^7y^3 + 205440x^7y^2$ $- 131328x^7y - 435744x^7 - 5120x^6y^7 + 5120x^6y^6 + 59392x^6y^5$ $- 45056x^6y^4 - 144384x^6y^3 + 242624x^6y^2 - 376704x^6y - 861627x^6$ $+ 2048x^5y^8 - 29696x^5y^7 + 12288x^5y^6 + 225280x^5y^5 - 15360x^5y^4$ $- 322560x^5y^3 + 54176x^5y^2 - 605677x^5y - 698584x^5 + 15360x^4y^8$ $- 62464x^4y^7 - 128000x^4y^6 + 301056x^4y^5 + 310272x^4y^4$ $- 228352x^4y^3 - 149605x^4y^2 - 550717x^4y + 33837x^4 - 2048x^3y^9$ $+ 34816x^3y^8 + 5120x^3y^7 - 224256x^3y^6 - 10240x^3y^5$ $+ 400384x^3y^4 + 109220x^3y^3 + 86988x^3y^2 - 563698x^3y + 77584x^3$ $- 6144x^2y^9 + 15360x^2y^8 + 61440x^2y^7 - 67584x^2y^6 - 157696x^2y^5$ $+ 84624x^2y^4 + 28732x^2y^3 + 77674x^2y^2 - 1022490x^2y + 346531x^2$ $- 4096xy^9 - 6144xy^8 + 18432xy^7 + 22528xy^6 - 31588xy^5 - 97928xy^4$ $- 301852xy^3 - 242008xy^2 - 832853xy + 514268x - 1156y^6 - 24820y^5 - 141096y^4$ $- 140300y^3 - 311349y^2 - 122761y + 43007$
$i = 2, m = 2$, neighbours 0, 65, 66, 98	
W	$-6x + 4$
A	$-12x - 20$
B	$24x^3 - 32x^2 - 4xy^2 + 24x - 4y^3 - 16y^2 + 8y - 12$
$i = 3, m = 16$, neighbours 0, 21, 22, 26, 45, 71, 75, 76, 77, 99	
W	$x^4 + 105x^3y - 72x^3 + 105x^2y^2 - 48x^2y + 72x^2 + 24xy^2 + 93xy - 49x$ $- 104y^4 + 73y^3 + 20y^2 - 155y + 78$
A	$768x^{12}y - 512x^{12} + 256x^{11}y - 4096x^{11} + 2048x^{10}y - 896x^{10} + 2432x^9y$ $+ 4160x^9 - 7616x^8y + 11232x^8 - 5408x^7y + 7376x^7 - 10736x^6y - 6664x^6$ $- 6856x^5y - 38060x^5 + 8196x^4y - 58016x^4 + 57612x^3y - 48900x^3 + 39756x^2y$ $- 2372x^2 - 7656xy + 15576x - 17776y + 13904$
B	$-1536x^{14}y + 1024x^{14} - 1536x^{13}y^2 + 2048x^{13}y + 7168x^{13} - 512x^{12}y^2$ $+ 4608x^{12}y - 6400x^{12} + 1536x^{11}y^4 - 5632x^{11}y^3 + 2816x^{11}y^2$ $- 3072x^{11}y - 9088x^{11} + 1536x^{10}y^5 - 8448x^{10}y^4 + 4864x^{10}y^3$ $+ 7936x^{10}y^2 - 2304x^{10}y - 7488x^{10} + 1024x^9y^6 + 2304x^9y^5$

$$\begin{aligned}
& -16128x^9y^4 + 34048x^9y^3 - 16512x^9y^2 + 128x^9y - 2272x^9 \\
& -5632x^8y^7 + 16384x^8y^6 - 17408x^8y^5 + 13824x^8y^4 - 6656x^8y^3 \\
& -4160x^8y^2 + 7360x^8y + 22192x^8 + 2816x^7y^8 - 1280x^7y^7 \\
& -9472x^7y^6 - 17152x^7y^5 + 70400x^7y^4 - 73984x^7y^3 + 33824x^7y^2 \\
& + 7520x^7y + 60360x^7 + 1280x^6y^9 - 11264x^6y^8 + 37888x^6y^7 \\
& -58112x^6y^6 + 35328x^6y^5 + 9728x^6y^4 - 26624x^6y^3 + 25584x^6y^2 \\
& + 31920x^6y + 44711x^6 - 1280x^5y^{10} - 512x^5y^9 + 6912x^5y^8 \\
& + 1280x^5y^7 - 28416x^5y^6 + 60416x^5y^5 - 74752x^5y^4 + 52224x^5y^3 \\
& - 11832x^5y^2 - 8315x^5y + 249x^5 + 256x^4y^{11} + 5376x^4y^{10} \\
& - 23552x^4y^9 + 39936x^4y^8 - 41472x^4y^7 + 30720x^4y^6 - 1280x^4y^5 \\
& - 29440x^4y^4 + 28672x^4y^3 + 2891x^4y^2 + 43702x^4y - 52208x^4 \\
& - 512x^3y^{11} + 1024x^3y^{10} + 3072x^3y^9 - 7936x^3y^8 - 7168x^3y^7 \\
& + 38144x^3y^6 - 45568x^3y^5 + 23808x^3y^4 + 49376x^3y^3 + 30080x^3y^2 \\
& + 37629x^3y - 6787x^3 - 1024x^2y^{12} + 4864x^2y^{11} - 12032x^2y^{10} \\
& + 20736x^2y^9 - 21504x^2y^8 + 4608x^2y^7 + 16640x^2y^6 - 20224x^2y^5 \\
& + 31568x^2y^4 + 21566x^2y^3 + 23160x^2y^2 + 48586x^2y - 39225x^2 \\
& + 256xy^{13} - 512xy^{12} - 1024xy^{11} + 3328xy^{10} - 1024xy^9 - 5376xy^8 + 8192xy^7 \\
& - 5376xy^6 - 9024xy^5 - 12528xy^4 + 5311xy^3 - 26141xy^2 - 38026xy + 15400x \\
& - 256y^{13} + 1536y^{12} - 3584y^{11} + 3584y^{10} - 3584y^8 + 3584y^7 - 12352y^6 \\
& - 6192y^5 - 3249y^4 - 35210y^3 - 10941y^2 - 5764y + 7820 \\
i = 4, m = 48, \text{neighbours } 0, 26, 27, 37, 55, 56, 60, 66, 67, 69, 72, 75, 79 \\
W \quad & -920x^3y + 3858x^3 + 919x^2y^2 + 3976x^2y - 14123x^2 - xy^4 + 2xy^3 \\
& - 7829xy^2 + 7728xy - 15629x + y^5 - 918y^4 + 3847y^3 + 6401y^2 + 1692y - 599 \\
A \quad & 13824x^{18} + 13824x^{17}y + 163584x^{17} + 168192x^{16}y + 847872x^{16} + 903168x^{15}y \\
& + 2492928x^{15} + 2730240x^{14}y + 4613760x^{14} + 4852224x^{13}y + 6448320x^{13} \\
& + 4755456x^{12}y + 10662624x^{12} + 2775744x^{11}y + 20925360x^{11} + 5386176x^{10}y \\
& + 26488728x^{10} + 12482352x^9y + 4747068x^9 + 5403888x^8y - 31274514x^8 \\
& - 20725092x^7y - 24422589x^7 - 22305204x^6y + 16080135x^6 + 18701883x^5y \\
& - 15696613x^5 + 14053009x^4y - 83128355x^4 - 63661897x^3y + 15810757x^3 \\
& - 1120294x^2y + 66539077x^2 + 40884650xy - 3692965x - 10864486y - 12775702 \\
B \quad & -27648x^{20} - 271872x^{19} + 27648x^{18}y^2 - 36864x^{18}y - 1124352x^{18} \\
& - 27648x^{17}y^3 + 253440x^{17}y^2 - 350208x^{17}y - 2465280x^{17} \\
& - 202752x^{16}y^3 + 963072x^{16}y^2 - 1324800x^{16}y - 3158784x^{16} \\
& + 27648x^{15}y^5 - 36864x^{15}y^4 - 596736x^{15}y^3 + 1764864x^{15}y^2 \\
& - 2241792x^{15}y - 3950208x^{15} - 27648x^{14}y^6 + 184320x^{14}y^5 \\
& - 258048x^{14}y^4 - 854784x^{14}y^3 + 993024x^{14}y^2 - 614016x^{14}y \\
& - 9418176x^{14} - 119808x^{13}y^6 + 495360x^{13}y^5 - 663552x^{13}y^4 \\
& - 702720x^{13}y^3 - 1481472x^{13}y^2 + 2590272x^{13}y - 16571232x^{13} \\
& + 27648x^{12}y^8 - 36864x^{12}y^7 - 154368x^{12}y^6 + 601344x^{12}y^5 \\
& - 580608x^{12}y^4 - 1251072x^{12}y^3 - 900864x^{12}y^2 - 2295072x^{12}y \\
& - 5803056x^{12} - 27648x^{11}y^9 + 138240x^{11}y^8 - 221184x^{11}y^7 \\
& + 32256x^{11}y^6 + 248832x^{11}y^5 + 407808x^{11}y^4 - 3451392x^{11}y^3 \\
& + 5099328x^{11}y^2 - 19297584x^{11}y + 25131528x^{11} - 36864x^{10}y^{10} \\
& + 73728x^{10}y^9 + 34560x^{10}y^8 - 258048x^{10}y^7 + 41472x^{10}y^6 \\
& + 324864x^{10}y^5 + 612864x^{10}y^4 - 4412160x^{10}y^3 + 5811264x^{10}y^2 \\
& - 26741160x^{10}y + 28751652x^{10} - 9216x^9y^{11} + 9216x^9y^{10} + 205056x^9y^9 \\
& - 442368x^9y^8 + 322560x^9y^7 - 571392x^9y^6 + 1532160x^9y^5 \\
& - 1645056x^9y^4 - 262656x^9y^3 - 6752304x^9y^2 - 5216796x^9y \\
& - 12864294x^9 - 36864x^8y^{11} + 248832x^8y^{10} - 297216x^8y^9 \\
& - 223488x^8y^8 + 582912x^8y^7 - 806400x^8y^6 + 2055168x^8y^5 \\
& - 3845376x^8y^4 + 6130944x^8y^3 - 12355056x^8y^2 + 13559454x^8y \\
& - 8694606x^8 - 36864x^7y^{12} + 117504x^7y^{11} + 76032x^7y^{10} \\
& - 746496x^7y^9 + 834048x^7y^8 - 361728x^7y^7 + 324864x^7y^6 \\
& + 110592x^7y^5 - 2262528x^7y^4 + 7577856x^7y^3 + 5615460x^7y^2 \\
& - 3484248x^7y + 66393800x^7 - 9216x^6y^{13} + 18432x^6y^{12} + 186624x^6y^{11} \\
& - 499968x^6y^{10} + 29952x^6y^9 + 838656x^6y^8 - 963072x^6y^7 \\
& + 1366272x^6y^6 - 2299392x^6y^5 + 1612800x^6y^4 + 3168000x^6y^3 \\
& + 9660852x^6y^2 - 20665x^6y + 10889202x^6 + 92160x^5y^{12} - 129024x^5y^{11} \\
& - 359424x^5y^{10} + 781056x^5y^9 - 283392x^5y^8 - 186624x^5y^7 \\
& + 831744x^5y^6 - 2439936x^5y^5 + 3400704x^5y^4 - 1283331x^5y^3 \\
& - 24070663x^5y^2 + 65606321x^5y - 144216307x^5 + 16128x^4y^{13} \\
& - 4608x^4y^{12} - 228096x^4y^{11} + 230400x^4y^{10} + 368640x^4y^9
\end{aligned}$$

	$ \begin{aligned} & -698112x^4y^8 + 562176x^4y^7 - 207360x^4y^6 - 1002240x^4y^5 \\ & + 2345475x^4y^4 - 2250997x^4y^3 - 12932259x^4y^2 - 27294425x^4y \\ & + 145661229x^4 + 2304x^3y^{13} - 62208x^3y^{12} - 6912x^3y^{11} \\ & + 288000x^3y^{10} - 170496x^3y^9 - 276480x^3y^8 + 444672x^3y^7 \\ & - 458496x^3y^6 - 18432x^3y^5 + 811454x^3y^4 - 5430501x^3y^3 \\ & + 104607653x^3y^2 - 17673365x^3y - 46443766x^3 - 6912x^2y^{13} \\ & - 16128x^2y^{12} + 76032x^2y^{11} + 62208x^2y^{10} - 172800x^2y^9 \\ & + 16128x^2y^8 + 115200x^2y^7 - 209667x^2y^6 + 131332x^2y^5 \\ & + 1759200x^2y^4 + 10130705x^2y^3 - 176643025x^2y^2 + 173571114x^2y \\ & - 189216413x^2 - 2304xy^{13} + 6912xy^{12} + 27648xy^{11} - 13824xy^{10} - 41472xy^9 \\ & + 29952xy^8 + 2307xy^7 - 42846xy^6 + 923055xy^5 - 21468494xy^4 + 89270095xy^3 \\ & - 32473222xy^2 + 72778092xy + 3852499x + 2304y^{12} + 2304y^{11} - 4608y^{10} - 2304y^9 \\ & + 4607y^8 - 467y^7 - 854558y^6 + 7905315y^5 - 10101314y^4 - 43945585y^3 \\ & + 9820572y^2 + 29848437y - 13134503 \end{aligned} $
$i = 5, m = 24$, neighbours 0, 7, 22, 23, 26, 28, 30, 47, 57, 70, 80, 81, 90	
W	$ \begin{aligned} & -224x^3y - 290x^3 - 224x^2y^2 + 555x^2y - 686x^2 + 845xy^2 - 1791xy + 894x \\ & + 224y^4 + 289y^3 - 1105y^2 - 271y + 304 \end{aligned} $
A	$ \begin{aligned} & -9216x^{16} + 105984x^{15} + 4608x^{14}y - 569088x^{14} - 46080x^{13}y + 1881216x^{13} \\ & + 210816x^{12}y - 4234752x^{12} - 569088x^{11}y + 6743808x^{11} + 951552x^{10}y \\ & - 7538688x^{10} - 877824x^9y + 5356800x^9 + 39168x^8y - 1064448x^8 + 1237248x^7y \\ & - 2931840x^7 - 2369664x^6y + 5068800x^6 + 2757888x^5y - 5982336x^5 \\ & - 1934208x^4y + 6653056x^4 + 690816x^3y - 6579392x^3 - 495168x^2y + 4010688x^2 \\ & + 1538624xy - 1833408x - 1167104y + 513152 \end{aligned} $
B	$ \begin{aligned} & 18432x^{18} + 18432x^{17}y - 193536x^{17} - 165888x^{16}y + 926208x^{16} \\ & - 18432x^{15}y^3 + 46080x^{15}y^2 + 649728x^{15}y - 2642688x^{15} - 18432x^{14}y^4 \\ & + 202752x^{14}y^3 - 396288x^{14}y^2 - 1410048x^{14}y + 4891392x^{14} \\ & + 119808x^{13}y^4 - 884736x^{13}y^3 + 1435392x^{13}y^2 + 1684224x^{13}y \\ & - 5847552x^{13} + 18432x^{12}y^6 - 101376x^{12}y^5 - 152064x^{12}y^4 \\ & + 1930752x^{12}y^3 - 2748672x^{12}y^2 - 503424x^{12}y + 3769344x^{12} \\ & + 18432x^{11}y^7 - 193536x^{11}y^6 + 700416x^{11}y^5 - 509184x^{11}y^4 \\ & - 2169216x^{11}y^3 + 2776320x^{11}y^2 - 2093184x^{11}y + 784512x^{11} \\ & - 9216x^{10}y^8 - 23040x^{10}y^7 + 488448x^{10}y^6 - 1633536x^{10}y^5 \\ & + 1541376x^{10}y^4 + 1145088x^{10}y^3 - 951552x^{10}y^2 + 4872960x^{10}y \\ & - 5090688x^{10} - 9216x^9y^9 + 133632x^9y^8 - 410112x^9y^7 - 25344x^9y^6 \\ & + 1684224x^9y^5 - 1933056x^9y^4 + 577152x^9y^3 - 1362816x^9y^2 \\ & - 6394752x^9y + 6732288x^9 - 9216x^8y^{10} + 41472x^8y^9 - 214272x^8y^8 \\ & + 694656x^8y^7 - 415872x^8y^6 - 1584000x^8y^5 + 2911104x^8y^4 \\ & - 3491712x^8y^3 + 3204864x^8y^2 + 5657472x^8y - 5853312x^8 + 9216x^7y^{11} \\ & - 23040x^7y^{10} + 62208x^7y^9 - 420480x^7y^8 + 1157760x^7y^7 \\ & - 2074752x^7y^6 + 3407616x^7y^5 - 5037696x^7y^4 + 6651648x^7y^3 \\ & - 4484736x^7y^2 - 2577024x^7y + 4735872x^7 - 27648x^6y^{11} + 103680x^6y^{10} \\ & - 213120x^6y^9 + 1153152x^6y^8 - 3773952x^6y^7 + 5958144x^6y^6 \\ & - 5602176x^6y^5 + 5531904x^6y^4 - 6776064x^6y^3 + 4859136x^6y^2 \\ & - 1252224x^6y - 3781376x^6 + 9216x^5y^{12} - 29952x^5y^{11} + 42624x^5y^{10} \\ & - 114048x^5y^9 - 572544x^5y^8 + 3518208x^5y^7 - 6382080x^5y^6 \\ & + 5377536x^5y^5 - 3611520x^5y^4 + 3738240x^5y^3 - 4033152x^5y^2 \\ & + 3767296x^5y + 1709696x^5 - 27648x^4y^{12} + 175104x^4y^{11} - 442368x^4y^{10} \\ & + 692352x^4y^9 - 430848x^4y^8 - 1327104x^4y^7 + 3485952x^4y^6 \\ & - 3234816x^4y^5 + 1613952x^4y^4 - 880128x^4y^3 + 2336384x^4y^2 \\ & - 3224704x^4y + 2454396x^4 + 29952x^3y^{12} - 220032x^3y^{11} + 588672x^3y^{10} \\ & - 775296x^3y^9 + 510336x^3y^8 + 152064x^3y^7 - 955008x^3y^6 \\ & + 1217664x^3y^5 - 632448x^3y^4 + 120704x^3y^3 + 480448x^3y^2 \\ & + 2367848x^3y - 1968700x^3 - 13824x^2y^{12} + 119808x^2y^{11} - 346752x^2y^{10} \\ & + 419328x^2y^9 - 167040x^2y^8 - 40320x^2y^7 + 87552x^2y^6 \\ & - 246528x^2y^5 + 321536x^2y^4 - 415424x^2y^3 - 2584281x^2y^2 \\ & - 164448x^2y + 1251580x^2 + 2304xy^{12} - 28800xy^{11} + 96768xy^{10} - 117504xy^9 \\ & + 17280xy^8 + 42624xy^7 + 9216xy^6 - 37504xy^5 - 708096xy^4 - 221979xy^3 \\ & + 2955773xy^2 + 3122372xy - 1956224x + 2304y^{11} - 10368y^{10} + 13824y^9 - 9216y^7 \\ & - 51328y^6 - 177344y^5 + 286655y^4 + 1224097y^3 + 903967y^2 - 1607904y + 420736 \end{aligned} $
$i = 6, m = 2$, neighbours 0, 7, 12, 29, 73, 80, 82, 102	
W	$ \begin{aligned} & -3x^3 + 4x^2y - 2x^2 - 2xy^2 - 3y^3 + y^2 - 3y + 1 \end{aligned} $
A	$ \begin{aligned} & -8x^8 - 8x^7y + 32x^7 + 12x^6y - 44x^6 - 12x^5y + 24x^5 - 4x^4y + 4x^4 - 12x^3 \end{aligned} $

$$\begin{aligned}
& +24x^2y + 4x^2 + 8xy - 12x - 20y + 12 \\
B \quad & 16x^{10} - 48x^9 - 16x^8y^2 + 56x^8y + 24x^8 + 16x^7y^3 - 24x^7y^2 \\
& - 32x^7y + 24x^7 - 40x^6y^3 + 120x^6y^2 - 72x^6y - 16x^6 - 16x^5y^5 \\
& + 104x^5y^4 - 224x^5y^3 + 132x^5y^2 - 4x^5y + 16x^5 + 16x^4y^6 \\
& - 80x^4y^5 + 84x^4y^4 + 120x^4y^3 - 188x^4y^2 + 64x^4y - 29x^4 \\
& - 88x^3y^6 + 372x^3y^5 - 596x^3y^4 + 416x^3y^3 - 108x^3y^2 + 9x^3y \\
& - 9x^3 - 32x^2y^8 + 144x^2y^7 - 248x^2y^6 + 172x^2y^5 + 24x^2y^4 \\
& - 112x^2y^3 + 72x^2y^2 - 4x^2y - 4x^2 - 16xy^9 + 104xy^8 - 272xy^7 \\
& + 384xy^6 - 328xy^5 + 176xy^4 - 59xy^3 + xy^2 - 9xy - x + 8y^9 - 44y^8 + 100y^7 \\
& - 120y^6 + 80y^5 - 37y^4 - 8y^3 - 14y^2 - 16y + 11
\end{aligned}$$

$i = 7, m = 4$, neighbours 5, 6, 8, 9, 10, 11, 86

$$\begin{aligned}
W \quad & 18x^3 - 6x^2 + 6x + 12 \\
A \quad & 72x^8 - 144x^7 + 216x^6 - 324x^5 + 360x^4 - 396x^3 + 144x^2 - 36x - 72 \\
B \quad & -576x^{11} - 1200x^{10}y + 2352x^{10} - 1728x^9y^2 + 5760x^9y - 5400x^9 \\
& - 1968x^8y^3 + 9000x^8y^2 - 14280x^8y + 9048x^8 - 2016x^7y^4 \\
& + 10464x^7y^3 - 22416x^7y^2 + 23664x^7y - 11352x^7 - 1872x^6y^5 \\
& + 10080x^6y^4 - 25248x^6y^3 + 35928x^6y^2 - 28152x^6y + 11316x^6 \\
& - 1440x^5y^6 + 7920x^5y^5 - 22128x^5y^4 + 37488x^5y^3 - 39780x^5y^2 \\
& + 25116x^5y - 9156x^5 - 816x^4y^7 + 4704x^4y^6 - 14640x^4y^5 \\
& + 28704x^4y^4 - 36912x^4y^3 + 31560x^4y^2 - 16968x^4y + 5916x^4 \\
& - 288x^3y^8 + 1872x^3y^7 - 6864x^3y^6 + 15696x^3y^5 - 23616x^3y^4 \\
& + 24480x^3y^3 - 17964x^3y^2 + 8532x^3y - 3540x^3 - 48x^2y^9 + 432x^2y^8 \\
& - 2160x^2y^7 + 5808x^2y^6 - 10080x^2y^5 + 12000x^2y^4 - 10656x^2y^3 \\
& + 7056x^2y^2 - 3264x^2y + 1308x^2 + 48xy^9 - 432xy^8 + 1344xy^7 - 2544xy^6 \\
& + 3360xy^5 - 3408xy^4 + 2736xy^3 - 1860xy^2 + 828xy - 468x - 48y^9 + 144y^8 \\
& - 288y^7 + 384y^6 - 432y^5 + 432y^4 - 384y^3 + 216y^2 - 216y - 168
\end{aligned}$$

$i = 8, m = 32$, neighbours 7, 15, 17, 25, 26, 28, 29, 30, 31, 89

$$\begin{aligned}
W \quad & -x^4 - 10x^3y + 235x^3 + 31x^2y^2 - 295x^2y - 399x^2 - 62xy^3 + 171xy^2 \\
& + 339xy + 153x + 93y^4 + 367y^3 + 184y^2 + 741y - 759 \\
A \quad & 24576x^{11} + 24576x^{10}y - 159744x^{10} - 135168x^9y + 577536x^9 + 344064x^8y \\
& - 1142784x^8 - 417792x^7y + 1265664x^7 + 49152x^6y - 872448x^6 + 374784x^5y \\
& - 491520x^5 - 752640x^4y + 921358x^4 + 94016x^3y - 301217x^3 + 1037866x^2y \\
& + 615605x^2 - 285551xy - 171467x - 45362y + 540288 \\
B \quad & -16384x^{13} + 106496x^{12} + 4096x^{11}y^2 - 12288x^{11}y - 376832x^{11} \\
& - 8192x^{10}y^3 - 18432x^{10}y^2 + 120832x^{10}y + 708608x^{10} + 4096x^9y^4 \\
& + 28672x^9y^3 + 16384x^9y^2 - 307200x^9y - 696320x^9 + 4096x^8y^5 \\
& - 14336x^8y^4 + 16384x^8y^3 - 4096x^8y^2 + 215040x^8y + 425984x^8 \\
& + 8192x^7y^6 + 14336x^7y^5 + 22528x^7y^4 - 284672x^7y^3 - 38912x^7y^2 \\
& + 249856x^7y + 389120x^7 + 4096x^6y^7 + 16384x^6y^6 - 61440x^6y^5 \\
& - 217088x^6y^4 + 618496x^6y^3 + 550912x^6y^2 - 913408x^6y - 864095x^6 \\
& + 18432x^5y^7 - 55296x^5y^6 - 83968x^5y^5 + 481280x^5y^4 \\
& - 307200x^5y^3 - 1243136x^5y^2 + 899715x^5y + 643336x^5 + 8192x^4y^8 \\
& - 55296x^4y^7 - 22528x^4y^6 + 342016x^4y^5 - 262144x^4y^4 \\
& - 487424x^4y^3 + 1182352x^4y^2 + 60075x^4y - 784118x^4 - 20480x^3y^8 \\
& + 34816x^3y^7 + 124928x^3y^6 - 278528x^3y^5 - 147456x^3y^4 \\
& + 818495x^3y^3 - 571236x^3y^2 + 136041x^3y - 4754x^3 + 16384x^2y^8 \\
& + 2048x^2y^7 - 92160x^2y^6 + 59392x^2y^5 + 77980x^2y^4 - 374495x^2y^3 \\
& - 103175x^2y^2 - 362436x^2y + 93893x^2 - 4096xy^8 - 4096xy^7 + 20480xy^6 \\
& + 41575xy^5 + 114102xy^4 - 31764xy^3 + 300931xy^2 - 148457xy + 60787x - 8649y^6 \\
& - 59613y^5 - 100651y^4 - 112618y^3 - 177945y^2 + 34693y - 35793
\end{aligned}$$

$i = 9, m = 4$, neighbours 7, 18, 44, 49, 56, 57, 58, 63, 64, 96

$$\begin{aligned}
W \quad & -4xy^2 - 4xy + 12x + 12y^2 - 16y + 20 \\
A \quad & -16xy^5 - 96xy^4 + 304xy^3 - 528xy^2 + 144xy + 16x + 48y^5 - 48y^4 - 16y^3 \\
& + 352y^2 - 656y + 288 \\
B \quad & 32x^{11}y^5 - 32x^{11}y^4 - 160x^{11}y^3 + 160x^{11}y^2 + 128x^{11}y - 128x^{11} \\
& + 16x^{10}y^5 + 48x^{10}y^4 - 304x^{10}y^3 + 272x^{10}y^2 + 480x^{10}y - 512x^{10} \\
& - 192x^9y^5 + 256x^9y^4 + 496x^9y^3 - 704x^9y^2 + 496x^9y - 352x^9 \\
& + 64x^8y^5 - 720x^8y^4 + 688x^8y^3 - 528x^8y^2 - 16x^8y + 512x^8 \\
& - 32x^7y^7 + 624x^7y^5 - 768x^7y^4 - 608x^7y^3 + 1408x^7y^2 - 976x^7y \\
& + 352x^7 + 80x^6y^7 - 272x^6y^6 - 320x^6y^5 + 1168x^6y^4 - 240x^6y^3 \\
& + 1312x^6y^2 - 1504x^6y - 224x^6 - 16x^5y^7 - 208x^5y^5 + 928x^5y^4 \\
& - 1776x^5y^3 + 608x^5y^2 + 528x^5y - 64x^5 - 32x^4y^8 - 32x^4y^7
\end{aligned}$$

$$\begin{aligned}
& +544x^4y^6 - 752x^4y^5 - 64x^4y^4 - 416x^4y^3 - 928x^4y^2 + 1648x^4y \\
& +32x^4 + 48x^3y^8 - 128x^3y^7 + 112x^3y^6 + 336x^3y^5 - 736x^3y^4 \\
& +1056x^3y^3 - 960x^3y^2 + 656x^3y - 32x^3 + 32x^2y^8 + 64x^2y^7 \\
& -352x^2y^6 + 928x^2y^5 - 992x^2y^4 + 1408x^2y^3 - 512x^2y^2 \\
& +448x^2y - 608x^2 - 32xy^8 + 160xy^7 - 400xy^6 + 512xy^5 - 240xy^4 + 352xy^3 \\
& +512xy^2 - 224xy - 576x - 48y^8 + 48y^7 - 80y^6 - 208y^5 + 496y^4 - 624y^3 \\
& +928y^2 - 592y - 112 \\
i = 10, m = 756, \text{neighbours } 7, 25, 30, 38, 46, 49, 68, 69, 79, 81, 90, 101 \\
W \quad & -97x^4y - 438x^4 + 290x^3y^2 + 794577x^3y + 3587076x^3 + x^2y^3 \\
& -792830x^2y^2 - 1810453x^2y + 637386x^2 - 680xy^4 - 3065xy^3 - 1776272xy^2 \\
& +163962xy + 358363x + 871y^5 + 795904y^4 + 3586609y^3 - 799691y^2 + 6074348y - 2981145 \\
A \quad & -54867456x^{16} - 109734912x^{15}y + 658409472x^{15} + 1344252672x^{14}y \\
& -3698980992x^{14} - 8179823232x^{13}y + 12816123264x^{13} + 33117081984x^{12}y \\
& -30094799616x^{12} - 101171016576x^{11}y + 48271930560x^{11} + 252385725312x^{10}y \\
& -44530655904x^{10} - 547264009152x^9y - 17215235856x^9 + 1088247980736x^8y \\
& +181644142680x^8 - 2070645779664x^7y - 514497421620x^7 + 3873816712080x^6y \\
& +1148492438032x^6 - 7216491815538x^5y - 2373564054728x^5 + 13437428235309x^4y \\
& +7330371967841x^4 - 24046914408919x^3y - 30077102252329x^3 \\
& +20498035557626x^2y + 43190814675701x^2 - 5031568350230xy - 37215379005995x \\
& -9194553464912y + 15799091644382 \\
B \quad & 109734912x^{18} + 109734912x^{17}y - 1097349120x^{17} - 219469824x^{16}y^2 \\
& -1261951488x^{16}y + 5093528832x^{16} + 109734912x^{15}y^3 + 2030095872x^{15}y^2 \\
& +7589998080x^{15}y - 14347839744x^{15} + 109734912x^{14}y^4 - 713276928x^{14}y^3 \\
& -9556081920x^{14}y^2 - 30899522304x^{14}y + 26345523456x^{14} - 292626432x^{13}y^5 \\
& -768144384x^{13}y^4 + 1764903168x^{13}y^3 + 30570317568x^{13}y^2 \\
& +94532054400x^{13}y - 28846564992x^{13} - 36578304x^{12}y^6 + 1920360960x^{12}y^5 \\
& +3100011264x^{12}y^4 - 992186496x^{12}y^3 - 75589065216x^{12}y^2 \\
& -233689639680x^{12}y + 1323677376x^{12} + 310915584x^{11}y^7 + 192036096x^{11}y^6 \\
& -6099432192x^{11}y^5 - 8723925504x^{11}y^4 - 5971408128x^{11}y^3 \\
& +158562375552x^{11}y^2 + 498729172032x^{11}y + 78740514720x^{11} \\
& +201180672x^{10}y^8 - 1655168256x^{10}y^7 - 914457600x^{10}y^6 \\
& +11636472960x^{10}y^5 + 17758766592x^{10}y^4 + 20794765824x^{10}y^3 \\
& -306091820160x^{10}y^2 - 977730064416x^{10}y - 237225732912x^{10} \\
& +36578304x^9y^9 - 566963712x^9y^8 + 4174498944x^9y^7 \\
& +2535333696x^9y^6 - 13947764544x^9y^5 - 26318089728x^9y^4 \\
& -37538484480x^9y^3 + 576929013696x^9y^2 + 1849884284304x^9y \\
& +524074793256x^9 + 365783040x^8y^9 + 598969728x^8y^8 - 7176206016x^8y^7 \\
& -4229366400x^8y^6 + 10029313728x^8y^5 + 28174438656x^8y^4 \\
& +45736596864x^8y^3 - 1087946209728x^8y^2 - 3472773044328x^8y \\
& -1056140212544x^8 + 237758976x^7y^{10} - 1426553856x^7y^9 \\
& -951035904x^7y^8 + 8854235712x^7y^7 + 5237555904x^7y^6 \\
& -2830246272x^7y^5 - 21329723520x^7y^4 - 40126399488x^7y^3 \\
& +2052041139792x^7y^2 + 6523290583796x^7y + 2093557655256x^7 \\
& +36578304x^6y^{11} - 438939648x^6y^{10} + 2084963328x^6y^9 \\
& +1664312832x^6y^8 - 7267651776x^6y^7 - 5210122176x^6y^6 \\
& -2331866880x^6y^5 + 10831750272x^6y^4 + 25636818816x^6y^3 \\
& -3853745117485x^6y^2 - 12265743094474x^6y - 9235844461790x^6 \\
& +91445760x^5y^{11} + 342921600x^5y^{10} - 1940936256x^5y^9 \\
& -1604873088x^5y^8 + 3634968960x^5y^7 + 3799571328x^5y^6 \\
& +3436074432x^5y^5 - 3113728128x^5y^4 - 11707401855x^5y^3 \\
& +7203573897478x^5y^2 + 26179702729880x^5y + 39077602782778x^5 \\
& +36578304x^4y^{12} - 283481856x^4y^{11} - 352066176x^4y^{10} \\
& +1353397248x^4y^9 + 994472640x^4y^8 - 925888320x^4y^7 \\
& -1812912192x^4y^6 - 2164978368x^4y^5 + 57877850x^4y^4 \\
& +10274500725x^4y^3 - 12114328918716x^4y^2 - 26435852392490x^4y \\
& -10708642765664x^4 - 73156608x^3y^{12} + 196608384x^3y^{11} \\
& +340635456x^3y^{10} - 576108288x^3y^9 - 443511936x^3y^8 - 6858432x^3y^7 \\
& +498379392x^3y^6 + 790207505x^3y^5 - 5368294160x^3y^4 \\
& -3195354882953x^3y^3 - 8335646332153x^3y^2 - 17025987287335x^3y \\
& +6989915522472x^3 + 45722880x^2y^{12} - 41150592x^2y^{11} - 155457792x^2y^{10} \\
& +105162624x^2y^9 + 121165632x^2y^8 + 70870464x^2y^7 - 61088446x^2y^6 \\
& +129362683x^2y^5 + 1255332245101x^2y^4 + 8902699553941x^2y^3
\end{aligned}$$

$$\begin{aligned}
& +4129763769623x^2y^2 + 22495088284772x^2y - 2559160539507x^2 - 9144576xy^{12} \\
& +25147584xy^{10} - 2286144xy^9 - 13716864xy^8 - 11773663xy^7 + 2475417998xy^6 \\
& +653661205289xy^5 + 8564597577752xy^4 + 25007044829671xy^3 - 2807989859292xy^2 \\
& +26729502557544xy - 21254982608011x - 758641y^8 - 1387223409y^7 - 641097514862y^6 \\
& -6347510128803y^5 - 17307803715710y^4 - 14887938297863y^3 - 27852298764426y^2 \\
& +2336153716601y + 6911866133357
\end{aligned}$$

$i = 11, m = 36$, neighbours 7, 19, 24, 42, 54, 83, 85, 90

$W \quad x^3y + 40x^3 - 3x^2y^2 - 27x^2y + 104x^2 + 3xy^3 + 12xy^2 - 47xy + 16x - 5y^4$
 $\quad + y^3 - 7y^2 - 10y + 78$

$A \quad 6840x^4 + 6912x^3y + 5112x^3 + 5508x^2y + 5724x^2 + 936xy + 3780x + 2520y + 936$

$B \quad 5184x^7y^2 + 5184x^7y - 10368x^7 + 12960x^6y^3 + 2592x^6y^2 - 10368x^6y$
 $\quad - 18864x^6 + 10368x^5y^4 + 5184x^5y^3 + 7776x^5y^2 - 20160x^5y - 13536x^5$
 $\quad + 2592x^4y^5 + 12960x^4y^4 - 8209x^4y^2 + 280x^4y - 7720x^4$
 $\quad + 5184x^3y^5 - 7776x^3y^4 + 2888x^3y^3 - 12507x^3y^2 + 10400x^3y$
 $\quad - 10064x^3 - 2592x^2y^5 - 7950x^2y^4 - 5997x^2y^3 - 1653x^2y^2$
 $\quad + 2856x^2y - 9472x^2 - 5104xy^5 + 199xy^4 - 226xy^3 + 4508xy^2 + 3584xy - 3864x$
 $\quad - 25y^6 - 15y^5 - 61y^4 - 132y^3 - 1840y^2 - 2940y - 5148$

$i = 12, m = 4$, neighbours 6, 14

$W \quad 4x^2 - 2x + 1$

$A \quad 0$

$B \quad -8x^2y + 4x^2 - 8xy + 4x - 8y^3 + 12y^2 - 2y - 1$

$i = 13, m = 144$, neighbours 0, 30, 55, 66, 70

$W \quad -96x^3y + 10x^3 + 96x^2y^2 - 77x^2y + 16x^2 + 67xy^2 - 32xy + 144x - 96y^4$
 $\quad + 9y^3 + 16y^2 + 48y - 96$

$A \quad 36864x^4 + 13824x^3y + 18432x^3 + 6912x^2y - 23040xy - 27648x + 9216y + 9216$

$B \quad -73728x^6 + 46080x^5y - 36864x^5 + 18432x^4y^2 + 24960x^4y - 100x^4$
 $\quad - 46080x^3y^3 - 4800x^3y^2 + 13928x^3y + 54976x^3 + 18432x^2y^4$
 $\quad - 29376x^2y^3 - 44137x^2y^2 - 93344x^2y + 15296x^2 + 9216xy^5 + 11136xy^4$
 $\quad - 10341xy^3 + 19456xy^2 + 27680xy + 15744x - 9216y^6 + 1728y^5 + 2991y^4 + 8928y^3$
 $\quad - 8032y^2 + 11712y - 17664$

$i = 14, m = 216$, neighbours 12, 20, 51, 52, 59, 63, 67, 68, 71, 72, 73, 74

$W \quad -x^5y + x^3y^3 + 293x^3y - 236x^3 + 293x^2y^2 + 3753x^2y - 1493x^2$
 $\quad + 3989xy^2 + 1458xy - 1163x + 2016y^{11} - 293y^4 + 236y^3 + 2952y^2 - 1054y - 230$

$A \quad 746496x^{18} + 2778624x^{17}y + 2239488x^{17} + 269568x^{16}y + 2882304x^{16}$
 $\quad - 2187648x^{15}y + 4167936x^{15} - 6899904x^{14}y + 186624x^{14} - 4077216x^{13}y$
 $\quad - 4863744x^{13} - 534672x^{12}y - 10535760x^{12} + 3689352x^{11}y - 9176400x^{11}$
 $\quad + 4673772x^{10}y - 9752724x^{10} - 13446x^9y - 1164780x^9 - 230769x^8y + 3332907x^8$
 $\quad + 8770395x^7y - 2563569x^7 + 16722348x^6y - 2004890x^6 + 3699814x^5y$
 $\quad + 15534531x^5 - 17437146x^4y + 28340515x^4 - 19464015x^3y + 13367750x^3$
 $\quad - 2733314x^2y + 1105073x^2 + 3838060xy + 534980x + 590640y + 52900$

$B \quad -1492992x^{20} - 7050240x^{19}y - 4478976x^{19} - 5557248x^{18}y^2 - 5018112x^{18}y$
 $\quad - 5764608x^{18} + 1492992x^{17}y^3 - 539136x^{17}y^2 - 2135808x^{17}y - 8335872x^{17}$
 $\quad + 7050240x^{16}y^4 + 3732480x^{16}y^3 + 850176x^{16}y^2 + 1731456x^{16}y$
 $\quad + 373248x^{16} + 5557248x^{15}y^5 + 1492992x^{15}y^4 + 746496x^{15}y^3$
 $\quad + 4240512x^{15}y^2 + 3945024x^{15}y + 11966976x^{15} - 1492992x^{14}y^6$
 $\quad - 2239488x^{14}y^5 - 373248x^{14}y^4 - 373248x^{14}y^3 + 5220288x^{14}y^2$
 $\quad + 3373344x^{14}y + 23953824x^{14} - 7050240x^{13}y^7 - 3732480x^{13}y^6$
 $\quad - 1119744x^{13}y^5 - 2426112x^{13}y^4 - 2892672x^{13}y^3 + 2681568x^{13}y^2$
 $\quad + 5864976x^{13}y + 22520736x^{13} - 5557248x^{12}y^8 - 1492992x^{12}y^7$
 $\quad - 746496x^{12}y^6 - 2052864x^{12}y^5 - 1213056x^{12}y^4 - 878400x^{12}y^3$
 $\quad - 211824x^{12}y^2 + 10763784x^{12}y + 19692072x^{12} + 1492992x^{11}y^9$
 $\quad + 2239488x^{11}y^8 + 373248x^{11}y^7 + 373248x^{11}y^6 + 1679616x^{11}y^5$
 $\quad + 517248x^{11}y^4 - 1213056x^{11}y^3 - 3067848x^{11}y^2 + 35904996x^{11}y$
 $\quad - 2534184x^{11} + 7050240x^{10}y^{10} + 2239488x^{10}y^9 - 373248x^{10}y^8$
 $\quad + 5038848x^{10}y^7 + 5132160x^{10}y^6 - 97344x^{10}y^5 - 1213056x^{10}y^4$
 $\quad - 2922624x^{10}y^3 + 13592916x^{10}y^2 + 27640602x^{10}y - 17201574x^{10}$
 $\quad + 4064256x^9y^{11} + 746496x^9y^{10} + 1492992x^9y^9 - 2799360x^9y^8$
 $\quad - 93312x^9y^7 + 7223616x^9y^6 + 746496x^9y^5 - 6156864x^9y^4$
 $\quad - 12575808x^9y^3 + 18122310x^9y^2 - 2947488x^9y - 4049262x^9$
 $\quad - 2239488x^8y^{12} - 1119744x^8y^{11} + 3732480x^8y^{10} + 373248x^8y^9$
 $\quad - 13436928x^8y^8 - 9941760x^8y^7 + 9237888x^8y^6 + 3857472x^8y^5$
 $\quad - 34015104x^8y^4 + 327744x^8y^3 + 4919987x^8y^2 - 12149958x^8y$

	$ \begin{aligned} & -5742944x^8 - 3317760x^7y^{13} - 746496x^7y^{12} - 7651584x^7y^{11} \\ & -373248x^7y^{10} + 17635968x^7y^9 + 1497024x^7y^8 - 17122752x^7y^7 \\ & + 963072x^7y^6 - 13041216x^7y^5 - 18247680x^7y^4 + 8892287x^7y^3 \\ & - 7843035x^7y^2 - 14159365x^7y - 32233842x^7 - 4064256x^6y^{14} \\ & + 373248x^6y^{13} + 4478976x^6y^{12} - 1399680x^6y^{11} - 11757312x^6y^{10} \\ & + 5000256x^6y^9 + 14603328x^6y^8 + 884736x^6y^7 + 6883776x^6y^6 \\ & - 17455680x^6y^5 + 5042879x^6y^4 + 927360x^6y^3 - 16724692x^6y^2 \\ & - 1693036x^6y - 53348123x^6 - 933120x^5y^{13} + 1679616x^5y^{12} \\ & + 4712256x^5y^{11} - 3868416x^5y^{10} - 7884864x^5y^9 + 4307328x^5y^8 \\ & + 36814464x^5y^7 - 1167552x^5y^6 - 5995584x^5y^5 + 279936x^5y^4 \\ & - 1172x^5y^3 - 3714826x^5y^2 + 1479344x^5y - 29299069x^5 + 4064256x^4y^{16} \\ & - 466560x^4y^{15} - 1166400x^4y^{14} + 1348992x^4y^{13} + 3032640x^4y^{12} \\ & - 4415616x^4y^{11} + 15000768x^4y^{10} + 19180800x^4y^9 - 8005824x^4y^8 \\ & - 1347264x^4y^7 + 586x^4y^6 - 8450x^4y^5 + 17600024x^4y^4 \\ & - 18498874x^4y^3 - 4270732x^4 + 4064256x^3y^{17} + 186624x^3y^{16} \\ & - 377280x^3y^{15} - 699840x^3y^{14} - 2564352x^3y^{13} - 12389184x^3y^{12} \\ & + 17082432x^3y^{11} - 4576320x^3y^{10} - 694080x^3y^9 + 586x^3y^8 \\ & - 472x^3y^7 + 423341x^3y^6 + 18774643x^3y^5 - 65010761x^3y^4 + 13759875x^3 \\ & + 93312x^2y^{13} + 46656x^2y^{12} - 1414656x^2y^{11} - 31309056x^2y^{10} \\ & + 327744x^2y^9 + 4735872x^2y^8 - 46656x^2y^7 + 171698x^2y^6 \\ & + 4398516x^2y^5 - 18583413x^2y^4 - 62331179x^2y^3 + 25456730x^2 \\ & - 4064256xy^{19} + 1181376xy^{18} - 17035200xy^{17} - 17781120xy^{16} + 8938944xy^{15} \\ & + 927360xy^{14} - 85849xy^{13} + 2475850xy^{12} + 645756xy^{11} - 30769798xy^{10} - 9127500xy^9 \\ & + 9786472x - 4064256y^{20} + 1181376y^{19} - 951552y^{18} - 11902464y^{17} + 4249728y^{16} \\ & + 927360y^9 - 85849y^8 + 138296y^7 + 1674176y^6 - 2010988y^5 - 8942236y^4 \\ & + 6278476y - 934276 \end{aligned} $
$i = 15, m = 2$, neighbours 1, 8	
W	$-6x^2 + 2y^2 + 2y + 2$
A	0
B	$6xy + 12x$
$i = 16, m = 1$, neighbours 1	
W	$-y - 2$
A	0
B	$-3x + 3y + 3$
$i = 17, m = 1$, neighbours 8	
W	$-x - y - 1$
A	0
B	$-2x - y$
$i = 18, m = 2$, neighbours 1, 9	
W	$3x^2 - 2xy - 4x + y - 1$
A	0
B	$-3x^2 - 6x + 3y + 3$
$i = 19, m = 2$, neighbours 11, 50, 97	
W	2
A	14
B	$-56x^3y - 28x^2y^2 - 28x^2y - 28x^2 - 28xy^2 - 14xy + 14y^3 - 28y^2 + 56y - 38$
$i = 20, m = 2$, neighbours 1, 14	
W	$4x^2 - 4xy - 2y$
A	0
B	$4x^2 - 12xy - 4x + 4y^2$
$i = 21, m = 2$, neighbours 1, 3	
W	$2x^2 - 4x + 2y - 2$
A	0
B	$-8x^2 + 8xy - 4x - 4y^2 + 4$
$i = 22, m = 1$, neighbours 3, 5	
W	$x^2 - x$
A	0
B	$-2x^3 + 3x^2y - 2xy^2 + xy + y^3 - 2y^2 + 2y - 1$
$i = 23, m = 2$, neighbours 1, 5	
W	$2x^2 - 6x + 4y$

A	0
B	$-8x^2 + 8xy + 4x - 4y^2 - 4y + 4$
$i = 24, m = 2$, neighbours 1, 11	
W	$-3x^2 - xy + x + 2y^2 - 2$
A	0
B	$3x^2 + 3xy - 3x + 6y$
$i = 25, m = 2$, neighbours 8, 10	
W	$-4x^2 - 4xy + 4x + 4y$
A	0
B	$-4x^2 - 12xy - 4y^2 + 4y + 4$
$i = 26, m = 8$, neighbours 3, 4, 5, 8, 33, 40, 43	
W	$216y^2 + 600y + 24$
A	$64y^{11} + 64y^{10} - 64y^9 + 320y^8 - 576y^7 - 2304y^6 + 1088y^5 + 17408y^4$ $+ 12736y^3 - 85376y^2 - 105408y + 20928$
B	$192x^{17} - 192x^{16}y + 960x^{16} + 448x^{15}y^2 - 1088x^{15}y + 2368x^{15}$ $- 128x^{14}y^3 + 1536x^{14}y^2 - 3456x^{14}y + 4160x^{14} + 448x^{13}y^4$ $- 256x^{13}y^3 + 1856x^{13}y^2 - 6272x^{13}y + 6720x^{13} - 64x^{12}y^5$ $+ 320x^{12}y^4 + 64x^{12}y^3 + 1792x^{12}y^2 - 8000x^{12}y + 9920x^{12}$ $+ 320x^{11}y^6 + 192x^{11}y^5 - 576x^{11}y^4 + 4928x^{11}y^3 - 2176x^{11}y^2$ $- 10624x^{11}y + 13120x^{11} + 128x^{10}y^7 - 128x^{10}y^6 + 448x^{10}y^5$ $- 64x^{10}y^4 + 7872x^{10}y^3 - 14400x^{10}y^2 - 12288x^{10}y + 15936x^{10}$ $+ 64x^9y^8 + 256x^9y^7 + 64x^9y^6 + 960x^9y^5 - 1088x^9y^4$ $+ 9280x^9y^3 - 23680x^9y^2 - 12992x^9y + 18112x^9 + 64x^8y^9$ $- 192x^8y^8 + 1024x^8y^7 - 704x^8y^6 - 1408x^8y^5 + 2560x^8y^4$ $+ 18240x^8y^3 - 30976x^8y^2 - 13696x^8y + 19904x^8 + 64x^7y^{10}$ $- 64x^7y^9 + 896x^7y^8 + 384x^7y^7 - 1664x^7y^6 + 128x^7y^5$ $+ 11712x^7y^4 + 22080x^7y^3 - 34048x^7y^2 - 12928x^7y + 20928x^7$ $+ 320x^6y^9 + 64x^6y^8 - 384x^6y^7 - 320x^6y^6 + 1728x^6y^5$ $+ 5248x^6y^4 + 27008x^6y^3 - 32960x^6y^2 - 12992x^6y + 21312x^6$ $+ 256x^5y^{10} - 192x^5y^9 + 1088x^5y^8 - 1664x^5y^7 + 4032x^5y^6$ $- 10752x^5y^5 + 320x^5y^4 + 31552x^5y^3 - 32704x^5y^2 - 13440x^5y$ $+ 21504x^5 + 192x^4y^{11} - 256x^4y^{10} + 704x^4y^9 + 576x^4y^8$ $- 1280x^4y^7 + 4160x^4y^6 - 17344x^4y^5 - 3712x^4y^4 + 33088x^4y^3$ $- 32256x^4y^2 - 13632x^4y + 21504x^4 + 64x^3y^{12} - 192x^3y^{11}$ $+ 832x^3y^{10} - 64x^3y^9 + 704x^3y^8 - 448x^3y^7 + 17728x^3y^6$ $- 26944x^3y^5 - 40320x^3y^4 + 9344x^3y^3 + 139136x^3y^2 + 196992x^3y$ $- 20352x^3 - 64x^2y^{12} + 192x^2y^{11} + 256x^2y^{10} + 64x^2y^9$ $- 2432x^2y^8 + 7616x^2y^7 + 19264x^2y^6 - 31872x^2y^5 - 39744x^2y^4$ $+ 10688x^2y^3 + 139328x^2y^2 + 196992x^2y - 20352x^2 + 128xy^{12} + 192xy^{11}$ $+ 256xy^{10} - 1472xy^9 - 1024xy^8 + 10624xy^7 + 18688xy^6 - 33856xy^5$ $- 38528xy^4 + 11072xy^3 + 139520xy^2 + 196992xy - 20352x + 128y^{13} - 192y^{12}$ $+ 704y^{11} - 1088y^{10} - 1088y^9 - 832y^8 + 13824y^7 + 16960y^6 - 51456y^5$ $- 50880y^4 + 96640y^3 + 244928y^2 + 176064y - 20352$
$i = 27, m = 1$, neighbours 4	
W	$x + 1$
A	0
B	$x + y + 1$
$i = 28, m = 2$, neighbours 5, 8	
W	$-3x^2 + xy - x + y^2 + 2$
A	0
B	$3x^2 + 3xy - 9x + 3y^2$
$i = 29, m = 2$, neighbours 6, 8	
W	$-2x^2 - 4x - 2y + 2$
A	0
B	$8x^2 + 8xy - 4x + 4y^2 - 4$
$i = 30, m = 4$, neighbours 5, 8, 10, 13, 97	
W	$280x^4y + 140x^4 + 840y^3 - 35y^2 + 2079y - 742$
A	$-39200x^4y^2 - 39200x^4y - 9632x^4 + 336x^3y - 504x^3 + 560x^2y^2$ $- 1120x^2y + 728x^2 + 20160xy^3 + 17640xy^2 + 6552xy - 672x + 112y^5 + 112y^4$ $- 716072y^3 + 52052y^2 - 1758736y + 615272$
B	$19488x^3y^5 - 215096x^3y^4 + 649992x^3y^3 - 606872x^3y^2 + 1664376x^3y$

$$\begin{aligned}
& -515816x^3 + 10304x^2y^5 - 109900x^2y^4 + 327712x^2y^3 - 305060x^2y^2 \\
& + 833056x^2y - 258328x^2 + 448xy^6 + 3472xy^5 - 62622xy^4 + 152880xy^3 \\
& - 153734xy^2 + 415352xy - 128744x + 224y^7 - 10472y^6 - 6524y^5 + 302729y^4 \\
& + 43470y^3 + 804545y^2 - 95956y - 64708
\end{aligned}$$

$i = 31, m = 2$, neighbours 8, 32

$W \quad 3x^2 + 4xy - x + 2y^2 - 2$

$A \quad 0$

$B \quad 3x^2 + 9x + 6y$

$i = 32, m = 48$, neighbours 31, 41, 48, 55, 58, 62, 66, 77, 85, 105

$W \quad -83x^3y - 612x^3 + 83x^2y^2 + 342x^2y - 568x^2 - xy^3 + 269xy^2 - 640xy - 448x$
 $-83y^4 - 611y^3 + 1207y^2 + 810y - 1318$

$A \quad 4608x^{15} + 2304x^{14}y + 48384x^{14} + 23040x^{13}y + 237312x^{13} + 110592x^{12}y$
 $+ 715392x^{12} + 328320x^{11}y + 1476288x^{11} + 649152x^{10}y + 2244384x^{10}$
 $+ 851616x^9y + 2790576x^9 + 613296x^8y + 3389832x^8 - 217368x^7y + 4619340x^7$
 $- 1369044x^6y + 6577362x^6 - 2243142x^5y + 8522811x^5 - 2596725x^4y$
 $+ 9523377x^4 - 2724669x^3y + 9235827x^3 - 3190086x^2y + 6814494x^2 - 3217644xy$
 $+ 4104540x - 2187702y + 1814400$

$B \quad -11520x^{17} + 2304x^{16}y - 110592x^{16} + 13824x^{15}y^2 - 11520x^{15}y - 470016x^{15}$
 $+ 117504x^{14}y^2 - 182016x^{14}y - 1156608x^{14} - 6912x^{13}y^4 + 43776x^{13}y^3$
 $+ 361728x^{13}y^2 - 718848x^{13}y - 1832832x^{13} - 2304x^{12}y^5 - 34560x^{12}y^4$
 $+ 225792x^{12}y^3 + 527616x^{12}y^2 - 1492992x^{12}y - 2040768x^{12}$
 $- 25344x^{11}y^5 - 27648x^{11}y^4 + 405504x^{11}y^3 + 383616x^{11}y^2$
 $- 1977984x^{11}y - 1942560x^{11} - 4608x^{10}y^6 - 41472x^{10}y^5 - 36864x^{10}y^4$
 $+ 414720x^{10}y^3 - 50112x^{10}y^2 - 1668672x^{10}y - 2299824x^{10} - 2304x^9y^6$
 $- 59904x^9y^5 - 76032x^9y^4 + 163584x^9y^3 - 388512x^9y^2 - 612000x^9y$
 $- 3503880x^9 + 2304x^8y^7 + 16128x^8y^6 - 138240x^8y^5 + 78336x^8y^4$
 $- 705024x^8y^3 + 347472x^8y^2 - 4176x^8y - 4989132x^8 + 20736x^7y^7$
 $- 62208x^7y^6 + 117504x^7y^5 - 248832x^7y^4 - 1331712x^7y^3$
 $+ 2090520x^7y^2 - 1075464x^7y - 5669874x^7 + 4608x^6y^8 - 11520x^6y^7$
 $+ 25344x^6y^6 + 202752x^6y^5 - 642816x^6y^4 - 1294848x^6y^3$
 $+ 3795156x^6y^2 - 3649860x^6y - 4970268x^6 - 13824x^5y^7 + 99072x^5y^6$
 $+ 138240x^5y^5 - 705024x^5y^4 - 995328x^5y^3 + 4676166x^5y^2$
 $- 6652818x^5y - 3212172x^5 - 4608x^4y^8 - 2304x^4y^7 + 80640x^4y^6$
 $+ 195840x^4y^5 - 831744x^4y^4 - 294912x^4y^3 + 4305749x^4y^2$
 $- 8343225x^4y + 756270x^4 - 4608x^3y^8 - 20736x^3y^7 + 158976x^3y^6$
 $- 36864x^3y^5 - 511488x^3y^4 + 117786x^3y^3 + 2850997x^3y^2$
 $- 11052580x^3y + 2074401x^3 - 9216x^2y^8 + 11520x^2y^7 + 76032x^2y^6$
 $- 105984x^2y^5 - 94345x^2y^4 + 214896x^2y^3 + 2827246x^2y^2$
 $- 8895444x^2y + 2192930x^2 - 4608xy^8 + 13824xy^7 - 16317xy^5 + 165359xy^4$
 $+ 780052xy^3 + 1159184xy^2 - 7829278xy + 2846336x - 6889y^6 - 108315y^5 - 274385y^4$
 $+ 1443344y^3 + 3219614y^2 - 3604066y + 77276$

$i = 33, m = 64$, neighbours 26, 34, 37, 38, 39, 41, 42, 90

$W \quad 52x^3y^2 - 236x^3y + 256x^3 - 98x^2y^2 + 490x^2y - 640x^2 - 102xy^4$
 $+ 470xy^3 + 226xy^2 - 854xy + 246x + 194y^4 - 980y^3 - 92y^2 - 98y + 208$

$A \quad -8192x^{19} + 16384x^{18} + 57344x^{17} + 32768x^{16}y - 122880x^{16} - 106496x^{15}y$
 $- 32768x^{15} + 40960x^{14}y + 57344x^{14} + 90112x^{13}y + 270336x^{13} + 106496x^{12}y$
 $- 346276x^{12} - 319488x^{11}y + 137880x^{11} + 133284x^{10}y - 363964x^{10} - 121496x^9y$
 $+ 895348x^9 + 437700x^8y - 753892x^8 - 319464x^7y + 117148x^7 - 128856x^6y$
 $+ 8852x^6 + 191016x^5y - 73816x^5 - 187348x^4y + 559228x^4 + 514164x^3y$
 $- 208384x^3 - 337148x^2y - 205724x^2 - 155544xy + 102336x + 45760y + 43264$

$B \quad 16384x^{19} - 32768x^{18} + 16384x^{17}y - 114688x^{17} - 90112x^{16}y + 245760x^{16}$
 $+ 8192x^{15}y^3 + 98304x^{15}y + 65536x^{15} - 16384x^{14}y^3 - 57344x^{14}y^2$
 $+ 73728x^{14}y - 98304x^{14} + 8192x^{13}y^4 - 73728x^{13}y^3 + 163840x^{13}y^2$
 $- 106496x^{13}y - 499712x^{13} - 49152x^{12}y^4 + 155648x^{12}y^3 - 81920x^{12}y^2$
 $- 8192x^{12}y + 569672x^{12} + 40960x^{11}y^4 + 114688x^{11}y^3 + 40960x^{11}y^2$
 $+ 40960x^{11}y - 300336x^{11} - 32768x^{10}y^5 + 106496x^{10}y^4 - 221184x^{10}y^3$
 $- 163840x^{10}y^2 - 24576x^{10}y + 785272x^{10} - 8192x^9y^6 + 98304x^9y^5$
 $- 65536x^9y^4 - 180224x^9y^3 + 204800x^9y^2 - 161628x^9y - 1520360x^9$
 $+ 32768x^8y^6 + 32768x^8y^5 - 98304x^8y^4 + 182436x^8y^3$
 $- 446792x^8y^2 + 562848x^8y + 1161508x^8 - 8192x^7y^7 - 16384x^7y^6$
 $- 172032x^7y^5 + 83304x^7y^3 + 677168x^7y^2 - 1112716x^7y - 96416x^7$
 $+ 32768x^6y^7 - 40960x^6y^6 - 73728x^6y^5 + 32768x^6y^4 - 17292x^6y^3$

$$\begin{aligned}
& -375688x^6y^2 + 1385084x^6y - 381668x^6 + 8192x^5y^8 + 32768x^5y^6 \\
& + 139264x^5y^5 + 40960x^5y^4 - 72936x^5y^3 + 204744x^5y^2 - 1636748x^5y \\
& + 1042980x^5 - 16384x^4y^8 - 65536x^4y^7 + 46940x^4y^5 + 106488x^4y^4 \\
& - 217412x^4y^3 - 332360x^4y^2 + 1821364x^4y - 1872348x^4 - 8192x^3y^8 \\
& - 8192x^3y^6 + 6808x^3y^5 - 438832x^3y^4 + 1200312x^3y^3 \\
& - 772548x^3y^2 - 547960x^3y + 533916x^3 + 24576x^2y^8 + 57344x^2y^7 \\
& + 5980x^2y^6 + 41860x^2y^5 + 243468x^2y^4 - 1693560x^2y^3 \\
& + 2187408x^2y^2 - 1149964x^2y + 354764x^2 + 39576xy^6 - 382280xy^5 \\
& + 814744xy^4 + 840800xy^3 - 1437608xy^2 + 71464xy + 49192x - 8192y^8 - 16384y^7 \\
& - 45828y^6 + 380240y^5 - 924704y^4 - 142296y^3 - 327008y^2 + 346384y - 62852
\end{aligned}$$

$i = 34, m = 2$, neighbours 33, 35

$$\begin{aligned}
W & x^3 + x^2y + 2x^2 - 2xy + y^2 - y \\
A & 0 \\
B & 3x^3 - 3x^2y + 3y^2 + 3y
\end{aligned}$$

$i = 35, m = 3$, neighbours 34, 36, 49, 64, 65, 84, 100

$$\begin{aligned}
W & -x^3y^2 + x^3y - 5x^3 + x^2y^2 - 4x^2y + 8x^2 - xy^4 + xy^2 + 10xy + 7x + 4y^3 \\
& + 32y + 8 \\
A & 9x^7 + 58x^6 + 167x^5 - 24x^4y + 254x^4 - 82x^3y + 22x^3 - 51x^2y - 417x^2 \\
& - 303xy - 510x - 440y - 312 \\
B & -18x^9 - 18x^8y - 98x^8 - 62x^7y - 218x^7 + 18x^6y^3 + 54x^6y^2 - x^6y \\
& - 192x^6 + 17x^5y^3 + 135x^5y^2 + 141x^5y + 375x^5 - 20x^4y^4 \\
& - 117x^4y^3 - 154x^4y^2 + 58x^4y + 648x^4 - x^3y^5 - 31x^3y^4 \\
& - 219x^3y^3 - 345x^3y^2 - 72x^3y + 208x^3 - x^2y^6 + x^2y^5 \\
& + 28x^2y^4 - 15x^2y^3 + 3x^2y^2 + 104x^2y + 28x^2 + 8xy^5 + 28xy^4 \\
& + 130xy^3 + 275xy^2 + 243xy - 246x - 16y^4 + 16y^3 + 184y^2 + 48y - 376
\end{aligned}$$

$i = 36, m = 1$, neighbours 35, 40

$$\begin{aligned}
W & x^2 + 3x + 2 \\
A & 0 \\
B & 2x^3 - 3x^2y + 6x^2 + 2xy^2 - 5xy + 6x - y^3 - 4y + 1
\end{aligned}$$

$i = 37, m = 2$, neighbours 4, 33

$$\begin{aligned}
W & x^3 + 2x^2y + x^2 - 2xy - y^2 - 2y \\
A & 0 \\
B & 3x^3 + 3x^2 + 3y^2
\end{aligned}$$

$i = 38, m = 2$, neighbours 10, 33

$$\begin{aligned}
W & x^3 + 2x^2y - x^2 - 2xy - y^2 + 2y \\
A & 0 \\
B & 3x^3 - 3x^2 + 3y^2
\end{aligned}$$

$i = 39, m = 6$, neighbours 33, 40

$$\begin{aligned}
W & 3x^3 + 3x^2y + 9x^2 - 6xy + 3y^2 - 9y \\
A & 0 \\
B & 27x^3 - 27x^2y + 27x^2 + 27y^2 + 27y
\end{aligned}$$

$i = 40, m = 24$, neighbours 26, 36, 39, 49, 53, 61, 76, 82, 83, 103

$$\begin{aligned}
W & x^3y^2 + 1393x^3y + 1631x^3 - x^2y^3 + 1392x^2y^2 + 193x^2y - 279x^2 \\
& - 2xy^4 - 2xy^3 - 1438xy^2 - 4321xy - 4552x - y^5 - 1392y^4 - 1627y^3 - 4040y^2 \\
& - 3911y - 2187 \\
A & 1728x^{13} + 23904x^{12} + 5760x^{11}y + 134352x^{11} + 80064x^{10}y + 355032x^{10} \\
& + 477216x^9y + 174420x^9 + 1587312x^8y - 1831338x^8 + 3226824x^7y - 6856875x^7 \\
& + 4239900x^6y - 14479395x^6 + 4214894x^5y - 25884480x^5 + 4827536x^4y \\
& - 42903415x^4 + 3593359x^3y - 53977640x^3 - 2926273x^2y - 42546858x^2 - 6305416xy \\
& - 18372447x - 2237763y - 2749827 \\
B & -3456x^{15} - 3456x^{14}y - 37440x^{14} - 38592x^{13}y - 146016x^{13} + 3456x^{12}y^3 \\
& - 1152x^{12}y^2 - 181728x^{12}y - 159696x^{12} + 3456x^{11}y^4 + 25344x^{11}y^3 \\
& - 32256x^{11}y^2 - 480528x^{11}y + 570024x^{11} + 16128x^{10}y^4 + 39168x^{10}y^3 \\
& - 226368x^{10}y^2 - 799992x^{10}y + 2455596x^{10} - 3456x^9y^6 - 4608x^9y^5 \\
& + 29952x^9y^4 - 36864x^9y^3 - 667296x^9y^2 - 774468x^9y + 4771242x^9 \\
& - 1152x^8y^7 - 5760x^8y^6 + 8064x^8y^4 - 102528x^8y^3 - 1131120x^8y^2 \\
& + 182754x^8y + 7633308x^8 + 1152x^7y^8 - 1152x^7y^7 + 13824x^7y^6 \\
& + 33408x^7y^5 - 124416x^7y^4 + 26496x^7y^3 - 1670472x^7y^2 \\
& + 2818568x^7y + 14738946x^7 + 2304x^6y^8 + 8064x^6y^7 + 24192x^6y^6 \\
& + 11520x^6y^5 - 268416x^6y^4 + 258048x^6y^3 - 2693908x^6y^2 \\
& + 8007059x^6y + 16895594x^6 - 1152x^5y^9 - 1152x^5y^8 + 19584x^5y^7
\end{aligned}$$

$$\begin{aligned}
& -4608x^5y^6 - 76032x^5y^5 - 118656x^5y^4 + 335247x^5y^3 \\
& -4205575x^5y^2 + 8565564x^5y + 6233953x^5 - 2304x^4y^9 - 3456x^4y^8 \\
& +14976x^4y^7 - 13824x^4y^6 - 89856x^4y^5 + 164742x^4y^4 \\
& +148858x^4y^3 - 2351778x^4y^2 + 17013818x^4y - 1446001x^4 + 1152x^3y^9 \\
& -3456x^3y^8 - 8064x^3y^7 - 1152x^3y^6 + 1145x^3y^5 + 224771x^3y^4 \\
& +9524387x^3y^3 + 11873684x^3y^2 + 33672498x^3y - 2637120x^3 + 2304x^2y^9 \\
& -18432x^2y^7 - 4618x^2y^6 + 68166x^2y^5 + 3968152x^2y^4 \\
& +12384934x^2y^3 + 10640307x^2y^2 + 21398244x^2y - 10408959x^2 + 4608xy^8 \\
& -13829xy^7 - 12961xy^6 - 1896812xy^5 - 2720802xy^4 - 5532634xy^3 \\
& -10130300xy^2 - 3329070xy - 10250033x - y^8 - 2784y^7 - 1940915y^6 - 4529296y^5 \\
& -8079566y^4 - 10450870y^3 - 8660141y^2 - 4614941y - 2510932
\end{aligned}$$

$i = 41, m = 2$, neighbours 32, 33

$$\begin{aligned}
W & x^3 - x^2y + 2x^2 + xy + 2y^2 - y \\
A & 0 \\
B & -3x^3 - 3x^2y + 3xy + 3y
\end{aligned}$$

$i = 42, m = 2$, neighbours 11, 33

$$\begin{aligned}
W & 2x^2 + 2xy + 2y^2 - 2y \\
A & 0 \\
B & -4x^3 - 4x^2y + 4x^2 + 4xy + 4y^2
\end{aligned}$$

$i = 43, m = 6$, neighbours 26, 44, 45, 46, 47, 48, 91

$$\begin{aligned}
W & -2x^3y^2 - 92x^3y + 62x^3 - 118x^2y^2 + 139x^2y - 50x^2 - 6xy^4 - 182xy^3 \\
& +120xy^2 - 101xy + 87x + 5y^5 - 52y^4 + 155y^3 - 21y^2 + 148y - 71 \\
A & 36x^{16} - 288x^{15} + 36x^{14}y + 972x^{14} - 216x^{13}y - 1590x^{13} + 515x^{12}y \\
& +2305x^{12} - 570x^{11}y - 4293x^{11} + 1100x^{10}y + 6008x^{10} + 144x^9y - 4502x^9 \\
& -3029x^8y - 5902x^8 - 4867x^7y + 14181x^7 + 29379x^6y - 18221x^6 - 58633x^5y \\
& +7136x^5 + 49837x^4y + 330x^4 - 18193x^3y - 10965x^3 - 17476x^2y + 11651x^2 \\
& +17409xy - 12354x - 10934y + 5041 \\
B & -72x^{16} + 504x^{15} - 1404x^{14} - 72x^{13}y + 1488x^{13} - 36x^{12}y^3 + 108x^{12}y^2 \\
& +122x^{12}y - 2294x^{12} + 288x^{11}y^3 - 612x^{11}y^2 + 710x^{11}y + 5632x^{11} \\
& -756x^{10}y^3 + 1390x^{10}y^2 - 1052x^{10}y - 4558x^{10} + 144x^9y^4 \\
& +798x^9y^3 - 1589x^9y^2 - 4405x^9y - 1444x^9 + 72x^8y^5 - 335x^8y^4 \\
& -1291x^8y^3 + 1357x^8y^2 + 821x^8y + 16395x^8 - 108x^7y^5 + 661x^7y^4 \\
& +1621x^7y^3 - 213x^7y^2 + 24626x^7y - 14600x^7 + 36x^6y^6 + 227x^6y^5 \\
& -929x^6y^4 - 2172x^6y^3 + 7720x^6y^2 - 58226x^6y + 11635x^6 \\
& -285x^5y^5 - 799x^5y^4 + 4834x^5y^3 - 16389x^5y^2 + 49253x^5y \\
& +7209x^5 + 72x^4y^6 + 435x^4y^5 + 2281x^4y^4 + 4256x^4y^3 \\
& +5857x^4y^2 + 4428x^4y - 10844x^4 + 36x^3y^7 - 279x^3y^6 + 323x^3y^5 \\
& +6902x^3y^4 - 31256x^3y^3 + 25562x^3y^2 - 41967x^3y + 18022x^3 \\
& -11x^2y^7 - 508x^2y^6 + 1982x^2y^5 - 43516x^2y^4 + 52251x^2y^3 \\
& -54168x^2y^2 + 62285x^2y - 16501x^2 + 121xy^7 + 532xy^6 - 13834xy^5 \\
& +53292xy^4 - 26789xy^3 + 58397xy^2 - 37451xy + 3521x - 25y^8 + 484y^7 - 4182y^6 \\
& +16294y^5 - 27689y^4 + 22612y^3 - 42771y^2 + 23185y - 3018
\end{aligned}$$

$i = 44, m = 4$, neighbours 9, 43

$$\begin{aligned}
W & -4xy + 4y \\
A & 0 \\
B & -16x^3 + 16x^2 - 16xy - 16x
\end{aligned}$$

$i = 45, m = 2$, neighbours 3, 43

$$\begin{aligned}
W & 2x^2y - 4x^2 - 2xy + 2y^2 - 2y \\
A & 0 \\
B & -4x^3 + 4x^2y - 4x^2 - 8xy + 4x
\end{aligned}$$

$i = 46, m = 2$, neighbours 10, 43

$$\begin{aligned}
W & -x^3 + x^2y - 4xy + x - y^2 + 2y \\
A & 0 \\
B & -3x^3 - 3x^2y + 6x^2 - 3x - 3y^2
\end{aligned}$$

$i = 47, m = 2$, neighbours 5, 43

$$\begin{aligned}
W & -x^3 - x^2y + x^2 - x + y^2 + y \\
A & 0 \\
B & -3x^3 + 3x^2y + 3x^2 - 3x + 3y^2 - 3y
\end{aligned}$$

$i = 48, m = 2$, neighbours 32, 43

$$\begin{aligned}
W & -x^3 + x^2y + 2xy + x + 2y^2 - y
\end{aligned}$$

A	0
B	$3x^3 + 3x^2y + 3x + 3y$
$i = 49, m = 2$	neighbours 9, 10, 35, 40
W	-4
A	$2x^4 - 2x^3y + 2x^3 + 4x^2y^2 + 10x^2y - 2x^2 - 2xy^3 - 4xy^2 + xy - 4x + 4y^3$ $+ 8y + 4$
B	$8x^3y^4 + 2x^3y^3 + 50x^3y^2 - 26x^3y + 2x^3 - 4x^2y^5 + 10x^2y^4$ $- 34x^2y^3 - 26x^2y^2 - 24x^2y - 6x^2 + 4xy^6 + 22xy^3 + 19xy^2 + 62xy + 2x$ $- 2y^6 - y^4 - 6y^3 - 10y^2 - 35y - 16$
$i = 50, m = 2$	neighbours 19, 51, 52, 53, 54, 59, 60, 61, 62, 92, 93
W	$12xy - 36x - 6y - 24$
A	$24xy^5 - 12xy^4 - 96xy^3 + 24xy^2 + 144xy - 144x - 12y^5 - 36y^4 - 120y^3 - 12y^2$ $+ 96y - 432$
B	$32x^9y^6 + 64x^9y^5 - 64x^9y^4 - 128x^9y^3 + 32x^9y^2 + 64x^9y$ $+ 16x^8y^7 + 200x^8y^6 + 336x^8y^5 - 464x^8y^4 - 720x^8y^3$ $+ 328x^8y^2 + 368x^8y - 64x^8 + 52x^7y^7 + 440x^7y^6 + 672x^7y^5$ $- 1344x^7y^4 - 1724x^7y^3 + 1048x^7y^2 + 808x^7y - 336x^7 - 16x^6y^8$ $- 8x^6y^7 + 396x^6y^6 + 696x^6y^5 - 2072x^6y^4 - 2480x^6y^3$ $+ 1484x^6y^2 + 944x^6y - 640x^6 - 68x^5y^8 - 172x^5y^7 + 232x^5y^6$ $+ 560x^5y^5 - 1944x^5y^4 - 2300x^5y^3 + 1228x^5y^2 + 832x^5y - 528x^5$ $- 76x^4y^8 - 156x^4y^7 + 460x^4y^6 + 540x^4y^5 - 1248x^4y^4$ $- 1064x^4y^3 + 912x^4y^2 + 568x^4y - 160x^4 + 12x^3y^8 - 4x^3y^7$ $+ 512x^3y^6 + 220x^3y^5 - 676x^3y^4 + 160x^3y^3 + 392x^3y^2 - 256x^3y$ $+ 288x^3 + 24x^2y^8 - 12x^2y^7 - 4x^2y^6 - 48x^2y^5 - 220x^2y^4$ $- 36x^2y^3 + 8x^2y^2 + 576x^2 - 36xy^8 - 24xy^7 - 84xy^6 - 12xy^5 + 48xy^4$ $- 324xy^3 + 24xy^2 + 336xy - 864x + 12y^8 + 36y^7 + 144y^6 + 72y^5 + 108y^4$ $+ 336y^3 - 240y^2 + 672y - 1008$
$i = 51, m = 2$	neighbours 14, 50
W	$2x^2 - xy + x - 2y$
A	0
B	$3xy + 3x$
$i = 52, m = 2$	neighbours 14, 50
W	$2x^2 - xy + x + y$
A	0
B	$-3xy + 3x - 3y$
$i = 53, m = 2$	neighbours 40, 50
W	$2x^2 - xy + 4x + y + 3$
A	0
B	$-3xy - 3y + 3$
$i = 54, m = 2$	neighbours 11, 50
W	$-2x^2 - 2xy - y + 2$
A	0
B	$-6x - 3y$
$i = 55, m = 2$	neighbours 4, 13, 32
W	$-2y + 2$
A	$2x^2 + 6xy + 6x + 4y^3 - 12y^2 + 18y$
B	$-8x^3y^3 + 48x^3y^2 - 12x^3y + 16x^3 + 36x^2y^3 - 12x^2y^2 + 30x^2y$ $- 4x^2 + 24xy^4 - 14xy^3 + 30xy^2 - 28xy + 10x + 12y^5 - 22y^4 + 28y^3 - 48y^2 + 26y$ $- 4$
$i = 56, m = 2$	neighbours 4, 9
W	$2x^2 + xy + x + y - 1$
A	0
B	$-3xy + 3x + 3y + 3$
$i = 57, m = 2$	neighbours 5, 9
W	$2x^2 - xy - 2x + 2y - 1$
A	0
B	$-3xy + 6x - 3$
$i = 58, m = 2$	neighbours 9, 32
W	$-2xy + 4x + 2y$
A	0

B	$-4x^2 - 4xy + 4x + 4$
$i = 59, m = 2$, neighbours 14, 50	
W	$2x^2 - xy + x - 2y$
A	0
B	$-3xy - 3x$
$i = 60, m = 2$, neighbours 4, 50	
W	$2x^2 + xy + 3x + 2y + 1$
A	0
B	$3xy - 3x - 3$
$i = 61, m = 2$, neighbours 40, 50	
W	$2x^2 - xy + 4x - 2y$
A	0
B	$-3xy - 6x - 6$
$i = 62, m = 2$, neighbours 32, 50	
W	$-2xy - 2y$
A	0
B	$-4x^2 - 4xy - 4x - 4$
$i = 63, m = 2$, neighbours 9, 14	
W	$2x^2 - xy - x - y$
A	0
B	$-3xy - 3x + 3y$
$i = 64, m = 2$, neighbours 9, 35	
W	$2x^2 - xy - y - 2$
A	0
B	$-3xy - 6x + 3y$
$i = 65, m = 2$, neighbours 2, 35	
W	$2x^2 - xy + 4x + 3$
A	0
B	$-3xy - 6y - 3$
$i = 66, m = 6$, neighbours 2, 4, 13, 32, 78	
W	$-14x^4y - 56x^4 + 11xy + 44x + 24y^4 + 24y^3 + 81y^2 + 49y + 74$
A	$18x^6 + 72x^5y + 162x^5 + 82x^4y^2 - 190x^4y - 902x^4 + 288x^3y^3$ $+ 1350x^3y^2 + 2151x^3y + 1620x^3 + 342x^2y^4 + 1935x^2y^3 + 4338x^2y^2$ $+ 4383x^2y + 2565x^2 + 288xy^5 + 2016xy^4 + 5404xy^3 + 8061xy^2 + 7371xy + 5245x$ $+ 126y^6 + 1303y^5 + 4917y^4 + 8454y^3 + 10854y^2 + 8082y + 4670$
B	$108x^3y^6 + 1008x^3y^5 + 3580x^3y^4 + 4176x^3y^3 + 4327x^3y^2$ $+ 1218x^3y + 1826x^3 + 36x^2y^7 + 216x^2y^6 + 92x^2y^5 + 326x^2y^4$ $- 2545x^2y^3 - 676x^2y^2 - 1190x^2y + 1414x^2 - 54xy^7 + 313xy^6 - 1192xy^5$ $- 2852xy^4 - 7018xy^3 - 4114xy^2 - 5240xy - 461x - 126y^8 - 1645y^7 - 4352y^6$ $- 10614y^5 - 14687y^4 - 17419y^3 - 11671y^2 - 7704y - 806$
$i = 67, m = 2$, neighbours 4, 14	
W	$2x^2 + 2x - 2y^2 - y$
A	0
B	$-6xy - 3y$
$i = 68, m = 2$, neighbours 10, 14	
W	$2x^2 - 2x - 2y^2 + y$
A	0
B	$-6xy + 3y$
$i = 69, m = 2$, neighbours 4, 10	
W	$2x^2 - xy^3 - 2xy^2 + 3y^3 - y^2 + 8y - 2$
A	$16x^3y - 64x^2y + 8xy - 32y$
B	$16x^6 - 8x^5y - 64x^5 + 4x^4y^2 + 32x^4y + 8x^4 - 2x^3y^3 - 16x^3y^2$ $+ 4x^3y - 32x^3 + x^2y^4 + 16x^2y^3 - 4x^2y^2 - 4x^2y - 12x^2 - 6xy^4$ $- 28xy^3 + 8xy^2 - 62xy + 48x + 9y^4 - 6y^3 + 13y^2 - 28y$
$i = 70, m = 2$, neighbours 5, 13	
W	$2x^2 - x + y^2 - 2y$
A	0
B	$-3x + 3y^2$
$i = 71, m = 2$, neighbours 3, 14	

W	$2x^2 - 2y^2 + 2y$
A	0
B	$-4x^2 + 8xy - 4x$
$i = 72, m = 2, \text{ neighbours } 4, 14$	
W	$2x^2 + 2x - 2y^2 - y$
A	0
B	$6xy + 3y$
$i = 73, m = 2, \text{ neighbours } 6, 14$	
W	$2x^2 - 2x$
A	0
B	$4x^2 + 4y^2 - 4y$
$i = 74, m = 2, \text{ neighbours } 14, 78, 86$	
W	$2x^2 + 2xy + 3x + 2y^2 + 3y$
A	0
B	$8x^3 + 24x^2 + 27x - 8y^3 - 24y^2 - 27y$
$i = 75, m = 2, \text{ neighbours } 3, 4$	
W	$2x^2 + 3x + y^2 - 2y + 1$
A	0
B	$3x - 3y^2 + 3$
$i = 76, m = 2, \text{ neighbours } 3, 40$	
W	$2x^2 + 4x - 2y^2 + 2$
A	0
B	$4x^2 - 8xy + 8x - 8y + 8$
$i = 77, m = 2, \text{ neighbours } 3, 32$	
W	$-x^2 - 3xy + 2x + y^2 - 3y + 2$
A	0
B	$-3x^2 + 3xy - 6x + 3y^2 - 3y$
$i = 78, m = 24, \text{ neighbours } 66, 74, 84, 104$	
W	$-15xy^2 - 42xy - 21x - 6y^4 - 27y^3 - 54y^2 - 54y - 15$
A	$18x^4 - 15x^3y + 60x^3 - 153x + 33y^4 + 102y^3 + 36y^2 - 117y - 180$
B	$-36x^6 + 66x^5y - 12x^5 - 30x^4y^2 - 78x^4y + 144x^4 - 36x^3y^3$ $-18x^3y^2 - 90x^3y - 90x^3 + 72x^2y^3 + 267x^2y^2 + 300x^2y + 216x^2$ $+36xy^5 + 69xy^4 - 291xy^3 - 657xy^2 - 135xy + 258x - 36y^6 - 249y^5 - 624y^4$ $-1047y^3 - 1629y^2 - 708y - 135$
$i = 79, m = 2, \text{ neighbours } 4, 10$	
W	$2x^2 - 2y^2 - 2$
A	0
B	$4x^2 + 8xy - 4$
$i = 80, m = 2, \text{ neighbours } 5, 6$	
W	$2x^2 - 2x - 2y^2 + 3y - 1$
A	0
B	$-6xy + 6x + 3y - 3$
$i = 81, m = 2, \text{ neighbours } 5, 10$	
W	$2x^2 - 3x - 2y^2 + 2y + 1$
A	0
B	$-6xy + 3x + 6y - 3$
$i = 82, m = 2, \text{ neighbours } 6, 40$	
W	$2x^2 + 4y - 4$
A	0
B	$-4x^2 - 8x - 4y^2 + 4$
$i = 83, m = 2, \text{ neighbours } 11, 40$	
W	$-2x^2 - xy - 2x + y^2 - 2y + 3$
A	0
B	$-3xy - 6x - 3y^2 - 3$
$i = 84, m = 1, \text{ neighbours } 35, 78$	
W	$x^2 + 2x + 1$
A	0
B	$x^2y + x^2 + 2xy + 2x + y^3 + 3y^2 + 4y + 2$
$i = 85, m = 2, \text{ neighbours } 11, 32$	

W	$x^2 + 3xy + 2y^2 - 2y + 2$
A	0
B	$-3x^2 - 3xy$
$i = 86, m = 3, \text{ neighbours } 7, 74, 92, 105$	
W	3
A	0
B	$-3x^3y + 3x^3 + 6x^2y^2 - 18x^2y + 12x^2 - 3xy^3 + 21xy^2 - 30xy + 12x - 6y^3$ $+18y^2 - 12y - 3$
$i = 87, m = 2, \text{ neighbours } 1, 88$	
W	$6x + 6$
A	0
B	$18x - 12y - 6$
$i = 88, m = 6, \text{ neighbours } 87, 90, 101$	
W	6
A	0
B	$72x^2 + 96xy - 96x - 96y^2 + 21$
$i = 89, m = 2, \text{ neighbours } 8$	
W	$-x + y + 3$
A	0
B	$-11x - 3y + 5$
$i = 90, m = 2, \text{ neighbours } 5, 10, 11, 33, 88, 97$	
W	2
A	$168x^6 + 336x^5y - 672x^5 + 336x^4y^2 - 1512x^4y + 1344x^4 + 336x^3y^3$ $-1848x^3y^2 + 3108x^3y - 1680x^3 + 336x^2y^4 - 2100x^2y^3 + 4368x^2y^2$ $-3864x^2y + 1428x^2 + 336xy^5 - 1848xy^4 + 4368xy^3 - 5460xy^2 + 3528xy - 1008x$ $+168y^6 - 1008y^5 + 3612y^4 - 5628y^3 + 5166y^2 - 3108y + 840$
B	$-504x^3y^6 + 2436x^3y^5 - 4704x^3y^4 + 5880x^3y^3 - 5460x^3y^2$ $+3192x^3y - 840x^3 - 336x^2y^7 + 1722x^2y^6 - 4284x^2y^5 + 6216x^2y^4$ $-7098x^2y^3 + 6132x^2y^2 - 3024x^2y + 714x^2 - 168xy^8 + 1176xy^7$ $-3360xy^6 + 5292xy^5 - 6930xy^4 + 7245xy^3 - 5376xy^2 + 2352xy - 504x - 84y^9$ $+672y^8 - 2478y^7 + 4914y^6 - 7875y^5 + 9996y^4 - 8904y^3 + 5859y^2 - 2394y + 418$
$i = 91, m = 2, \text{ neighbours } 43$	
W	$-2x - 2y$
A	0
B	$-4x^2 + 4x$
$i = 92, m = 2, \text{ neighbours } 50, 86$	
W	$3x$
A	0
B	$3x + 6$
$i = 93, m = 2, \text{ neighbours } 50, 94$	
W	$6x - 3$
A	0
B	$-12x - 15$
$i = 94, m = 21, \text{ neighbours } 93, 95$	
W	21
A	0
B	$336x + 399$
$i = 95, m = 1, \text{ neighbours } 94$	
W	1
A	0
B	-5
$i = 96, m = 2, \text{ neighbours } 9, 97$	
W	$-4x + 5$
A	0
B	$6x + 3$
$i = 97, m = 7, \text{ neighbours } 19, 30, 90, 96, 98$	
W	7
A	0
B	$56x^3 + 224x^2 - 14x - 63$

$i = 98, m = 2$, neighbours 2, 97

W 14

A 0

B $-28x + 28$

$i = 99, m = 2$, neighbours 3

W $x + y - 1$

A 0

B $3x - 3y + 3$

$i = 100, m = 4$, neighbours 35

W $2x - 4y - 2$

A 0

B $6x + 6$

$i = 101, m = 2$, neighbours 10, 88

W $6x - 6$

A 0

B $6x + 12y - 6$

$i = 102, m = 2$, neighbours 6

W $x - y + 1$

A 0

B $3x + 3y - 3$

$i = 103, m = 4$, neighbours 40

W $2x - 4y$

A 0

B $6x + 12$

$i = 104, m = 2$, neighbours 78

W $x - y$

A 0

B $3x + 3y + 6$

$i = 105, m = 2$, neighbours 32, 86

W $-3x - 3y$

A 0

B $3x - 3y + 6$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 1\}, k_{ij} = 7$

α -9

β $-18x^3 + 6x^2y + 6x^2 - 3x - 9y^2 - 2y - 20$

γ $6x^2 - 2$

$\{i, j\} = \{0, 2\}, k_{ij} = 7^3$

α $18x + 17$

β $18xy^2 - x + 17y^2 - 20$

γ $-36x^3 + 74x^2 - 114x - 18y + 120$

$\{i, j\} = \{0, 3\}, k_{ij} = 7$

α $-2x - 3$

β $-4x^4 + 4x^3y - 10x^3 + 2x^2y - 2xy^2 - 6xy + x - 3y^2 + 4y - 14$

γ $4x^3 + 2x^2 - 6x + 4$

$\{i, j\} = \{0, 4\}, k_{ij} = 3$

α $24x + 27$

β $-48x^4 - 48x^3y - 6x^3 + 42x^2y + 30x^2 + 24xy^2 - 36xy + 69x + 27y^2 + 30y + 84$

γ $-48x^3 + 42x^2 - 36x + 30$

$\{i, j\} = \{0, 5\}, k_{ij} = 7$

α $2x - 3$

β $4x^4 - 4x^3y + 2x^3 + 2x^2y - 6x^2 + 2xy^2 + 6xy - 9x - 3y^2 + 4y - 10$

γ $-4x^3 + 2x^2 + 6x + 4$

$\{i, j\} = \{0, 6\}, k_{ij} = 7$

α $2x - 3$

β $-4x^4 - 4x^3y + 10x^3 + 2x^2y + 4x^2 + 2xy^2 + 6xy - 7x - 3y^2 + 4y - 14$

$$\begin{aligned}
&\gamma \quad -4x^3 + 2x^2 + 6x + 4 \\
&\{i, j\} = \{0, 13\}, k_{ij} = 1 \\
&\alpha \quad -1 \\
&\beta \quad 2x^3 + 2x^2y - x - y^2 - 2 \\
&\gamma \quad 2x^2 \\
&\{i, j\} = \{1, 15\}, k_{ij} = 2^3 \cdot 3^5 \\
&\alpha \quad 948x^2 - 394xy + 1348x - 972y - 972 \\
&\beta \quad -632x^4 - 370x^3y - 900x^3 - 162x^2y^2 - 72x^2y + 808x^2 - 44xy^3 + 302xy^2 \\
&\quad + 992xy + 608x + 972y^2 + 972 \\
&\gamma \quad x^4 + 4x^3y + 38x^3 + 8x^2y^2 + 58x^2y - 231x^2 + 15xy^3 + 66xy^2 - 192xy - 694x \\
&\quad + 22y^4 + 24y^3 + 111y^2 - 293y + 185 \\
&\{i, j\} = \{1, 16\}, k_{ij} = 3^4 \\
&\alpha \quad -13y - 35 \\
&\beta \quad 13y^2 - 17y + 29 \\
&\gamma \quad -26x^3y - 70x^3 + 26x^2y + 70x^2 + 39xy^2 - 77xy + 17x - 39y^3 - y^2 - 45y - 17 \\
&\{i, j\} = \{1, 18\}, k_{ij} = 2 \cdot 3^7 \cdot 7^2 \\
&\alpha \quad 309x^3 + 25533x + 103y^3 + 824y^2 - 751y - 1946 \\
&\beta \quad -206x^5 - 206x^4y + 206x^3y - 12970x^3 - 12229x^2y + 9264x^2 - 309xy^3 \\
&\quad - 2575xy^2 - 27867xy + 50199x - 103y^4 - 1339y^3 + 9890y^2 - 75891y + 125413 \\
&\gamma \quad 412x^3y^2 + 412x^3y + 11950x^3 + 17022xy^2 - 111177xy + 149268x + 618y^4 \\
&\quad - 8521y^3 + 76262y^2 - 133348y + 86967 \\
&\{i, j\} = \{1, 20\}, k_{ij} = 3^5 \cdot 7 \cdot 13 \\
&\alpha \quad 3866x^2 - 5478xy + 490x - 9683y - 14243 \\
&\beta \quad -2577x^4 + 1057x^3y + 56x^3 + 1842x^2y^2 + 8180x^2y + 9186x^2 + 1341xy^3 \\
&\quad + 8025xy^2 + 5169xy + 6178x + 503y^4 + 2312y^3 + 3057y^2 + 5245y + 7870 \\
&\gamma \quad x^4 - 55x^3y + 1147x^3 - 166x^2y^2 + 1338x^2y - 4652x^2 - 335xy^3 + 568xy^2 \\
&\quad - 2611xy + 23242x - 503y^4 - 2312y^3 + 6123y^2 - 2494y + 5688 \\
&\{i, j\} = \{1, 21\}, k_{ij} = 3^6 \cdot 7 \cdot 13^3 \\
&\alpha \quad -241056x^3 + 2236563x + 282660y^2 - 746274y - 1873380 \\
&\beta \quad 160704x^5 + 160704x^4y + 160704x^3y^2 - 40176x^3y - 1219854x^3 \\
&\quad - 148264x^2y^2 - 1304890x^2y + 2223188x^2 - 148264xy^3 - 901702xy^2 + 1890392xy \\
&\quad + 3634161x - 148264y^4 + 190944y^3 + 2717793y^2 - 6803291y + 9337911 \\
&\gamma \quad 160704x^3y^2 + 40176x^3y + 974268x^3 - 823854xy^2 - 1766262xy + 9923685x \\
&\quad - 148264y^4 + 177076y^3 + 2191275y^2 - 3011339y + 465866 \\
&\{i, j\} = \{1, 23\}, k_{ij} = 2 \cdot 3^5 \cdot 7^2 \\
&\alpha \quad -960x^3 + 4224x - 80y^2 - 2945y - 4952 \\
&\beta \quad 640x^5 + 640x^4y + 640x^3y^2 - 160x^3y - 2856x^3 - 1026x^2y + 3648x^2 \\
&\quad - 1266xy^2 + 2403xy + 5299x + 1390y^3 + 7346y^2 - 11969y + 11064 \\
&\gamma \quad 640x^3y^2 + 160x^3y + 520x^3 - 2816xy^2 - 7991xy + 14521x + 1310y^3 + 7101y^2 \\
&\quad + 66y + 7798 \\
&\{i, j\} = \{1, 24\}, k_{ij} = 2^2 \cdot 3^5 \cdot 7 \\
&\alpha \quad -37334x^2 + 17434xy - 22825x + 13115y + 5115 \\
&\beta \quad 24889x^4 + 13307x^3y + 15385x^3 + 5175x^2y^2 - 1189x^2y - 11266x^2 \\
&\quad + 1109xy^3 - 9940xy^2 - 7650xy - 25465x + 2218y^3 - 12173y^2 - 3754y + 11919 \\
&\gamma \quad x^4 - 123x^3y - 504x^3 - 370x^2y^2 - 1018x^2y - 1825x^2 - 739xy^3 - 1545xy^2 \\
&\quad - 391xy + 59185x - 1109y^4 - 1580y^3 + 4297y^2 - 11111y - 2640 \\
&\{i, j\} = \{1, 87\}, k_{ij} = 2 \cdot 3^7 \\
&\alpha \quad -246x - 178y - 1088 \\
&\beta \quad 162x^3 + 276x^2y + 810x^2 + 208xy^2 + 810xy - 556x + 96y^3 + 438y^2 - 780y + 2514 \\
&\gamma \quad x^4 + 3x^3y - 42x^3 + 9x^2y^2 - 54x^2y + 237x^2 + 16xy^3 - 28xy^2 + 112xy \\
&\quad - 1313x + 24y^4 + 77y^3 - 372y^2 + 640y - 386 \\
&\{i, j\} = \{2, 65\}, k_{ij} = 2^2 \cdot 3^2 \cdot 7^3 \\
&\alpha \quad 648x + 612 \\
&\beta \quad -1296x^3 + 2664x^2 - xy^3 - 4104x + 682y^3 - 37y^2 - 648y + 4320 \\
&\gamma \quad -x^2y^2 + 681xy^2 - 36xy + 1329y^2 - 720y \\
&\{i, j\} = \{2, 66\}, k_{ij} = 2 \cdot 3^3 \cdot 7^3 \\
&\alpha \quad 972x + 918 \\
&\beta \quad -1944x^3 + 3996x^2 - xy^2 - xy - 6156x - y^3 + 970y^2 + 53y + 6480 \\
&\gamma \quad x^2y^2 + x^2y - 969xy^2 - 1023xy - 915y^2 - 1995y
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{2, 98\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
&\alpha \quad 0 \\
&\beta \quad 6 \\
&\gamma \quad 6x - 6 \\
&\{i, j\} = \{3, 21\}, k_{ij} = 2 \cdot 7 \cdot 13 \\
&\alpha \quad -60x^2 + 40xy + 17x - 125y + 216 \\
&\beta \quad 120x^4 - 25x^3 - 80x^2y^2 + 199x^2y - 133x^2 - 40xy^3 + 111xy^2 + 7xy + 315x \\
&\quad + 40y^4 - 89y^3 + 166y^2 + 7y + 398 \\
&\gamma \quad -40x^3y + 129x^3 - 40x^2y^2 + 112x^2y + 7x^2 - 18xy^2 + 136xy + 136x + 40y^4 \\
&\quad - 129y^3 + 130y^2 - 68y + 232 \\
&\{i, j\} = \{3, 22\}, k_{ij} = 2^2 \\
&\alpha \quad -x + 2 \\
&\beta \quad 2x^3 - 4x^2 + 1 \\
&\gamma \quad 2x^4 - 2x^3y - 2x^3 + 4x^2y - 3x^2 + xy^2 + xy + 2x - 2y^2 - 3y - 5 \\
&\{i, j\} = \{3, 26\}, k_{ij} = 3^2 \cdot 7 \cdot 13 \\
&\alpha \quad -81y + 171 \\
&\beta \quad 81y^2 + 72y + 216 \\
&\gamma \quad 162x^3y - 342x^3 + 162x^2y - 342x^2 - 81xy^2 + 90xy - 558x + 81y^3 - 9y^2 + 63y - 774 \\
&\{i, j\} = \{3, 45\}, k_{ij} = 2^3 \cdot 7^3 \\
&\alpha \quad -55160x^6 - 74301x^4 - 110320x^3y^2 + 193060x^3y - 129711x^3 - 162436xy^2 \\
&\quad + 284263xy - 122513x + 44002y^2 - 80495y + 37865 \\
&\beta \quad 165480x^7 + 55160x^6y - 110320x^6 + 360891x^5 + 110320x^4y^2 - 8439x^4y \\
&\quad - 318250x^4 + 110320x^3y^3 - 137900x^3y^2 + 129623x^3y + 316978x^3 \\
&\quad + 162436x^2y^2 - 121827x^2y - 4079x^2 + 162436xy^3 - 247047xy^2 + 682969xy \\
&\quad - 406085x - 44002y^3 + 300171y^2 - 317970y + 40609 \\
&\gamma \quad -110320x^7 + 55160x^6 - 148602x^5 + 63187x^4 - 165480x^3y^2 - 88x^3y \\
&\quad - 97934x^3 + 79846x^2 - 243654xy^2 + 40609xy + 126592x + 307680y^2 - 528765y + 242963 \\
&\{i, j\} = \{3, 71\}, k_{ij} = 2 \cdot 3 \cdot 7 \cdot 13 \\
&\alpha \quad -522xy - 963x + 666y - 393 \\
&\beta \quad 4x^3y + 2446x^3 + 8x^2y^2 - 1236x^2 + 16xy^3 - xy^2 + 484xy - 377x + 33y^4 \\
&\quad - 3y^3 + 967y^2 + 1019y + 153 \\
&\gamma \quad 520x^4 + 258x^3y + 424x^3 + 123x^2y^2 - 1078x^2y + 833x^2 + 49xy^3 - 575xy^2 \\
&\quad + 1302xy - 1460x + 33y^4 - 36y^3 + 1636y^2 + 325y + 739 \\
&\{i, j\} = \{3, 75\}, k_{ij} = 2^3 \cdot 7^3 \\
&\alpha \quad -2106xy + 556x - 1702y + 3074 \\
&\beta \quad 4211x^3y + 3099x^3 + x^2y^3 + x^2y^2 - 1921x^2y - 1633x^2 - 16xy + 3923x + y^2 \\
&\quad + 3420y + 8881 \\
&\gamma \quad x^4y - 4211x^4 - x^3y^2 + 4212x^3y - 3403x^3 + x^2y^3 - x^2y^2 \\
&\quad - 1921x^2y - 1986x^2 - 2106xy^2 + 539xy - 9185x - 1701y^2 + 6492y - 3063 \\
&\{i, j\} = \{3, 76\}, k_{ij} = 2^4 \cdot 7 \cdot 13 \\
&\alpha \quad -3060xy + 1726x - 1820y + 2548 \\
&\beta \quad 24x^3y - 3452x^3 + 48x^2y^2 + 142x^2y + 1404x^2 + 96xy^3 + 287xy^2 + 2232xy \\
&\quad - 989x + 192y^4 + 574y^3 + 4465y^2 + 3226y - 571 \\
&\gamma \quad -3048x^4 - 1512x^3y + 3037x^3 - 720x^2y^2 + 1377x^2y + 156x^2 - 288xy^3 \\
&\quad + 1623xy^2 - 1345xy + 3286x - 192y^4 - 574y^3 - 2645y^2 - 2134y + 4575 \\
&\{i, j\} = \{3, 77\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7^2 \\
&\alpha \quad 22x^3 - 36x^2y + 15381x^2 - 24928xy + 38139x - 18503y + 20267 \\
&\beta \quad x^4y + 639x^3y + 6058x^3 - 635x^2y^2 + 182x^2y + 1865x^2 - xy^4 + 1275xy^3 \\
&\quad + 5874xy^2 - 7038xy + 18382x + 2y^5 - 1908y^4 - 5694y^3 + 8909y^2 - 9428y + 23795 \\
&\gamma \quad 44x^5 - 27x^4y + 30674x^4 + 16x^3y^2 - 18532x^3y + 20988x^3 - 10x^2y^3 \\
&\quad + 10861x^2y^2 - 14842x^2y + 6481x^2 + 5xy^4 - 5096xy^3 + 12757xy^2 + 10404xy \\
&\quad + 13321x - 2y^5 + 1910y^4 + 3786y^3 + 3900y^2 + 16573y + 2019 \\
&\{i, j\} = \{3, 99\}, k_{ij} = 2 \\
&\alpha \quad -6y^2 + 9y - 2 \\
&\beta \quad 12x^4 + 6x^2y - 8x^2 + 6xy^2 - 14xy + 8x + 29y^2 - 11y + 8 \\
&\gamma \quad -12x^5 - 29xy^2 + 11xy - 8x - 6y^4 + 32y^3 - 51y^2 + 32y - 8 \\
&\{i, j\} = \{4, 26\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \\
&\alpha \quad -4y + 4 \\
&\beta \quad 4y^2 - 12y + 20 \\
&\gamma \quad 8x^3y - 8x^3 + 8x^2y - 8x^2 - 4xy^2 + 20xy - 28x - 4y^3 + 4y^2 + 20y - 68
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{4, 27\}, k_{ij} = 3 \\
&\alpha \quad 24x + 27 \\
&\beta \quad -48x^3 + 42x^2 - 36x + 30 \\
&\gamma \quad 48x^4 + 48x^3y + 6x^3 - 42x^2y - 30x^2 - 24xy^2 + 36xy - 69x - 27y^2 - 30y - 84 \\
&\{i, j\} = \{4, 37\}, k_{ij} = 2^2 \cdot 3^3 \\
&\alpha \quad -168x^4 - 48x^3y + 132x^3 + 372x^2y + 669x^2 + 60xy + 60x - 468y - 201 \\
&\beta \quad -12x^2y^2 - 1233x^2y - 627x^2 - 24xy^3 - 312xy^2 - 66xy + 246x - 48y^4 - 624y^3 \\
&\quad + 105y^2 + 27y - 93 \\
&\gamma \quad 336x^5 + 96x^4y - 600x^4 - 840x^3y - 738x^3 - 336x^2y^2 + 888x^2y + 618x^2 \\
&\quad + 72xy^3 + 996xy^2 + 1485xy + 243x + 48y^4 + 672y^3 + 1035y^2 + 693y + 897 \\
&\{i, j\} = \{4, 55\}, k_{ij} = 1 \\
&\alpha \quad -y + 1 \\
&\beta \quad y^2 - y + 1 \\
&\gamma \quad 2x^3y - 2x^3 - 2x^2y + 2x^2 - xy^2 + 3xy - 3x - y^3 + y^2 - 3y + 1 \\
&\{i, j\} = \{4, 56\}, k_{ij} = 1 \\
&\alpha \quad -6x^2 - 4xy - 3x - 3y + 4 \\
&\beta \quad 12x^4 - 4x^3y - 18x^3 - 4x^2y^2 + 17x^2y + 15x^2 + 4xy^3 + 2xy^2 - 15xy - 13x \\
&\quad - 4y^3 + y^2 + 5y + 29 \\
&\gamma \quad 4x^3y + x^3 - 4x^2y^2 - 11x^2y + 9xy^2 + 8xy - 24x + 4y^4 + 2y^3 - 8y^2 - 21y - 24 \\
&\{i, j\} = \{4, 60\}, k_{ij} = 3^2 \\
&\alpha \quad -84x^2 - 56xy - 127x - 96y - 34 \\
&\beta \quad 168x^4 - 56x^3y - 82x^3 - 56x^2y^2 + 186x^2y + 40x^2 + 56xy^3 - 31xy^2 - 74xy \\
&\quad - 43x + 40y^2 - 31y + 294 \\
&\gamma \quad -56x^3y - 24x^3 + 56x^2y^2 + 151x^2y - 21x^2 - 127xy^2 - 89xy + 353x - 56y^4 \\
&\quad - 25y^3 + 111y^2 + 291y + 319 \\
&\{i, j\} = \{4, 66\}, k_{ij} = 2 \cdot 3 \cdot 7 \\
&\alpha \quad 252x - 126y + 294 \\
&\beta \quad -504x^3 + 504x^2y + 311x^2 - 470xy - 311x - 252y^3 + 17y^2 + 361y + 1814 \\
&\gamma \quad 252x^3y + 109x^3 - 252x^2y^2 - 706x^2y + 84x^2 + 596xy^2 + 445xy - 1587x \\
&\quad + 252y^4 + 109y^3 - 529y^2 - 1411y - 1478 \\
&\{i, j\} = \{4, 67\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad -848x^2 - 566xy - 1408x - 315y - 524 \\
&\beta \quad 1696x^4 - 566x^3y - 576x^3 - 564x^2y^2 + 951x^2y + 504x^2 + 564xy^3 + 30xy^2 \\
&\quad - 309xy - 432x + 282y^3 + 188y^2 - 447y - 488 \\
&\gamma \quad -x^4 + 283x^3y + 155x^3 - 283x^2y^2 - 786x^2y - 24x^2 + 631xy^2 + 525xy - 1561x \\
&\quad + 282y^4 + 155y^3 - 501y^2 - 1377y - 1668 \\
&\{i, j\} = \{4, 69\}, k_{ij} = 2 \cdot 7^2 \\
&\alpha \quad -1350xy + 784x + 284y + 882 \\
&\beta \quad 10x^3y - 1568x^3 - 21x^2y^2 - 206x^2y + 1372x^2 + 42xy^3 + 411xy^2 - 102xy \\
&\quad - 1176x - 85y^4 - 821y^3 + 203y^2 - 2460y + 980 \\
&\gamma \quad -1345x^4 + 667x^3y - 593x^3 - 318x^2y^2 - 401x^2y - 350x^2 + 127xy^3 \\
&\quad - 76xy^2 + 534xy - 945x - 85y^4 - 906y^3 - 419y^2 - 2196y - 709 \\
&\{i, j\} = \{4, 72\}, k_{ij} = 2^2 \cdot 3^2 \\
&\alpha \quad -848x^2 - 566xy - 1408x - 315y - 524 \\
&\beta \quad 1696x^4 - 566x^3y - 576x^3 - 564x^2y^2 + 951x^2y + 504x^2 + 564xy^3 + 30xy^2 \\
&\quad - 309xy - 432x + 282y^3 + 188y^2 - 447y - 488 \\
&\gamma \quad x^4 - 283x^3y - 155x^3 + 283x^2y^2 + 786x^2y + 24x^2 - 631xy^2 - 525xy + 1561x \\
&\quad - 282y^4 - 155y^3 + 501y^2 + 1377y + 1668 \\
&\{i, j\} = \{4, 75\}, k_{ij} = 3 \cdot 7^3 \\
&\alpha \quad -2055xy - 972x - 3084y + 57 \\
&\beta \quad 4109x^3y - 2165x^3 + x^2y^3 - x^2y^2 - 3996x^2y + 2160x^2 - xy^3 + xy^2 \\
&\quad + 3774xy + 2068x + y^3 + y^2 - 5617y + 7976 \\
&\gamma \quad x^4y + 4109x^4 + x^3y^2 + 4110x^3y - 2053x^3 + x^2y^3 - x^2y^2 \\
&\quad - 3996x^2y - 2170x^2 - xy^3 - 2054xy^2 + 2805xy - 12102x + y^3 - 3082y^2 - 5557y - 6890 \\
&\{i, j\} = \{4, 79\}, k_{ij} = 2 \cdot 7^2 \\
&\alpha \quad -1350xy + 784x + 284y + 882 \\
&\beta \quad 10x^3y - 1568x^3 - 21x^2y^2 - 206x^2y + 1372x^2 + 42xy^3 + 411xy^2 - 102xy \\
&\quad - 1176x - 85y^4 - 821y^3 + 203y^2 - 2460y + 980 \\
&\gamma \quad 1345x^4 - 667x^3y + 593x^3 + 318x^2y^2 + 401x^2y + 350x^2 - 127xy^3 + 76xy^2 \\
&\quad - 534xy + 945x + 85y^4 + 906y^3 + 419y^2 + 2196y + 709
\end{aligned}$$

$$\begin{aligned}
&\{i, j\} = \{5, 7\}, k_{ij} = 5^2 \\
&\alpha \quad -9x + 14 \\
&\beta \quad 18x^3 + 8x^2 - 2x - 3 \\
&\gamma \quad -18x^4 + 18x^3y - 8x^3 + 8x^2y - 7x^2 - 9xy^2 - 11xy - 10x + 14y^2 + 11y + 42 \\
&\{i, j\} = \{5, 22\}, k_{ij} = 5^2 \\
&\alpha \quad -9x + 14 \\
&\beta \quad 18x^3 + 8x^2 - 2x - 3 \\
&\gamma \quad 18x^4 - 18x^3y + 8x^3 - 8x^2y + 7x^2 + 9xy^2 + 11xy + 10x - 14y^2 - 11y - 42 \\
&\{i, j\} = \{5, 23\}, k_{ij} = 2 \cdot 5^2 \cdot 7^2 \\
&\alpha \quad -581x^2 + 387xy + 1037x - 1547y + 1194 \\
&\beta \quad 1162x^4 + x^3y - 49x^3 - 775x^2y^2 + 1917x^2y - 167x^2 - 387xy^3 + 595xy^2 \\
&\quad + 495xy + 3550x + 387y^4 - 861y^3 + 750y^2 + 517y + 3656 \\
&\gamma \quad -387x^3y + 863x^3 - 388x^2y^2 + 209x^2y + 1706x^2 - 653xy^2 + 1684xy + 1119x \\
&\quad + 387y^4 - 861y^3 - 23y^2 - 784y - 12 \\
&\{i, j\} = \{5, 26\}, k_{ij} = 2 \cdot 3 \cdot 5^2 \cdot 13 \\
&\alpha \quad -468y + 858 \\
&\beta \quad 468y^2 + 78y + 1092 \\
&\gamma \quad 936x^3y - 1716x^3 + 936x^2y - 1716x^2 - 468xy^2 + 858xy - 2808x + 468y^3 + 546y^2 \\
&\quad + 1170y - 1716 \\
&\{i, j\} = \{5, 28\}, k_{ij} = 2 \cdot 5^2 \cdot 7 \\
&\alpha \quad -204x^2 + 136xy + 146x + 68y - 76 \\
&\beta \quad 408x^4 + 204x^3y + 23x^3 - 68x^2y^2 + 184x^2y - 66x^2 - 136xy^3 + 58xy^2 \\
&\quad + 104xy + 201x - 68y^4 + 25y^3 + 157y^2 + 494y + 274 \\
&\gamma \quad -68x^3y + 93x^3 - 68x^2y^2 + 18x^2y + 112x^2 - 76xy^2 - 21xy + 46x + 68y^4 \\
&\quad - 93y^3 - 132y^2 - 286y - 73 \\
&\{i, j\} = \{5, 30\}, k_{ij} = 1 \\
&\alpha \quad -y + 1 \\
&\beta \quad xy - x + y^2 + 3 \\
&\gamma \quad x^3y - x^3 - 2x + 2y^2 + 1 \\
&\{i, j\} = \{5, 47\}, k_{ij} = 3^2 \cdot 5^2 \cdot 7 \cdot 13 \\
&\alpha \quad -4407x^3 - 209xy^2 + 209xy + 5850x - 7587y^4 + 12108y^3 - 28370y^2 + 39683y - 13182 \\
&\beta \quad -23103x^6 + 23103x^5y + 55438x^5 + 15288x^4y^2 - 32031x^4y + 4945x^4 \\
&\quad + 26442x^3y - 11700x^3 - 114x^2y^4 - 418x^2y^2 + 16296x^2y + 25155x^2 \\
&\quad + 7473xy^4 + 18563xy^3 - 5287xy^2 + 19119xy - 55096x + 7473y^5 - 4293y^4 + 461y^3 \\
&\quad - 864y^2 + 5101y + 7293 \\
&\gamma \quad -23103x^5 + 23103x^4y + 46624x^4 - 7815x^3y^2 - 17742x^3y - 3869x^3 + 10655xy \\
&\quad + 19305x + 114y^5 - 228y^4 + 31089y^3 - 34779y^2 + 40825y - 68239 \\
&\{i, j\} = \{5, 57\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7^2 \\
&\alpha \quad -107x^3 - 1494x + 535y^3 - 2436y + 4384 \\
&\beta \quad 214x^5 + 214x^4y + 214x^4 + 642x^3y + 2988x^3 - 642x^2y^2 + 2302x^2y \\
&\quad + 1484x^2 - 1177xy^3 + 598xy^2 + 7630xy - 5350x - 535y^4 - 535y^3 + 724y^2 - 494y \\
&\quad + 12018 \\
&\gamma \quad 1284x^3y^2 + 1284x^3y - 3739x^3 + 2988xy^2 - 2988xy - 9426x - 642y^4 + 2096y^3 \\
&\quad + 8364y^2 - 1308y + 3224 \\
&\{i, j\} = \{5, 70\}, k_{ij} = 3 \cdot 5^2 \\
&\alpha \quad -54xy + 30x + 30y + 9 \\
&\beta \quad 108x^3y + 48x^3 - 12x^2y + 78x^2 + 18xy + 66x + 96y + 84 \\
&\gamma \quad 108x^4 - 108x^3y + 48x^3 + 12x^2y + 36x^2 + 54xy^2 - 48xy + 96x - 30y^2 - 105y - 48 \\
&\{i, j\} = \{5, 80\}, k_{ij} = 2^5 \cdot 5^2 \\
&\alpha \quad -1216x^2 + 810xy + 1334x - 1205y + 437 \\
&\beta \quad 2432x^4 + 812x^3y - 236x^3 - 810x^2y^2 + 2554x^2y - 1488x^2 - 810xy^3 \\
&\quad + 1647xy^2 - 856xy + 1171x + 405y^3 + 179y^2 + 88y + 1664 \\
&\gamma \quad 405x^3y - 1027x^3 + 406x^2y^2 + 36x^2y - 1725x^2 + 1061xy^2 - 1451xy - 973x \\
&\quad - 405y^4 + 1026y^3 + 275y^2 + 930y + 427 \\
&\{i, j\} = \{5, 81\}, k_{ij} = 3 \cdot 5^2 \cdot 7 \cdot 13 \\
&\alpha \quad -7712x^3 - 1632x - 53984y^3 + 92544y^2 - 24774y - 3077 \\
&\beta \quad 15424x^5 + 15424x^4y + 15424x^4 + 46272x^3y + 3264x^3 + 95262x^2y - 23375x^2 \\
&\quad + 53984xy^3 - 546xy^2 + 55837xy - 16982x + 53984y^4 - 38560y^3 + 164136y^2 - 112266y \\
&\quad + 25491 \\
&\gamma \quad -46272x^3y^2 + 46272x^3y + 17369x^3 + 1632xy^2 - 4080xy - 19098x - 61969y^3
\end{aligned}$$

	$+8499y^2 - 75651y + 21743$
$\{i, j\} = \{5, 90\}, k_{ij} = 2$	
α	0
β	0
γ	1
$\{i, j\} = \{6, 7\}, k_{ij} = 2^2$	
α	$-x + 2$
β	$2x^3 - 4x^2 + 1$
γ	$2x^4 + 2x^3y - 6x^3 - 4x^2y + 3x^2 - xy^2 - xy + 2y^2 + 3y + 5$
$\{i, j\} = \{6, 12\}, k_{ij} = 3$	
α	$-16x + 16$
β	$32x^3 - 32x^2y + 32xy - 32x + 4$
γ	$-16x^2y^2 + 40x^2y - 36x^2 + 8xy^2 - 44xy + 20x + 8y^2 + 6y + 15$
$\{i, j\} = \{6, 29\}, k_{ij} = 2^3 \cdot 7$	
α	$-20x^2 - 13xy - 18x - 53y + 81$
β	$40x^4 - x^3y + 29x^3 - 27x^2y^2 + 75x^2y - 48x^2 + 13xy^3 - 51xy^2 + 10xy - 67x$ $+ 13y^4 - 34y^3 + 85y^2 - 3y + 137$
γ	$-13x^3y + 47x^3 + 14x^2y^2 - 52x^2y + 16x^2 + 4xy^2 + 51xy + 38x - 13y^4 + 47y^3$ $- 66y^2 + 60y - 62$
$\{i, j\} = \{6, 73\}, k_{ij} = 2 \cdot 3$	
α	$7xy - 10x - 9y + 12$
β	$-14x^3y + 6x^3 - 2x^2y + 4xy + 11y + 18$
γ	$14x^4 + 14x^3y - 18x^3 + 2x^2y - 6x^2 - 7xy^2 - xy - 18x + 9y^2 - 14y + 8$
$\{i, j\} = \{6, 80\}, k_{ij} = 2^4$	
α	$-24x^2 - 16xy + 12x - 8y + 16$
β	$48x^4 - 16x^3y + 24x^3 - 16x^2y^2 + 10x^2y - 10x^2 + 16xy^3 - 6xy^2 + 23xy - 9x$ $- 8y^3 + 23y^2 - 4y + 41$
γ	$8x^3y - 23x^3 - 8x^2y^2 + 34x^2y - 24x^2 - 10xy^2 + 6xy - 44x + 8y^4 - 23y^3$ $+ 19y^2 - 37y + 9$
$\{i, j\} = \{6, 82\}, k_{ij} = 2^6 \cdot 3 \cdot 7^3 \cdot 13$	
α	$-413x^2 + 423647xy - 1483517x - 1070160y + 1498224$
β	$-413x^3y^2 - 846468x^3y + 1273287x^3 + 413x^2y^3 - 612x^2y^2 + 18895x^2y$ $+ 13274x^2 - 9xy^3 - 4xy^2 + 5442xy - 777356x - 2y^3 + 1279y^2 + 1843173y + 1376320$
γ	$-413x^5 - 846667x^4 - 847502x^3y + 1291551x^3 + 413x^2y^3 - 621x^2y^2$ $+ 18485x^2y - 404447x^2 - 9xy^3 + 423641xy^2 - 629500xy + 1285905x - 2y^3$ $- 1068881y^2 + 1201077y - 978032$
$\{i, j\} = \{6, 102\}, k_{ij} = 2$	
α	$x - y + 2$
β	$-2x^3 + 3x^2y - 3x^2 - y^3 + 3y^2 + 4$
γ	$x^3y - 3x^3 - 2x^2y^2 + 6x^2y - 4x^2 - 2xy^2 - 5x + y^4 - 3y^3 + 2y^2 - 3y$
$\{i, j\} = \{7, 8\}, k_{ij} = 3 \cdot 7$	
α	$-9x - 3$
β	$18x^4 + 6x^3y - 4x^2y - 23x^2 - 9xy^2 - 11xy - 7x - 3y^2 + 6y + 9$
γ	$6x^3 - 4x^2 - 2x + 9$
$\{i, j\} = \{7, 9\}, k_{ij} = 3$	
α	$x + 1$
β	$xy^2 + xy + 5x + y^2 + y + 2$
γ	$-2x^3 - 4x^2 - 2x - y + 2$
$\{i, j\} = \{7, 10\}, k_{ij} = 3$	
α	$x + 1$
β	$-2x^4 - 2x^3y - 4x^3 - 4x^2y - x^2 + xy^2 - xy + 5x + y^2 + 2y + 3$
γ	$-2x^3 - 4x^2 - 2x + 1$
$\{i, j\} = \{7, 11\}, k_{ij} = 7$	
α	$-3x - 1$
β	$12x^4 + 6x^3y - 8x^3 - 4x^2y - 6x^2 - 3xy^2 - 3xy + 7x + 7y - 2$
γ	$6x^3 - 4x^2 - 2x + y + 8$
$\{i, j\} = \{7, 86\}, k_{ij} = 3$	
α	0
β	0

γ	1
$\{i, j\} = \{8, 15\}, k_{ij} = 2^4 \cdot 3^2 \cdot 5^2 \cdot 13^2$	
α	$-69282x^3 + 186246x - 10264y^4 + 7698y^3 - 71310y^2 - 365246y - 383208$
β	$46188x^5 - 46188x^4y + 30792x^3y^2 - 15396x^3y - 139560x^3 + 273468x^2y$ $+ 546348x^2 - 10264xy^4 + 17962xy^3 - 232080xy^2 - 448508xy + 23426x + 10264y^5$ $- 17962y^4 + 99536y^3 + 258012y^2 + 175978y + 225192$
γ	$10264x^3y^3 - 51320x^3y^2 + 220138x^3y - 218157x^3 + 62082xy^2 - 248328xy$ $- 305481x - 10264y^5 + 48754y^4 - 138470y^3 + 163767y^2 - 80297y + 40705$
$\{i, j\} = \{8, 17\}, k_{ij} = 13^2$	
α	$-15y - 43$
β	$15y^2 - 17y + 53$
γ	$-30x^3y - 86x^3 - 30x^2y - 86x^2 + 45xy^2 - 81xy + 73x + 45y^3 - 21y^2 + 35y + 73$
$\{i, j\} = \{8, 25\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \cdot 13^2$	
α	$6408x^2 + 9108xy + 14058x - 3852y - 14868$
β	$-4272x^4 - 1769x^3y - 9014x^3 + 3069x^2y^2 + 12045x^2y + 7218x^2 - 2231xy^3$ $- 7564xy^2 + 721xy - 10077x + 836y^4 + 77y^3 - 7534y^2 + 13000y + 30716$
γ	$-93x^3y - 1074x^3 + 279x^2y^2 + 1623x^2y + 10206x^2 - 559xy^3 - 1646xy^2$ $- 6971xy + 60731x + 836y^4 + 77y^3 - 9714y^2 + 466y + 18298$
$\{i, j\} = \{8, 26\}, k_{ij} = 13^2$	
α	$-15y - 43$
β	$15y^2 - 17y + 53$
γ	$30x^3y + 86x^3 + 30x^2y + 86x^2 - 45xy^2 + 81xy - 73x - 45y^3 + 21y^2 - 35y - 73$
$\{i, j\} = \{8, 28\}, k_{ij} = 2^2 \cdot 3^3 \cdot 7 \cdot 13^2$	
α	$-6x^3 + 6x^2y - 28853x^2 - 5333xy + 45891x - 2668y - 42850$
β	$4x^5 - 8x^4y + 19235x^4 + 7x^3y^2 - 15742x^3y - 27934x^3 - 4x^2y^3$ $+ 9454x^2y^2 + 22806x^2y + 23847x^2 + 2xy^4 - 4454xy^3 - 7437xy^2 + 14431xy$ $- 50307x - y^5 + 1673y^4 + 10007y^3 - 45630y^2 - 1882y + 84914$
γ	$x^4 - x^3y^2 + 182x^3y - 7981x^3 - 549x^2y^2 + 10026x^2y - 29019x^2$ $+ 1108xy^3 - 6136xy^2 + 16804xy + 91792x + y^5 - 1672y^4 - 11679y^3 + 36618y^2$ $+ 80355y - 32066$
$\{i, j\} = \{8, 29\}, k_{ij} = 3 \cdot 7^2 \cdot 13^2$	
α	$1152x^3 - 12843x + 288y^2 - 6156y - 3684$
β	$-768x^5 + 768x^4y - 768x^3y^2 + 192x^3y + 7266x^3 - 6714x^2y + 7112x^2$ $+ 5850xy^2 - 3038xy - 9987x + 3288y^3 + 9197y^2 - 18995y + 21159$
γ	$768x^3y^2 + 192x^3y + 4656x^3 - 8562xy^2 - 17328xy + 23847x - 3576y^3 - 9905y^2$ $- 287y + 952$
$\{i, j\} = \{8, 30\}, k_{ij} = 3 \cdot 7$	
α	$-4x - 4y - 6$
β	$2x^3 + 5x^2 - 6xy - x + 4y^2 + y + 7$
γ	$x^4 + x^3y - x^3 - x^2y^2 + x^2y - 13x - y^3 + 4y^2 - 8$
$\{i, j\} = \{8, 31\}, k_{ij} = 2^4 \cdot 3 \cdot 7 \cdot 13^2$	
α	$-2215x^3 + 8109x + 2658y^2 + 6984y - 6926$
β	$-1772x^3y + 13735x^3 - 52947x - 2658y^3 + 3648y^2 - 69726y + 49858$
γ	$4430x^5 - 13290x^4 - 3544x^3y - 17553x^3 - 15746x^2y + 66511x^2 + 3987xy^3$ $- 4875xy^2 + 74811xy - 56642x - 3987y^3 + 35173y^2 + 2455y + 20352$
$\{i, j\} = \{8, 89\}, k_{ij} = 13^2$	
α	$9x - 22$
β	$-6x^3 + 6x^2y + 16x^2 - 4xy^2 - 14xy + 21x + 2y^3 + 7y^2 - 27y + 49$
γ	$4x^3 - 4x^2y + 53x^2 + 5xy^2 - 36xy + 110x + 2y^4 + 7y^3 - 47y^2 + 76y - 98$
$\{i, j\} = \{9, 18\}, k_{ij} = 3^2 \cdot 7^2$	
α	$39x - 6$
β	$-78x^3 - 66x^2 - 39xy^2 - 21xy + 195x - 111y^2 + 3y + 531$
γ	$-39xy^2 - 21xy + 183x + 6y^2 + 105y - 96$
$\{i, j\} = \{9, 44\}, k_{ij} = 7$	
α	$2x - 3$
β	$-2x^3 + 2x^2y^2 + 3x^2 - 3xy^2 + 4$
γ	$2x^2 + 2xy^2 + x - 3y^2 - 6$
$\{i, j\} = \{9, 49\}, k_{ij} = 1$	
α	0

β	1
γ	$-x + 1$
$\{i, j\}$	$\{9, 56\}, k_{ij} = 1$
α	$x - 1$
β	$-2x^3 + 2x - 2y^2 + 2y + 1$
γ	$-2xy + x + y^2 + y - 1$
$\{i, j\}$	$\{9, 57\}, k_{ij} = 7^2$
α	$-8x + 5$
β	$16x^3 + 6x^2 - 10x - 5y^3 - 3y^2 - 8y + 68$
γ	$-5xy^2 - 13xy - 42x - 3y^2 + 2y + 14$
$\{i, j\}$	$\{9, 58\}, k_{ij} = 3 \cdot 7$
α	$2x - 3$
β	$-4x^3 + 2x^2 + 2xy^2 + 6xy + 13x + y^2 + 8y + 18$
γ	$2xy^2 + 6xy + 7x - 3y^2 - 2y$
$\{i, j\}$	$\{9, 63\}, k_{ij} = 3^2$
α	-3
β	$6x^2 + 6x + 3y^3 + 3y^2 + 6y + 6$
γ	$3xy^2 + 6x - 3y$
$\{i, j\}$	$\{9, 64\}, k_{ij} = 2 \cdot 7$
α	$-4x + 6$
β	$8x^3 - 4x^2 - 12x - 2y^3 - 12y^2 + 2y + 20$
γ	$-2xy^2 - 8xy + 14x - 4y^2 + 12y - 14$
$\{i, j\}$	$\{9, 96\}, k_{ij} = 7$
α	0
β	4
γ	$-2x + 3$
$\{i, j\}$	$\{10, 25\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5 \cdot 7$
α	$10x^3 + 18x^2y - 7760x^2 - 23188xy - 4670x + 10158y + 14940$
β	$-20x^5 - 15x^4y + 15480x^4 + 32x^3y^2 + 23135x^3y + 46394x^3 - 16x^2y^3$ $-23150x^2y^2 - 30244x^2y + 30171x^2 - 7xy^4 + 7xy^3 + 10053xy^2 + 7919xy$ $+42739x + 8y^5 + 7699y^4 + 9289y^3 - 20455y^2 - 19498y + 6732$
γ	$-x^4y + 2x^3y^2 + 7709x^3y - 6074x^3 - 7704x^2y^2 + 34634x^2y + 5109x^2$ $-7xy^4 + 5xy^3 - 28559xy^2 - 6985xy - 12489x + 8y^5 + 7683y^4 - 6077y^3 + 1873y^2$ $+17236y + 10728$
$\{i, j\}$	$\{10, 30\}, k_{ij} = 3^2 \cdot 7$
α	$-120y + 12$
β	$-80x^2y + 8x^2 + 80xy^2 - 8xy + 80y^2 + 72y + 292$
γ	$160x^3y - 16x^3 - 150x - 40y^4 + 4y^3 + 60y^2 - 76y + 217$
$\{i, j\}$	$\{10, 38\}, k_{ij} = 2^3 \cdot 7$
α	$24x^4 + 7x^3y - 86x^3 - 67x^2y + 121x^2 + 17xy - 133x - 125y + 130$
β	$2x^2y^2 - 309x^2y + 474x^2 + 3xy^3 + 29xy^2 - 192xy + 239x + 7y^4 + 56y^3$ $-196y^2 - 199y + 186$
γ	$-48x^5 - 14x^4y + 124x^4 + 120x^3y - 118x^3 + 48x^2y^2 + 62x^2y + 148x^2$ $-10xy^3 - 85xy^2 + 659xy - 562x - 7y^4 - 49y^3 + 370y^2 - 183y + 61$
$\{i, j\}$	$\{10, 46\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13$
α	$20312x^3 + 62043xy^2 - 62043xy - 126456x - 53556y^4 - 42206y^3 + 24207y^2 - 78720y$ $+114334$
β	$-201292x^5 + 148474x^4y^2 + 196304x^4y + 99182x^4 + 36582x^3y^2$ $+85290x^3y + 131040x^3 - 41362x^2y^4 - 124086x^2y^2 + 798353x^2y$ $-679338x^2 - 12194xy^4 - 50693xy^3 + 228154xy^2 - 328984xy + 776635x + 12194y^5$ $+13038y^4 - 177690y^3 + 532302y^2 - 184864y + 122524$
γ	$-160668x^4 + 148474x^3y^2 + 155680x^3y + 180430x^3 + 563127xy - 793998x$ $-41362y^5 - 41362y^4 - 261439y^3 + 613454y^2 - 689902y + 658043$
$\{i, j\}$	$\{10, 49\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5$
α	$-90y$
β	$90y^2 + 90y + 180$
γ	$180x^3y - 90xy^2 - 90xy - 180x - 90y^3 + 180y^2 + 360$
$\{i, j\}$	$\{10, 68\}, k_{ij} = 3^2 \cdot 5 \cdot 7$
α	$808x^3 - 4944x - 3232y^3 + 9696y^2 - 5448y + 4451$

β	$-1616x^5 + 1616x^4y - 3232x^4 + 4848x^3y + 5040x^3 - 12308x^2y + 4410x^2$
	$-3232xy^3 + 2420xy^2 - 19348xy + 4588x + 3232y^4 - 6464y^3 + 7470y^2 - 10744y + 4766$
γ	$-2424x^3y^2 + 9696x^3y - 11102x^3 - 4944xy^2 + 5439x - 5254y^3 - 2397y^2 - 2388y$
	$+1160$
$\{i, j\} = \{10, 69\}, k_{ij} = 2 \cdot 7^2$	
α	$-600x^2 - 750xy + 416x + 100y + 282$
β	$1200x^4 + 300x^3y + 1568x^3 - 898x^2y^2 - 520x^2y + 1372x^2 + 299xy^3$
	$-42xy^2 - 404xy + 1176x + 299y^4 + 23y^3 - 248y^2 - 84y + 380$
γ	$301x^3y - 276x^3 - 300x^2y^2 + 1389x^2y + 70x^2 - 1114xy^2 - 197xy - 388x$
	$+299y^4 - 276y^3 + 128y^2 + 469y + 416$
$\{i, j\} = \{10, 79\}, k_{ij} = 2 \cdot 7^2$	
α	$-600x^2 - 750xy + 416x + 100y + 282$
β	$1200x^4 + 300x^3y + 1568x^3 - 898x^2y^2 - 520x^2y + 1372x^2 + 299xy^3$
	$-42xy^2 - 404xy + 1176x + 299y^4 + 23y^3 - 248y^2 - 84y + 380$
γ	$-301x^3y + 276x^3 + 300x^2y^2 - 1389x^2y - 70x^2 + 1114xy^2 + 197xy + 388x$
	$-299y^4 + 276y^3 - 128y^2 - 469y - 416$
$\{i, j\} = \{10, 81\}, k_{ij} = 2 \cdot 3 \cdot 7 \cdot 13$	
α	$-20x^3 - 1026x + 60y^3 - 140y^2 - 1638y + 1592$
β	$40x^5 - 40x^4y + 80x^4 - 120x^3y + 2172x^3 - 3610x^2y + 3447x^2 + 60xy^3$
	$+1598xy^2 - 3721xy + 2335x - 60y^4 + 80y^3 - 140y^2 + 7305y - 3020$
γ	$40x^3y^2 - 120x^3y - 2367x^3 - 1026xy^2 - 513xy + 4243x + 899y^3 - 3939y^2 + 4831y$
	-5158
$\{i, j\} = \{10, 90\}, k_{ij} = 2$	
α	0
β	0
γ	1
$\{i, j\} = \{10, 101\}, k_{ij} = 2 \cdot 3^2 \cdot 5$	
α	$-650x - 130y + 740$
β	$1300x^3 - 780x^2y + 938x^2 - 260xy^2 - 686xy + 1042x + 520y^3 + 26y^2 - 948y + 410$
γ	$-130x^3y + 91x^3 + 130x^2y^2 - 568x^2y - 142x^2 + 477xy^2 + 167xy + 226x$
	$-130y^4 + 91y^3 - 26y^2 - 294y - 210$
$\{i, j\} = \{11, 19\}, k_{ij} = 1$	
α	0
β	0
γ	-1
$\{i, j\} = \{11, 24\}, k_{ij} = 2^3 \cdot 3^2 \cdot 7$	
α	$xy^2 + 283xy - 1565x + y^3 + 280y^2 - 2420y + 2967$
β	$-165x^3 + x^2y^2 + 164x^2y - 1746x^2 - 164xy^2 + 1744xy - 2458x - y^4 - 283y^3$
	$+1240y^2 + 3845y + 3471$
γ	$-2x^3y^2 - 566x^3y + 3295x^3 + 3x^2y^2 + 1145x^2y - 1058x^2 - xy^3 - 8xy^2$
	$-2640xy - 181x + y^4 + 448y^3 - 2924y^2 + 2127y - 4364$
$\{i, j\} = \{11, 42\}, k_{ij} = 2 \cdot 7$	
α	$x^2y - 9x^2 + xy^2 - 13xy + 3x - 4y^2 + 4y - 6$
β	$x^2y^2 - x^2y - 6x^2 - xy^3 + 4xy^2 + 7xy + 11x + 4y^3 + 7y + 8$
γ	$2x^3y - 18x^3 - 8x^2y + 6x^2 + xy^2 + xy - 18x + 3y^2 - 13y + 17$
$\{i, j\} = \{11, 54\}, k_{ij} = 2 \cdot 3 \cdot 7^2$	
α	-42
β	$4x^2 - 8xy - 84x + 16y^2 + 168y + 252$
γ	$-40x^3 + 20x^2y - 44x^2 - 8xy^2 + 24xy + 170x + 16y^3 + 110y^2 + 100y - 168$
$\{i, j\} = \{11, 83\}, k_{ij} = 2 \cdot 7^2$	
α	$-18x - 18y - 36$
β	$-12x^3y + 12x^3 + 12x^2y^2 - 8x^2y + 56x^2 - 12xy^3 + 8xy^2 - 75xy + 30x + 12y^4$
	$-8y^3 + 75y^2 + 36y + 45$
γ	$-12x^4 - 8x^3 + 12x^2y - 3x^2 - 12xy^2 + 20xy + 37x + 12y^4 - 20y^3 + 65y^2 - 63y - 17$
$\{i, j\} = \{11, 85\}, k_{ij} = 2 \cdot 3$	
α	$-3x - 3y$
β	$-2x^2 + xy + 3x + 3y^2 + 3y + 6$
γ	$-6x^3 - 2x^2 - xy + 5x + 4y^2 + 2y + 1$
$\{i, j\} = \{11, 90\}, k_{ij} = 2$	

α	0
β	0
γ	1
$\{i, j\}$	$\{12, 14\}, k_{ij} = 3$
α	$-16x + 16$
β	$-16x^4 + 16x^3y + 16x^3 - 8x^2y - 4x^2 - 8xy^2 - 12xy - 16x + 8y^2 + 6y + 19$
γ	$32x^3 - 16x^2 - 16x + 4$
$\{i, j\}$	$\{13, 30\}, k_{ij} = 5 \cdot 7^2$
α	$-66x - 200y - 127$
β	$132x^3 + 134x^2y + 180x^2 - 134xy^2 - 275xy - 21x + 267y^2 + 164y + 369$
γ	$67x^3y + 37x^3 - 66x^2y^2 + 81x^2y + 29x^2 - 119xy^2 + 2xy - 170x + 67y^4$ $+ 37y^3 - 31y^2 - 54y - 118$
$\{i, j\}$	$\{13, 55\}, k_{ij} = 1$
α	$-y + 1$
β	$x^2y - 2x^2 - xy^2 + xy + 2$
γ	$x^3y + xy^2 - 2x + y^4 - y^3 - 3y + 2$
$\{i, j\}$	$\{13, 66\}, k_{ij} = 1$
α	-1
β	$2x^2 - 2xy - 2y^2 + y$
γ	$2xy^2 - xy + 2y^3 - 2y^2 + 2y$
$\{i, j\}$	$\{13, 70\}, k_{ij} = 1$
α	$-2xy - 2x - 2y - 1$
β	$4x^3y + 2x^2 - 2xy + 4x + 2y + 2$
γ	$-4x^4 - 4x^3y + 4x^2 + 2xy^2 + 4xy + 2x + 2y^2 - y + 2$
$\{i, j\}$	$\{14, 20\}, k_{ij} = 3 \cdot 7 \cdot 13$
α	$16x - 1088y^3 - 1088y^2 - 562y - 273$
β	$1088x^5 + 2176x^4y + 1088x^4 - 544x^3y - 322x^3 - 32x^2y + 256x^2 + 1088xy^3$ $+ 1650xy^2 + 1915xy + 277x + 1088y^4 + 1648y^3 + 3426y^2 + 1698y + 261$
γ	$1088x^5 - 544x^3y - 290x^3 + 526xy^2 + 1369xy + 277x - 560y^3 - 128y^2 - 209y - 16$
$\{i, j\}$	$\{14, 51\}, k_{ij} = 3 \cdot 7$
α	$-x - 26y^2 - 42y - 21$
β	$2x^3 - 24x^2y - 16x^2 + 10xy + 26y^3 + 42y^2 + 73y + 58$
γ	$52x^3y + 58x^3 - 26x^2y - 2xy^2 - 14xy - 58x + 26y^3 + 33y^2 + 82y - 1$
$\{i, j\}$	$\{14, 52\}, k_{ij} = 3 \cdot 7$
α	$-38x^2 + 25xy - 45x + 3y - 21$
β	$76x^4 + 26x^3y + 90x^3 - 25x^2y^2 + 62x^2y + 39x^2 - 25xy^3 - 27xy^2 + 59xy$ $+ 19x - 25y^3 - 5y^2 + 38y + 47$
γ	$25x^3y + 3x^3 + 26x^2y^2 + 2x^2y - 19x^2 - 2xy - 85x - 25y^4 - 2y^3 + 17y^2 + 53y$ $- 45$
$\{i, j\}$	$\{14, 59\}, k_{ij} = 3 \cdot 7$
α	$-x - 26y^2 - 42y - 21$
β	$2x^3 - 24x^2y - 16x^2 + 10xy + 26y^3 + 42y^2 + 73y + 58$
γ	$-52x^3y - 58x^3 + 26x^2y + 2xy^2 + 14xy + 58x - 26y^3 - 33y^2 - 82y + 1$
$\{i, j\}$	$\{14, 63\}, k_{ij} = 3$
α	$6x^2 - 4xy - 2x - 4y - 3$
β	$-12x^4 - 4x^3y + 4x^3 + 4x^2y^2 + 7x^2y + 5x^2 + 4xy^3 + 4xy^2 + 4xy + 3x - 4y^3$ $+ 4y^2 + 4y + 19$
γ	$-4x^3y - x^3 - 4x^2y^2 + x^2y + 3x^2 + xy^2 + xy + 13x + 4y^4 - y^2 - 11y + 2$
$\{i, j\}$	$\{14, 67\}, k_{ij} = 3^2$
α	$4x^2 - 2xy - 9y - 9$
β	$-8x^4 - 2x^3y + 4x^2y^2 + 11x^2y + 18x^2 + 4xy^3 + 10xy^2 + 18xy + 2y^3$ $+ 12y^2 + 10y + 4$
γ	$x^4 - x^3y - 4x^3 - x^2y^2 + 4x^2 + 4xy^2 + 5xy + 7x + 2y^4 + 3y^3 + y^2 - 14y - 9$
$\{i, j\}$	$\{14, 68\}, k_{ij} = 3 \cdot 7$
α	$-100y^2 + 10y - 21$
β	$200x^2y^2 - 20x^2y + 42x^2 + 200xy^3 - 20xy^2 + 42xy + 250y$
γ	$125x + 100y^4 - 10y^3 - 29y^2 - 120y + 52$
$\{i, j\}$	$\{14, 71\}, k_{ij} = 2^2 \cdot 7 \cdot 13$
α	$1548xy - 2008x + 1280y - 364$

β	$-12x^3y + 2474x^3 - 24x^2y^2 - 207x^2y + 1177x^2 - 49xy^3 - 413xy^2 - 177xy$
	$+98x - 97y^4 - 827y^3 - 354y^2 - 1550y + 1360$
γ	$-1542x^4 - 765x^3y + 449x^3 - 364x^2y^2 - 709x^2y + 98x^2 - 146xy^3 + 65xy^2$
	$-838xy + 1360x - 97y^4 - 924y^3 + 2y^2 - 1912y + 2008$
$\{i, j\} = \{14, 72\}, k_{ij} = 3^2$	
α	$4x^2 - 2xy - 9y - 9$
β	$-8x^4 - 2x^3y + 4x^2y^2 + 11x^2y + 18x^2 + 4xy^3 + 10xy^2 + 18xy + 2y^3$
	$+12y^2 + 10y + 4$
γ	$-x^4 + x^3y + 4x^3 + x^2y^2 - 4x^2 - 4xy^2 - 5xy - 7x - 2y^4 - 3y^3 - y^2 + 14y + 9$
$\{i, j\} = \{14, 73\}, k_{ij} = 3^2$	
α	$27xy + 12x - 30y - 9$
β	$-54x^3y + 30x^3 - 18x^2y + 12x^2 - 6x + 27y + 60$
γ	$-54x^4 + 54x^3y + 6x^3 + 18x^2y + 6x^2 - 27xy^2 - 39xy - 60x + 30y^2 + 12y + 12$
$\{i, j\} = \{14, 74\}, k_{ij} = 1$	
α	$-4y^2 - 2y - 1$
β	$8x^4 + 4x^2y + 2x^2 + 4xy^2 + 2xy + 4y^2 + 11y + 1$
γ	$-8x^5 - 4xy^2 - 11xy - x - 4y^4 + 2y^3 + 2y^2 + y$
$\{i, j\} = \{19, 50\}, k_{ij} = 1$	
α	0
β	-1
γ	0
$\{i, j\} = \{19, 97\}, k_{ij} = 1$	
α	0
β	-1
γ	0
$\{i, j\} = \{26, 33\}, k_{ij} = 3$	
α	$x^2 - y$
β	$-x^3y^2 - 4x^3 - x^2y^2 - x^2y - 4x^2 + xy^2 + xy - 3x + 2y^2 + y - 3$
γ	$-2x^2 + y^2 + 4$
$\{i, j\} = \{26, 40\}, k_{ij} = 3^2 \cdot 13$	
α	$3y - 9$
β	$-6x^3y + 18x^3 - 6x^2y + 18x^2 + 3xy^2 - 12xy - 3y^3 + 15y^2 - 3y - 54$
γ	$-3y^2 + 6y + 18$
$\{i, j\} = \{26, 43\}, k_{ij} = 3$	
α	$-2y - 3$
β	$-2x^3y^2 + 5x^3y + 4x^2y + 6x^2 - 2xy^3 - xy^2 - xy - 4y^3 + 2y^2 - 4y$
γ	$2y^2 - y + 6$
$\{i, j\} = \{30, 97\}, k_{ij} = 7$	
α	0
β	0
γ	1
$\{i, j\} = \{31, 32\}, k_{ij} = 2^4 \cdot 3^2 \cdot 7$	
α	$261x^3 - 761x + 174y^2 - 2320y + 1982$
β	$-522x^5 + 1566x^4 + 1392x^3y - 693x^3 - 98x^2y + 598x^2 + 87xy^3 - 649xy^2$
	$-977xy - 1017x - 87y^3 - 177y^2 + 11y - 1906$
γ	$-348x^3y - 2483x^3 + 1967x - 174y^3 + 2320y^2 + 518y + 2990$
$\{i, j\} = \{32, 41\}, k_{ij} = 2 \cdot 3^3 \cdot 7$	
α	$573x^2y - 4557x^2 - 6369xy - 1305x - 573y^3 - 3531y^2 + 5082y - 4542$
β	$552x^3 - 549x^2y - 3252x^2 - 573xy^3 + 1119xy^2 + 9588xy + 2859x + 573y^4$
	$+4104y^3 - 978y^2 + 9843y - 4164$
γ	$1146x^3y - 9114x^3 - 1146x^2y^2 - 3624x^2y - 2058x^2 - 2292xy^2 + 3111xy$
	$-11784x + 573y^3 - 3894y^2 + 1290y + 159$
$\{i, j\} = \{32, 48\}, k_{ij} = 2 \cdot 3^2$	
α	$-4x^2y + 28x^2 + 44xy - 6x + 4y^3 + 20y^2 - 36y + 32$
β	$-24x^3 + 24x^2y + 44x^2 + 4xy^3 - 16xy^2 - 10xy - 38x - 4y^4 - 24y^3 + 4y^2 - 32y + 50$
γ	$8x^3y - 56x^3 - 8x^2y^2 - 32x^2y + 36x^2 - 8xy^2 + 28xy - 28x + 4y^3 - 8y^2 + 6y$
	-18
$\{i, j\} = \{32, 55\}, k_{ij} = 3^3 \cdot 5^2$	
α	$-300y + 525$

β	$300y^2 + 75y + 150$
γ	$600x^3y - 1050x^3 - 600x^2y + 1050x^2 - 300xy^2 + 525xy - 1200x - 300y^3 + 225y^2 - 1275y + 1050$
$\{i, j\}$	$\{32, 58\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5 \cdot 7$
α	$-340x^2 - 1020xy + 620x - 40y + 1240$
β	$680x^4 + 680x^3y - 1204x^3 - 680x^2y^2 + 716x^2y - 1057x^2 + 1020xy^2 + 212xy + 1771x - 680y^3 + 756y^2 - 609y + 1913$
γ	$-680x^3y + 716x^3 + 680x^2y^2 + 112x^2y + 183x^2 - 828xy^2 - 93xy + 43x - 680y^4 + 716y^3 - 89y^2 + 3109y + 3107$
$\{i, j\}$	$\{32, 62\}, k_{ij} = 2 \cdot 3^2 \cdot 7$
α	$250x^3 + 560x^2 + 524x + 256$
β	$-250xy^2 - 654xy - 990x - 60y^2 + 588y - 914$
γ	$500x^5 + 620x^4 + 500x^3y - 72x^3 + 370x^2y - 36x^2 + 250xy^3 + 250xy^2 - 536xy - 1132x + 60y^3 - 844y^2 + 718y - 1296$
$\{i, j\}$	$\{32, 66\}, k_{ij} = 3^2 \cdot 7^3$
α	$108x - 54y + 99$
β	$-216x^3 + 216x^2y + 285x^2 - 331xy - 750x - 108y^3 + 104y^2 + 824y + 1845$
γ	$108x^3y - 267x^3 - 108x^2y^2 - 227x^2y + 948x^2 + 493xy^2 - 503xy - 2328x + 108y^4 - 266y^3 - 445y^2 - 572y + 1341$
$\{i, j\}$	$\{32, 77\}, k_{ij} = 3^3 \cdot 7^2$
α	$2x^3 - 1119x^2 - 3360xy + 40x - 3948y + 2190$
β	$-4x^5 + 4x^4y + 2241x^4 - x^3y^2 + 3354x^3y - 393x^3 - x^2y^3 - 3355x^2y^2 + 2865x^2y - 2987x^2 + xy^4 + 1790xy^2 + 1326xy + 2819x + 1117y^4 - 1924y^3 + 3802y^2 - 1669y + 3513$
γ	$-x^4 - x^3y^2 - 1118x^3y + 1926x^3 + x^2y^3 + 1118x^2y^2 - 142x^2y - 611x^2 - xy^4 - 1782xy^2 + 1583xy + 659x - 1117y^4 + 1924y^3 - 971y^2 + 4234y + 3146$
$\{i, j\}$	$\{32, 85\}, k_{ij} = 2 \cdot 3^3$
α	$84xy - 45x + 84y^2 - 111y + 84$
β	$-72x^3 + 72x^2y + 68x^2 - 72xy^2 - 67xy - 183x - 84y^3 + 27y^2 - 141y + 138$
γ	$168x^3y - 162x^3 - 60x^2y + 164x^2 - 12xy^2 - 19xy - 187x - 72y^3 - 13y^2 - 245y + 23$
$\{i, j\}$	$\{32, 105\}, k_{ij} = 2^3 \cdot 3^3$
α	$32x + 40y + 64$
β	$-64x^3 - 65x^2 + 32xy^2 + 31xy + 65x - 16y^3 - 22y^2 - 95y - 50$
γ	$-16x^3y + x^3 + 16x^2y^2 + 3x^2y + 62x^2 - 5xy^2 - 42xy - 74x - 16y^4 + 2y^3 - 21y^2 + 61y + 165$
$\{i, j\}$	$\{33, 34\}, k_{ij} = 2^2 \cdot 3 \cdot 13^2$
α	$754x^5y - 324x^5 - 1276x^4y + 1234x^4 + 1652x^3y + 1878x^3 - 1914x^2y + 334xy - 2178y - 2028$
β	$-1508x^5y + 648x^5 + 2552x^4y - 2468x^4 - 1508x^3y^2 - 2656x^3y - 3756x^3 + 2174x^2y^2 + 1332x^2y + 2508x^2 - 754xy^4 - 52xy^3 - 3446xy^2 - 6178xy - 3216x + 522y^4 - 648y^3 + 300y^2 + 1770y + 2928$
γ	$-376x^3y^2 + 352x^3y - 2508x^3 + 262x^2y^2 - 586x^2y + 2892x^2 + 754xy^4 - 702xy^3 + 3814xy^2 + 2364xy + 2362x - 522y^4 + 1170y^3 + 708y^2 - 450y + 1878$
$\{i, j\}$	$\{33, 37\}, k_{ij} = 2^2 \cdot 3$
α	$-10x^5 - 3x^4y + 10x^4 + 24x^3y + 35x^3 + 2x^2y + 24x^2 - 12xy - 24y - 12$
β	$32x^2y + 28x^2 + xy^3 + 20xy^2 + 11xy + 37x + 3y^4 + 42y^3 + 9y^2 + 34y + 24$
γ	$20x^6 + 6x^5y - 40x^5 - 54x^4y - 30x^4 - 20x^3y^2 + 28x^3y - 46x^3 + 3x^2y^3 + 43x^2y^2 - 5x^2y - x^2 - xy^3 - 8xy^2 - 11xy + 11x - 3y^4 - 42y^3 + 15y^2 - 22y + 24$
$\{i, j\}$	$\{33, 38\}, k_{ij} = 2^4 \cdot 7$
α	$389x^5 + 111x^4y - 1401x^4 - 973x^3y + 714x^3 + 691x^2y - 140x^2 - 112xy - 224y - 112$
β	$-28x^2y^2 + 1388x^2y - 668x^2 - 56xy^3 - 259xy^2 + 40xy + 1022x - 111y^4 - 518y^3 + 102y^2 + 1391y + 308$
γ	$-778x^6 - 222x^5y + 2024x^5 + 1724x^4y + 596x^4 + 806x^3y^2 - 1435x^3y + 1544x^3 - 111x^2y^3 - 1070x^2y^2 - 628x^2y + 911x^2 + 56xy^3 + 371xy^2 - 40xy - 574x + 111y^4 + 518y^3 + 122y^2 - 1279y + 140$
$\{i, j\}$	$\{33, 39\}, k_{ij} = 2 \cdot 3 \cdot 7$
α	$-17x^5 + 17x^4y - 34x^4 + 17x^3y + 111x^3 - 64x^2y + 105x^2 - 36xy + 21y - 42$
β	$-67x^3 - 67x^2y + 65x^2 - 17xy^3 - 58xy^2 - 75xy + 47x - 17y^4 + 47y^3 + 3y^2 + 171y$

$$\begin{aligned}
& +22 \\
\gamma \quad & 34x^6 + 34x^5 - 189x^4 - 34x^3y^2 - 11x^3y - 86x^3 + 30x^2y^2 + 39x^2y \\
& + 22x^2 + 17xy^3 + 77xy^2 - 2xy + 6x + 17y^4 - 64y^3 + 40y^2 - 169y + 105 \\
\{i, j\} = \{33, 41\}, k_{ij} = 2^4 \cdot 5^2 \cdot 7 \\
\alpha \quad & -6x^2y^3 + 6209x^2y^2 - 9408x^2y + 14xy^3 - 18623xy^2 + 24405xy - 4y^4 \\
& + 6184y^3 + 20008y^2 - 5600y - 2800 \\
\beta \quad & -x^5 + 4x^4 - x^3y^2 - 8x^3y + 16695x^3 + 2x^2y^3 + 8x^2y^2 - 16710x^2y \\
& + 25608x^2 - 4xy^4 + 6202xy^3 + 2252xy^2 - 59094xy - 3808x + 4y^5 - 6174y^4 \\
& - 32407y^3 + 1263y^2 - 65731y + 32805 \\
\gamma \quad & -x^6 + x^5y + 4x^5 - 2x^4y^2 - 12x^4y + 16695x^4 - 8x^3y^3 + 12439x^3y^2 \\
& - 52222x^3y + 25608x^3 + 4x^2y^4 - 6210x^2y^3 + 17224x^2y^2 - 35888x^2y \\
& - 3808x^2 + 18xy^4 - 24834xy^3 + 41647xy^2 - 56428xy + 27205x - 10y^5 + 12407y^4 \\
& - 8035y^3 + 34699y^2 - 1597y \\
\{i, j\} = \{33, 42\}, k_{ij} = 2^3 \cdot 3 \cdot 7 \\
\alpha \quad & 198x^2y^2 - 120x^2y - 198xy^2 + 330xy + 198y^3 - 270y^2 - 336y - 168 \\
\beta \quad & 477x^3 - 477x^2y + 780x^2 + 198xy^3 + 66xy^2 - 54xy + 714x - 198y^4 - 126y^3 \\
& + 132y^2 + 453y + 834 \\
\gamma \quad & 477x^4 + 396x^3y^2 - 1194x^3y + 780x^3 - 198x^2y^3 + 864x^2y^2 - 651x^2y \\
& + 714x^2 - 594xy^3 + 720xy^2 - 1236xy + 498x + 396y^4 - 192y^3 + 453y^2 - 432y \\
\{i, j\} = \{33, 90\}, k_{ij} = 2 \\
\alpha \quad & 0 \\
\beta \quad & 0 \\
\gamma \quad & 1 \\
\{i, j\} = \{34, 35\}, k_{ij} = 13^2 \\
\alpha \quad & 690x^3 - 45x^2 - 655x - 21y^2 - 686y - 754 \\
\beta \quad & -1380x^3y - 6286x^3 + 90x^2y - 2139x^2 + 2070xy^2 - 14701xy - 3931x - 6934y^3 \\
& - 5018y^2 - 5154y + 3077 \\
\gamma \quad & -1380x^5 + 7756x^4 + 2760x^3y - 2927x^3 - 6934x^2y^2 - 199x^2y + 610x^2 \\
& + 4885xy^2 + 7150xy - 70x + 6955y^3 + 5683y^2 - 1670y - 585 \\
\{i, j\} = \{35, 36\}, k_{ij} = 1 \\
\alpha \quad & 9x + 8 \\
\beta \quad & -18x^3 + 20x^2 - 22x + 33 \\
\gamma \quad & 18x^4 - 18x^3y - 20x^3 + 20x^2y + 31x^2 + 9xy^2 - 31xy + 2x + 8y^2 + 25y + 24 \\
\{i, j\} = \{35, 49\}, k_{ij} = 2 \cdot 3^2 \cdot 13 \\
\alpha \quad & -117y - 117 \\
\beta \quad & 117y^2 + 117 \\
\gamma \quad & 234x^3y + 234x^3 - 117xy^2 - 117x + 117y^3 + 234y - 117 \\
\{i, j\} = \{35, 64\}, k_{ij} = 2^2 \cdot 3 \cdot 7 \\
\alpha \quad & -360x^2 + 240xy + 192x + 72y + 300 \\
\beta \quad & 720x^4 + 240x^3y - 1104x^3 - 240x^2y^2 - 1329x^2y + 582x^2 - 240xy^3 \\
& - 638xy^2 + 789xy - 760x + 240y^3 + 86y^2 - 628y + 384 \\
\gamma \quad & 240x^3y + 399x^3 + 240x^2y^2 - 653x^2y - 641x^2 - 1051xy^2 - 639xy + 283x \\
& - 240y^4 - 398y^3 + 2y^2 + 14y - 284 \\
\{i, j\} = \{35, 65\}, k_{ij} = 2^2 \cdot 3 \cdot 7^3 \\
\alpha \quad & 42425x^3 - 34434x + 93335y^3 + 132366y^2 - 87534y - 122690 \\
\beta \quad & -84850x^5 - 84850x^4y + 84850x^4 + 254550x^3y + 68868x^3 + 197892x^2y \\
& + 91662x^2 - 93335xy^3 - 3342xy^2 - 2454xy + 132826x - 93335y^4 - 39031y^3 \\
& - 82062y^2 - 177068y - 224488 \\
\gamma \quad & 101820x^3y^2 + 10182x^3y + 250931x^3 + 68868xy^2 + 206604xy - 57228x - 57646y^3 \\
& - 95904y^2 - 347574y - 105914 \\
\{i, j\} = \{35, 84\}, k_{ij} = 1 \\
\alpha \quad & 9x + 8 \\
\beta \quad & -18x^3 + 20x^2 - 22x + 33 \\
\gamma \quad & -18x^4 + 18x^3y + 20x^3 - 20x^2y - 31x^2 - 9xy^2 + 31xy - 2x - 8y^2 - 25y - 24 \\
\{i, j\} = \{35, 100\}, k_{ij} = 2 \cdot 3 \\
\alpha \quad & 8x - 24y - 22 \\
\beta \quad & -16x^3 + 16x^2y + 28x^2 + 16xy^2 + 48xy + 12x + 40y^2 + 14y + 4 \\
\gamma \quad & 8x^3y + 16x^3 + 8x^2y^2 - 18x^2y - 28x^2 - 34xy^2 - 33xy + 6x - 8y^4 - 16y^3 \\
& - 5y^2 + y - 10 \\
\{i, j\} = \{36, 40\}, k_{ij} = 2^4 \cdot 3^3 \cdot 5
\end{aligned}$$

α	$156x + 384$
β	$312x^4 - 312x^3y - 168x^3 + 480x^2y + 348x^2 + 156xy^2 - 828xy + 600x + 384y^2$ $+ 540y + 228$
γ	$-312x^3 + 480x^2 - 672x + 924$
$\{i, j\} = \{39, 40\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7$	
α	$1663x^3 + 4063x^2 + 9861x + 3630y^2 - 8053y + 5721$
β	$-3326x^3y + 76195x^3 - 8126x^2y + 69727x^2 + 11641xy^2 + 326040xy - 206122x$ $+ 60620y^3 - 138095y^2 + 182008y - 427063$
γ	$-3326x^5 - 74343x^4 + 13304x^3y + 67363x^3 + 60620x^2y^2 - 206293x^2y$ $+ 145893x^2 - 75891xy^2 - 72894xy - 4920x - 64250y^3 + 85528y^2 + 135856y + 8207$
$\{i, j\} = \{40, 49\}, k_{ij} = 2^5 \cdot 3^2 \cdot 7$	
α	$504y - 504$
β	$-504y^2 + 504y + 504$
γ	$-1008x^3y + 1008x^3 + 504xy^2 - 504xy - 504x - 504y^3 + 1512y^2 - 504y - 1512$
$\{i, j\} = \{40, 53\}, k_{ij} = 2^2 \cdot 3^4 \cdot 5^2 \cdot 7$	
α	$-414x^3 - 10974x - 2070y^3 + 7314y^2 + 612y - 32142$
β	$828x^5 + 828x^4y - 2484x^4 - 4968x^3y + 26916x^3 + 2760x^2y^2 + 9949x^2y$ $- 9012x^2 + 4830xy^3 - 2063xy^2 - 71576xy - 25549x + 2070y^4 - 4554y^3 + 1189y^2$ $- 75070y + 103683$
γ	$-3312x^3y^2 + 19872x^3y - 28577x^3 + 21948xy^2 + 65844xy - 245376x + 2760y^4$ $+ 6631y^3 - 97545y^2 + 3615y - 286491$
$\{i, j\} = \{40, 61\}, k_{ij} = 2^3 \cdot 3^3 \cdot 5^2 \cdot 7$	
α	$-x^3 + x^2y - 1601x^2 + 1057xy + 73x - 3380y + 2312$
β	$2x^5 + 3196x^4 - 2x^3y^2 + 1082x^3y - 9740x^3 - 2x^2y^3 - 1043x^2y^2$ $- 5770x^2y + 11658x^2 - xy^4 - 1053xy^3 - 1495xy^2 + 12688xy + 293x - 2y^3$ $+ 1278y^2 + 1024y + 12862$
γ	$1065x^3y - 1675x^3 - x^2y^3 + 1067x^2y^2 - 10733x^2y + 4699x^2 - xy^4$ $+ 10xy^3 - 9061xy^2 + 9681xy + 20459x - y^5 - 1051y^4 + 1665y^3 + 4968y^2 + 5538y - 763$
$\{i, j\} = \{40, 76\}, k_{ij} = 2^5 \cdot 3 \cdot 5^2 \cdot 7 \cdot 13$	
α	$21x^2y - 81255xy - 9153x - 163322y - 11754$
β	$-x^3y^2 + 638x^3y + 18285x^3 - x^2y^3 + 1274x^2y^2 + 513x^2y + 49610x^2$ $- xy^4 + 2549xy^3 + 1066xy^2 + 84747xy - 113396x + 5101y^4 + 2214y^3 + 169428y^2$ $- 136465y + 101860$
γ	$21x^5 + 10x^4y - 80957x^4 + 5x^3y^2 - 40172x^3y - 90289x^3 + 2x^2y^3$ $- 19136x^2y^2 + 6369x^2y + 89498x^2 + xy^4 - 7648xy^3 + 26974xy^2 + 58861xy$ $+ 25455x - 5101y^4 - 12416y^3 - 41140y^2 + 71713y - 98934$
$\{i, j\} = \{40, 82\}, k_{ij} = 2^6 \cdot 3^5 \cdot 5^2 \cdot 7^2$	
α	$483x^2 - 496059xy + 471219x - 234474y + 2333190$
β	$483x^3y^2 + 989220x^3y - 2923797x^3 - 483x^2y^3 + 714x^2y^2$ $- 1460297x^2y + 4157344x^2 + 713xy^3 - 764xy^2 + 2592510xy - 4912758x - 1265y^3$ $+ 270y^2 - 4359624y + 8074061$
γ	$483x^5 - 966x^4y - 990921x^4 + 966x^3y^2 + 990668x^3y + 521627x^3$ $- 483x^2y^3 - 965x^2y^2 - 1462498x^2y - 866872x^2 + 713xy^3 - 494132xy^2$ $+ 2075356xy + 3337104x - 1265y^3 - 236734y^2 - 2502432y + 2837793$
$\{i, j\} = \{40, 83\}, k_{ij} = 2^3 \cdot 3^2 \cdot 5^2 \cdot 7^3$	
α	$-78x^3 + 33x^2y - 75236x^2 + 14665xy - 198026x - 144296y - 248646$
β	$156x^5 + 90x^4y + 150004x^4 - 62x^3y^2 + 120411x^3y - 54430x^3 - 144x^2y^3$ $+ 560x^2y^2 - 73396x^2y + 198455x^2 - 100xy^4 - 59459xy^3 - 86909xy^2 + 67002xy$ $- 148550x - 26y^5 - 29789y^4 + 6056y^3 - 13497y^2 + 58257y + 128521$
γ	$4x^4y - x^4 + 4x^3y^2 + 29960x^3y - 6057x^3 - 22x^2y^3 + 29966x^2y^2$ $- 120295x^2y + 144830x^2 - 48xy^4 + 93xy^3 - 114345xy^2 - 72466xy + 104321x - 26y^5$ $- 29789y^4 + 6004y^3 - 217371y^2 - 178225y + 16809$
$\{i, j\} = \{40, 103\}, k_{ij} = 2^7 \cdot 3^3 \cdot 5^2$	
α	$98x - 288y + 2788$
β	$-196x^3 + 188x^2y - 2892x^2 + 190xy^2 - 1510xy - 3590x + 668y^2 - 6940y + 8868$
γ	$96x^3y - 1048x^3 + 97x^2y^2 - 1165x^2y + 8523x^2 + xy^3 - 122xy^2 + 5939xy$ $- 11757x - 95y^4 + 1038y^3 - 2574y^2 + 3739y + 15646$
$\{i, j\} = \{43, 44\}, k_{ij} = 2 \cdot 3^2 \cdot 7$	
α	$106x^5 - 308x^4 + 452x^3 - 418x^2 + 252x - 126$
β	$106x^2y^2 + 14x^2y + 18x^2 - 96xy^2 - 84xy - 90x + 154y^2 + 162y + 212$

γ	$212x^6 - 404x^5 + 288x^4 + 106x^3y^2 - 92x^3y - 126x^3 + 106x^2y^3$ $- 96x^2y^2 + 118x^2y - 124x^2 - 96xy^3 + 168xy^2 + 36xy + 378x + 154y^3 + 36y^2$ $+ 212y - 252$
$\{i, j\} = \{43, 45\}, k_{ij} = 3^2 \cdot 7^3$	
α	$13x^7 - 111x^6 - 5x^5y + 4258x^5 + 2757x^4y + 2377x^4 - 21254x^3y + 15325x^3$ $+ 67980x^2y - 32464x^2 - 55964xy + 16530x + 10356y - 3087$
β	$7x^4 + x^3y^2 + 21x^3y - 69x^3 + 2x^2y^3 - 441x^2y^2 - 2021x^2y$ $- 76716x^2 + 5xy^4 - 851xy^3 - 28554xy^2 + 96064xy + 16386x + 13y^5 - 2602y^4$ $- 61046y^3 + 110803y^2 - 74850y + 16231$
γ	$26x^8 - 196x^7 - 36x^6y + 8346x^6 - 26x^5y^2 + 5713x^5y + 12911x^5$ $+ 13x^4y^3 + 207x^4y^2 - 45437x^4y + 51818x^4 - 34x^3y^3 - 14361x^3y^2$ $+ 93672x^3y - 76728x^3 + 13x^2y^4 - 2561x^2y^3 - 1259x^2y^2 + 6342x^2y$ $+ 96925x^2 + 5xy^4 - 846xy^3 - 85364xy^2 + 16874xy - 65225x + 13y^5 - 2589y^4$ $- 63635y^3 + 57524y^2 - 20413y + 16530$
$\{i, j\} = \{43, 46\}, k_{ij} = 2 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13$	
α	$5x^7 + 5x^6y - 17x^6 - 12x^5y + 3532x^5 + 3519x^4y - 3500x^4 - 3432x^3y$ $- 587x^3 - 12903x^2y + 24436x^2 + 39502xy - 8110x - 16380y - 8190$
β	$3x^4 - 2x^3y^2 - 29x^3y - 48176x^3 + 2x^2y^3 + 59x^2y^2 + 48189x^2y$ $+ 31583x^2 - 3xy^4 + 3556xy^3 + 10081xy^2 - 24601xy - 23659x + 5y^5 - 3488y^4$ $- 26702y^3 + 7839y^2 - 14852y + 40870$
γ	$10x^8 - 14x^7 - 10x^6y + 7026x^6 - 3x^5y + 7069x^5 + 5x^4y^3 + 20x^4y^2$ $- 13965x^4y - 42246x^4 - 15x^3y^3 + 7038x^3y^2 + 40227x^3y + 37090x^3$ $+ 14x^2y^3 - 19754x^2y^2 + 18917x^2y - 6744x^2 + 2xy^4 + 63xy^3 + 26369xy^2$ $+ 1830xy + 32654x + 5y^5 - 3488y^4 - 26702y^3 - 8541y^2 - 23042y + 8110$
$\{i, j\} = \{43, 47\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \cdot 13$	
α	$-9x^7 + 9x^6y + 31x^6 - 13x^5y - 2788x^5 + 2700x^4y + 16724x^4 - 8301x^3y$ $- 4073x^3 + 1635x^2y - 1492x^2 + 18915xy + 16841x + 7013y - 4914$
β	$3x^4 + x^3y^2 + 17x^3y - 22900x^3 + 2x^2y^3 + 28x^2y^2 - 22909x^2y$ $+ 24787x^2 + 4xy^4 - 2778xy^3 - 34635xy^2 + 14991xy - 8946x + 9y^5 - 2690y^4$ $- 23719y^3 + 13857y^2 - 14758y + 28017$
γ	$-18x^8 + 44x^7 + 18x^6y - 5514x^6 - 87x^5y + 27893x^5 - 9x^4y^3 - 33x^4y^2$ $+ 11172x^4y + 2367x^4 + 9x^3y^3 + 5639x^3y^2 - 3248x^3y + 19149x^3$ $+ 3x^2y^3 - 1567x^2y^2 + 40847x^2y - 3381x^2 - 13xy^4 + 5446xy^3$ $+ 47566xy^2 + 28277xy + 12668x - 9y^5 + 2681y^4 + 26400y^3 + 5530y^2 + 25202y - 16841$
$\{i, j\} = \{43, 48\}, k_{ij} = 2^4 \cdot 3^3 \cdot 7$	
α	$167x^2y^2 - 1514x^2y - 3192x^2 + 6215xy - 1177x + 168y^3 + 3862y^2 - 7225y - 3024$
β	$x^4 - 2436x^3 + x^2y^2 + 2436x^2y + 2351x^2 + 168xy^3 - 2184xy^2 - 2519xy$ $- 5292x - 168y^4 - 3863y^3 + 1681y^2 - 5626y + 3527$
γ	$-x^5 + x^4y + 2437x^4 - 336x^3y^2 - 1846x^3y + 1597x^3 + 168x^2y^3$ $+ 4031x^2y^2 - 6637x^2y + 9996x^2 - 337xy^3 - 7055xy^2 + 4619xy - 6719x + y^4$ $+ 5208y^3 + 1094y^2 + 14785y - 1177$
$\{i, j\} = \{43, 91\}, k_{ij} = 3^6$	
α	$-567x^3 + 1053x^2 - 729$
β	$-567xy^2 - 1539xy - 2106x - 81y^2 + 243y - 405$
γ	$-1134x^5 + 2106x^4 - 567x^3y^2 - 405x^3y - 2106x^3 + 486x^2y^2 - 891x^2y$ $+ 243x^2 - 567xy^3 - 1620xy^2 - 1863xy - 1134x - 81y^3 - 486y^2 - 405y - 1458$
$\{i, j\} = \{50, 51\}, k_{ij} = 3 \cdot 7$	
α	$-x - 5$
β	$2x^3 + 8x^2 - 10x + 5y^3 + 10y + 16$
γ	$-5xy^2 + 5xy - 14x - 4y^2 + 4y - 7$
$\{i, j\} = \{50, 52\}, k_{ij} = 3^2 \cdot 7$	
α	$-12x - 15$
β	$24x^3 + 6x^2 - 30x + 3y^3 - 15y^2 + 48$
γ	$3xy^2 - 12xy - 24x - 12y^2 - 15y - 30$
$\{i, j\} = \{50, 53\}, k_{ij} = 3$	
α	$-x - 2$
β	$2x^3 + 2x^2 - 4x + y^3 + 2y^2 + 2y + 2$
γ	$xy^2 + 2xy + x - y^2 + y - 1$
$\{i, j\} = \{50, 54\}, k_{ij} = 7$	
α	-1

β	$2x^2 - 2x + 4y + 6$
γ	$2xy + 3x - y^2 + 2$
$\{i, j\}$	$\{50, 59\}, k_{ij} = 3 \cdot 7$
α	$-x - 5$
β	$2x^3 + 8x^2 - 10x + 5y^3 + 10y + 16$
γ	$5xy^2 - 5xy + 14x + 4y^2 - 4y + 7$
$\{i, j\}$	$\{50, 60\}, k_{ij} = 3^2$
α	$-3x - 6$
β	$6x^3 + 6x^2 - 12x + 6y^3 - 3y^2 + 6y$
γ	$-6xy^2 - 3xy - 6x - 3y^2 - 6y - 3$
$\{i, j\}$	$\{50, 61\}, k_{ij} = 2 \cdot 3 \cdot 7$
α	$-8x + 2$
β	$16x^3 - 20x^2 + 4x - 2y^3 - 20y^2 + 4y + 72$
γ	$-2xy^2 - 16xy + 28x - 10y^2 + 4y + 14$
$\{i, j\}$	$\{50, 62\}, k_{ij} = 7$
α	$-2x - 3$
β	$4x^3 + 2x^2 + 2xy^2 - x + 3y^2 + 2y + 8$
γ	$2xy^2 + 5x + 3y^2 + 4$
$\{i, j\}$	$\{50, 92\}, k_{ij} = 3$
α	0
β	1
γ	$-x + 1$
$\{i, j\}$	$\{50, 93\}, k_{ij} = 3 \cdot 7$
α	0
β	16
γ	$4x - 1$
$\{i, j\}$	$\{66, 78\}, k_{ij} = 1$
α	-1
β	$2x^3 - 2x^2y - 2x^2 - 3xy - 2y^3 - 10y^2 - 19y - 6$
γ	$2y^2 + 7y + 2$
$\{i, j\}$	$\{74, 78\}, k_{ij} = 3^2$
α	$20x + 25y + 36$
β	$-10x^3y - 25x^3 + 10x^2y^2 + 53x^2y + 35x^2 - 28xy^2 - 72xy + 168x - 10y^4$ $-26y^3 + 38y^2 + 287y + 377$
γ	$-40x^3 + 73x^2 + 20xy^2 - 31xy - 134x - 10y^3 - 21y^2 + 95y + 15$
$\{i, j\}$	$\{74, 86\}, k_{ij} = 3$
α	0
β	0
γ	1
$\{i, j\}$	$\{78, 84\}, k_{ij} = 1$
α	$9x + 8$
β	$-18x^3 + 20x^2 - 22x + 33$
γ	$18x^4 + 18x^3y + 16x^3 - 20x^2y - 27x^2 - 9xy^2 + 31xy - 24x - 8y^2 - 25y - 90$
$\{i, j\}$	$\{78, 104\}, k_{ij} = 2 \cdot 3$
α	$52x - 26y + 20$
β	$-104x^3 + 104x^2y + 119x^2 - 98xy - 98x - 52y^3 - 75y^2 + 140y + 1106$
γ	$52x^3y + 153x^3 - 52x^2y^2 - 278x^2y - 253x^2 + 124xy^2 + 340xy - 663x + 52y^4$ $+153y^3 - 85y^2 - 1079y - 1646$
$\{i, j\}$	$\{86, 92\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$\{86, 105\}, k_{ij} = 3$
α	0
β	1
γ	0
$\{i, j\}$	$\{87, 88\}, k_{ij} = 2^2 \cdot 3$
α	0
β	0

γ	1
$\{i, j\} = \{88, 90\}, k_{ij} = 2^2 \cdot 3$	
α	0
β	1
γ	0
$\{i, j\} = \{88, 101\}, k_{ij} = 2^2 \cdot 3$	
α	0
β	1
γ	0
$\{i, j\} = \{90, 97\}, k_{ij} = 2$	
α	0
β	1
γ	0
$\{i, j\} = \{93, 94\}, k_{ij} = 3 \cdot 7$	
α	0
β	0
γ	1
$\{i, j\} = \{94, 95\}, k_{ij} = 3 \cdot 7$	
α	0
β	1
γ	0
$\{i, j\} = \{96, 97\}, k_{ij} = 7$	
α	0
β	0
γ	1
$\{i, j\} = \{97, 98\}, k_{ij} = 7$	
α	0
β	1
γ	0

Table 3: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle -y, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 0, 0 \rangle$
1	$\langle -2x - y, -x + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -2x, -x + 1 \rangle$
2	$\langle -x^2 - y, -x^2 \rangle$	$\langle -2, -2 \rangle$	$y = \langle -x^2, -x^2 \rangle$
3	$\langle x^2 - y, x^2 + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2, x^2 + 1 \rangle$
4	$\langle -y, -x^2 - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x^2 - 1 \rangle$
5	$\langle -y - 1, -x^2 - 1 \rangle$	$\langle -2, -1 \rangle$	$y = \langle -1, -x^2 - 1 \rangle$
6	$\langle x^2 - x - y + 1, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 - x + 1, 0 \rangle$
7	$\langle -x - y, -x^2 + 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle -x, -x^2 + 1 \rangle$
8	$\langle -2y + 1, -x^2 + x \rangle$	$\langle -1, -1 \rangle$	—
9	$\langle x, 0 \rangle$	$\langle 2, 0 \rangle$	$x = \langle 0, 0 \rangle$
10	$\langle x + 1, 1 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -1, -1 \rangle$
11	$\langle x + 1, 0 \rangle$	$\langle -1, -1 \rangle$	$x = \langle -1, 0 \rangle$
12	$\langle x, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 0, -1 \rangle$
13	$\langle x, -1 \rangle$	$\langle -24, -15 \rangle$	$x = \langle 0, 1 \rangle$
14	$\langle x - 1, 0 \rangle$	$\langle -2, -1 \rangle$	$x = \langle 1, 0 \rangle$
15	$\langle x - 1, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 1 \rangle$
16	$\langle x - y, -y \rangle$	$\langle -1, -1 \rangle$	$y = \langle 0, -x \rangle$
17	$\langle x - y, 0 \rangle$	$\langle -1, -2 \rangle$	$y = \langle x, 0 \rangle$
18	$\langle x, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, -x \rangle$
19	$\langle x, y \rangle$	$\langle 0, -1 \rangle$	$y = \langle x, x \rangle$
20	$\langle x + y, y \rangle$	$\langle -2, -1 \rangle$	$y = \langle 0, x \rangle$
21	$\langle x - y + 1, -y + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle 1, -x \rangle$
22	$\langle x + 1, -y + 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x, -x - 1 \rangle$

i	J_i	c_i	chart
23	$\langle x + y + 1, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x - 1, 0 \rangle$
24	$\langle x + y + 1, y \rangle$	$\langle -2, -2 \rangle$	$y = \langle 0, x + 1 \rangle$
25	$\langle x - y, -y - 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -1, -x - 1 \rangle$
26	$\langle x - y - 1, 0 \rangle$	$\langle 0, -3 \rangle$	$y = \langle x - 1, 0 \rangle$
27	$\langle x + y - 1, y - 1 \rangle$	$\langle 0, -1 \rangle$	$y = \langle 1, x \rangle$
28	$\langle x + 2, y + 1 \rangle$	$\langle 0, -12 \rangle$	$y = \langle x + 1, x + 2 \rangle$
29	$\langle x - y + 1, -y - 1 \rangle$	$\langle -4, -8 \rangle$	$y = \langle -1, -x - 2 \rangle$
30	$\langle x + 1, -y - 1 \rangle$	$\langle 0, -4 \rangle$	$y = \langle -x - 2, -x - 1 \rangle$
31	$\langle 2x - 1, 0 \rangle$	$\langle -4, 8 \rangle$	—
32	$\langle -y + 1, x + 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle 1, x + 1 \rangle$
33	$\langle 2x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	—
34	$\langle -x - y + 1, 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x + 1, 1 \rangle$
35	$\langle -y, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x^2 \rangle$
36	$\langle x - y + 1, 0 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x + 1, 0 \rangle$
37	$\langle -x^2 - y + 1, 0 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x^2 + 1, 0 \rangle$
38	$\langle x + y + 1, y - 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle 1, x + 2 \rangle$
39	$\langle -x^2 + x - y - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + x - 1, 0 \rangle$

Table 4: The data of the identities (70) and (71); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

$i = 0, D_i = 2, \rho_{ij}: 4:1, 5:2, 14:1, 21:1, 24:1, 34:1$
$w_i \quad \langle 2t^2 + 2, 2t^2 \rangle$
$i = 1, D_i = 4, \rho_{ij}: 3:1, 5:1, 11:2, 12:1, 13:2, 15:1, 22:1, 38:2, 39:2$
$w_i \quad \langle -32t^3 - 86t^2 - 6t + 4, -46t^3 - 20t^2 + 36t + 12 \rangle$
$i = 2, D_i = 4, \rho_{ij}: 6:2, 7:2, 11:1, 13:1, 14:1, 16:2, 24:2, 34:2, 37:2, 38:2, 39:1$
$w_i \quad \langle 40t^5 - 24t^4 + 24t^3 + 48t^2 - 8t + 8, 40t^5 + 24t^4 - 16t^3 + 8t^2 + 8t - 24 \rangle$
$i = 3, D_i = 16, \rho_{ij}: 1:2, 4:2, 5:1, 7:2, 10:2, 11:1, 12:2, 16:2, 17:2, 18:2, 28:2, 29:2, 31:1, 32:2, 37:2, 38:1, 39:2$
$w_i \quad \langle 2592t^5 + 10560t^4 - 6320t^3 + 6848t^2 - 10032t - 11872, 9600t^5 + 2448t^4 + 5408t^3 + 2608t^2 - 12960t - 5696 \rangle$
$i = 4, D_i = 4, \rho_{ij}: 0:2, 3:1, 7:1, 10:2, 11:2, 15:2, 29:2, 30:2$
$w_i \quad \langle -4t^5 + 8t^4 - 8t^3, 4t^5 + 16t^4 + 16t^2 \rangle$
$i = 5, D_i = 4, \rho_{ij}: 0:1, 1:2, 3:2, 6:1, 8:1, 11:1, 16:2, 17:2, 20:2, 21:1, 27:2, 29:2, 31:2$
$w_i \quad \langle -96t^5 + 80t^4 - 128t^3 - 64t^2 - 84t - 68, -36t^5 - 8t^4 - 4t^3 - 80t^2 + 24t + 32 \rangle$
$i = 6, D_i = 1, \rho_{ij}: 2:1, 5:2, 13:1, 15:2, 24:2, 28:1, 31:1, 37:2$
$w_i \quad \langle -7t^5 + 13t^4 - 14t^3 + 16t^2 - 8t + 4, -4t^5 + 25t^4 - 16t^3 + 23t^2 - 11t + 6 \rangle$
$i = 7, D_i = 1, \rho_{ij}: 2:1, 3:1, 4:2, 8:1, 11:2, 24:1, 28:2, 35:2, 39:2$
$w_i \quad \langle -t^5 + 37t^4 + 25t^3 - 22t^2 - 27t - 12, 16t^5 + 39t^4 - 23t^3 - 32t^2 - 14t - 10 \rangle$
$i = 8, D_i = 4, \rho_{ij}: 5:2, 7:2, 9:1, 10:1, 16:2, 20:1, 21:2, 27:1, 31:2, 32:1, 34:1$
$T \quad \langle -19x^5 - 178x^4 + 11x^3 - 330x^2 + 407x - 5, -187x^5 - 525x^4 - 237x^3 - 189x^2 + 208x - 39 \rangle$
$U \quad \langle -3584x^{18} - 512x^{17}y + 768x^{17} - 512x^{16}y^2 + 256x^{16}y + 20608x^{16} + 1024x^{15}y^2 + 20352x^{15}y - 32192x^{15} + 2816x^{14}y^3 + 17024x^{14}y^2 - 66112x^{14}y + 51264x^{14} + 256x^{13}y^4 + 3072x^{13}y^3 - 27520x^{13}y^2 + 81856x^{13}y + 80512x^{13} + 256x^{12}y^5 - 512x^{12}y^4 - 27712x^{12}y^3 + 43584x^{12}y^2 - 13248x^{12}y + 571392x^{12} - 512x^{11}y^5 - 8384x^{11}y^4 + 30144x^{11}y^3 - 60608x^{11}y^2 - 9152x^{11}y + 2930880x^{11} - 1024x^{10}y^6 - 5824x^{10}y^5 + 23744x^{10}y^4 - 16896x^{10}y^3 + 11136x^{10}y^2 - 23424x^{10}y + 14451840x^{10} - 512x^9y^6 + 7360x^9y^5 - 24960x^9y^4 + 33792x^9y^3 + 24000x^9y^2 - 59904x^9y + 35873816x^9 + 6912x^8y^6 - 10240x^8y^5 + 7872x^8y^4 - 19712x^8y^3 + 7808x^8y^2 + 62208x^8y + 43769920x^8 + 1024x^7y^7 - 5248x^7y^6 + 14656x^7y^5 - 9856x^7y^4 - 12416x^7y^3 + 5120x^7y^2 + 19840x^7y + 23959208x^7 + 1024x^6y^8 - 3072x^6y^7 + 1408x^6y^6 - 6464x^6y^5 + 14720x^6y^4 + 3712x^6y^3 - 32064x^6y^2 - 3904x^6y - 17358456x^6 - 512x^5y^8 + 3072x^5y^7 - 5696x^5y^6 + 1472x^5y^5 + 2496x^5y^4 + 1216x^5y^3 - 1472x^5y^2 + 768x^5y + 73532416x^5 - 256x^4y^9 + 256x^4y^8 - 1280x^4y^7 + 5056x^4y^6 - 1984x^4y^5 - 8704x^4y^4 + 4928x^4y^3 + 19200x^4y^2 - 20544x^4y + 20415288x^4 + 256x^3y^9 - 896x^3y^8 + 960x^3y^7 - 256x^3y^6 + 320x^3y^4 - 3904x^3y^2 + 3904x^3y + 58107192x^3 - 64x^2y^9 + 640x^2y^8 - 768x^2y^7 - 832x^2y^6 + 896x^2y^5 + 2240x^2y^4 - 2816x^2y^3 \rangle$

$$\begin{aligned}
& -2624x^2y^2 + 6400x^2y - 37866520x^2 - 128xy^8 + 192xy^7 + 192xy^6 - 256xy^5 - 576xy^4 \\
& + 832xy^3 + 768xy^2 - 2048xy + 6045296x + 293032, \\
& -4096x^{18} - 512x^{17}y + 9216x^{17} + 15104x^{16}y - 12032x^{16} + 3328x^{15}y^2 - 5504x^{15}y \\
& + 11008x^{15} + 3072x^{14}y^3 - 9472x^{14}y^2 - 31680x^{14}y + 832x^{14} + 256x^{13}y^4 \\
& - 5632x^{13}y^3 + 1088x^{13}y^2 + 25088x^{13}y - 57920x^{13} - 7168x^{12}y^4 - 3328x^{12}y^3 \\
& + 33024x^{12}y^2 + 1152x^{12}y - 192512x^{12} - 1280x^{11}y^5 + 2624x^{11}y^4 + 896x^{11}y^3 \\
& - 24768x^{11}y^2 + 35776x^{11}y - 1315968x^{11} - 1024x^{10}y^6 + 4096x^{10}y^5 \\
& + 10304x^{10}y^4 - 8064x^{10}y^3 - 19456x^{10}y^2 - 12992x^{10}y - 8326400x^{10} + 1536x^9y^6 \\
& - 1600x^9y^5 - 1344x^9y^4 + 16896x^9y^3 + 832x^9y^2 - 34048x^9y - 37458048x^9 \\
& + 1536x^8y^7 + 768x^8y^6 - 5696x^8y^5 - 8064x^8y^4 + 9152x^8y^3 + 32320x^8y^2 \\
& - 37824x^8y - 66800312x^8 - 512x^7y^7 + 1920x^7y^6 + 128x^7y^5 - 4032x^7y^4 \\
& - 4736x^7y^3 - 6464x^7y^2 + 39424x^7y - 70050632x^7 - 1664x^6y^7 - 2432x^6y^6 \\
& + 8768x^6y^5 + 5504x^6y^4 - 21952x^6y^3 - 2176x^6y^2 + 37504x^6y - 33449848x^6 \\
& + 256x^5y^8 - 768x^5y^7 + 320x^5y^6 - 1280x^5y^5 + 192x^5y^4 + 9984x^5y^3 \\
& - 7040x^5y^2 - 25536x^5y + 55462720x^5 - 256x^4y^9 + 2560x^4y^7 - 1152x^4y^6 \\
& - 5184x^4y^5 + 2752x^4y^4 + 8192x^4y^3 - 6016x^4y^2 - 13312x^4y + 52050736x^4 \\
& + 64x^3y^8 - 512x^3y^7 + 768x^3y^6 + 960x^3y^5 - 2048x^3y^4 - 2880x^3y^3 \\
& + 5632x^3y^2 + 6528x^3y - 4547816x^3 + 192x^2y^9 - 192x^2y^8 - 576x^2y^7 \\
& + 192x^2y^6 + 1408x^2y^5 - 448x^2y^4 - 2240x^2y^3 + 1536x^2y^2 + 3072x^2y \\
& - 1636192x^2 - 64xy^9 + 64xy^8 + 192xy^7 - 128xy^6 - 448xy^5 + 320xy^4 + 832xy^3 - 1024xy^2 \\
& - 1280xy + 8533264x - 159120) \\
V \quad & \langle -1024x^{20} + 15872x^{19} + 1024x^{18}y^2 + 13312x^{18}y - 40960x^{18} + 3584x^{17}y^2 \\
& - 17408x^{17}y + 34688x^{17} - 19968x^{16}y^2 + 29696x^{16}y - 59712x^{16} + 25216x^{15}y^2 \\
& - 45312x^{15}y - 368128x^{15} + 6592x^{14}y^2 - 60992x^{14}y - 1856000x^{14} + 76352x^{13}y^2 \\
& - 162688x^{13}y - 3926347x^{13} + 288896x^{12}y^2 - 1640320x^{12}y + 2026143x^{12} \\
& + 1390080x^{11}y^2 - 6540816x^{11}y + 9984901x^{11} + 7253632x^{10}y^2 - 3792206x^{10}y \\
& + 34527299x^{10} + 17950540x^9y^2 + 8103058x^9y + 51465225x^9 + 21906272x^8y^2 \\
& + 20193188x^8y + 76829287x^8 + 11978196x^7y^2 + 28219736x^7y + 22474341x^7 \\
& - 8717308x^6y^2 - 5199186x^6y + 40538859x^6 + 36775584x^5y^2 + 4238298x^5y \\
& + 62214568x^5 + 10221788x^4y^2 + 5807920x^4y + 56595781x^4 + 29048444x^3y^2 \\
& + 17073882x^3y + 49401609x^3 - 18934156x^2y^2 - 11632814x^2y - 41805810x^2 + 3023032xy^2 \\
& + 1548864xy + 6987158x + 146516y^2 + 73258y + 329661, \\
& 7168x^{20} + 1024x^{19}y + 5632x^{19} + 1024x^{18}y^2 - 1536x^{18}y - 34560x^{18} - 1024x^{17}y^2 \\
& - 7168x^{17}y + 13440x^{17} - 9984x^{16}y^2 + 36224x^{16}y - 52480x^{16} + 8960x^{15}y^2 \\
& - 62144x^{15}y - 247040x^{15} - 20736x^{14}y^2 - 189760x^{14}y - 1307776x^{14} + 17344x^{13}y^2 \\
& - 830208x^{13}y - 2277123x^{13} - 71424x^{12}y^2 - 4684352x^{12}y - 3868130x^{12} - 694208x^{11}y^2 \\
& - 12912470x^{11}y - 14787438x^{11} - 4208320x^{10}y^2 - 11432664x^{10}y - 34746167x^{10} \\
& - 18706528x^9y^2 - 5282366x^9y - 87894829x^9 - 33369980x^8y^2 + 2775294x^8y \\
& - 72688135x^8 - 35028708x^7y^2 - 17975218x^7y - 107404518x^7 - 16723388x^6y^2 \\
& + 4093712x^6y + 19249681x^6 + 27717600x^5y^2 - 9725910x^5y + 47361655x^5 \\
& + 26017336x^4y^2 + 37718146x^4y + 87208474x^4 - 2265780x^3y^2 - 9565760x^3y \\
& - 15050175x^3 - 816048x^2y^2 - 1140494x^2y + 1654129x^2 + 4265224xy^2 + 2244882xy + 9633304x \\
& - 79560y^2 - 39780y - 179010) \\
i = 9, D_i = 4, \rho_{ij}: & 8:2, 22:2, 23:2, 29:1, 34:2, 35:2, 38:1, 39:1 \\
w_i \quad & \langle -16t^2 - 8t + 8, \\
& 0) \\
i = 10, D_i = 2, \rho_{ij}: & 3:1, 4:1, 8:2, 18:1, 38:2, 39:1 \\
w_i \quad & \langle -4t^2 - 4t + 10, \\
& -4t^2 + 2t + 8) \\
i = 11, D_i = 4, \rho_{ij}: & 1:1, 2:2, 3:2, 4:1, 5:2, 7:1, 21:1, 36:2 \\
w_i \quad & \langle 8t^2 - 12t + 4, \\
& 8t^2 - 12t + 4) \\
i = 12, D_i = 2, \rho_{ij}: & 1:2, 3:1, 26:2, 34:1 \\
w_i \quad & \langle 2t + 2, \\
& -4) \\
i = 13, D_i = 4, \rho_{ij}: & 1:1, 2:2, 6:2, 16:1, 19:2, 25:1, 38:1 \\
w_i \quad & \langle -4t^2 + 20t + 32, \\
& -20t^2 + 16t - 8) \\
i = 14, D_i = 2, \rho_{ij}: & 0:2, 2:2, 24:1, 26:2, 32:2, 35:2 \\
w_i \quad & \langle 4t^2 + 4t + 2, \\
& 4t^2) \\
i = 15, D_i = 4, \rho_{ij}: & 1:2, 4:1, 6:1, 19:1, 26:1, 27:1, 29:1, 35:2, 38:1, 39:1 \\
w_i \quad & \langle -12t^2 + 4t + 8, \\
& -8t^2 + 12t - 4) \\
i = 16, D_i = 4, \rho_{ij}: & 2:1, 3:1, 5:1, 8:1, 13:2, 17:2, 24:1, 34:2, 37:1 \\
w_i \quad & \langle 2t^3 - 9t^2 + 6t + 4, \\
& -9t^3 - t^2 + 16t - 3) \\
i = 17, D_i = 2, \rho_{ij}: & 3:1, 5:1, 16:1, 30:1, 31:2, 37:1, 39:1 \\
w_i \quad & \langle 6t^2, \\
& 0)
\end{aligned}$$

$$\begin{aligned}
i &= 18, D_i = 2, \rho_{ij}: 3:1, 10:2, 33:2, 37:2 \\
w_i &\langle 2, \\
&\quad -2t^2 - 2t \rangle \\
i &= 19, D_i = 4, \rho_{ij}: 13:1, 15:2, 21:2, 27:2, 31:2, 35:2, 37:1, 39:2 \\
w_i &\langle -3t^3 + 8t^2 + 2t + 5, \\
&\quad -t^3 + 23t^2 - 16t + 9 \rangle \\
i &= 20, D_i = 4, \rho_{ij}: 5:1, 8:2, 21:2, 30:1, 33:2, 34:2, 35:2, 37:2 \\
w_i &\langle 5t^3 - t^2 - 20t - 3, \\
&\quad -11t^3 - 2t^2 - 10t - 9 \rangle \\
i &= 21, D_i = 8, \rho_{ij}: 0:2, 5:2, 8:1, 11:2, 19:1, 20:1, 27:2, 35:2, 39:2 \\
w_i &\langle 39t^3 + 227t^2 + 196t - 10, \\
&\quad -63t^3 + 39t^2 + 251t + 66 \rangle \\
i &= 22, D_i = 1, \rho_{ij}: 1:2, 9:1, 23:2, 25:1, 26:2, 31:1, 32:1 \\
w_i &\langle -4t^3 + 2t^2 - 2t + 4, \\
&\quad -2t^3 + 4t^2 + 2t + 2 \rangle \\
i &= 23, D_i = 1, \rho_{ij}: 9:1, 22:1, 31:2, 32:1, 37:1, 38:2 \\
w_i &\langle 2t^3 - t^2 + 4, \\
&\quad 2t^3 - t^2 + 4 \rangle \\
i &= 24, D_i = 2, \rho_{ij}: 0:2, 2:1, 6:1, 7:2, 14:2, 16:2 \\
w_i &\langle 2t^3 - 4t^2 - 4t, \\
&\quad 2t^3 - 6t^2 - 10t - 4 \rangle \\
i &= 25, D_i = 1, \rho_{ij}: 13:2, 22:2, 31:1 \\
w_i &\langle -2t^3 + t^2, \\
&\quad -2t^3 + t^2 \rangle \\
i &= 26, D_i = 2, \rho_{ij}: 12:1, 14:1, 15:2, 22:1, 28:2, 29:2, 31:2, 33:1, 37:2 \\
w_i &\langle 20t^3 + 4t^2 - 16t, \\
&\quad 10t^3 + 2t^2 - 8t \rangle \\
i &= 27, D_i = 2, \rho_{ij}: 5:1, 8:2, 15:2, 19:1, 21:1, 28:1, 39:1 \\
w_i &\langle -2t^3 - 2t^2 + 3t + 4, \\
&\quad -3t^3 + t^2 + 10t - 2 \rangle \\
i &= 28, D_i = 2, \rho_{ij}: 3:1, 6:2, 7:1, 26:1, 27:2, 29:1, 31:1, 34:2, 38:1 \\
w_i &\langle -67t^3 + 43t^2 + 124t + 50, \\
&\quad -38t^3 - 70t^2 + 77t + 70 \rangle \\
i &= 29, D_i = 4, \rho_{ij}: 3:1, 4:1, 5:1, 9:2, 15:2, 26:1, 28:2, 31:2, 35:1, 39:2 \\
w_i &\langle 125t^3 + 227t^2 + 46t - 26, \\
&\quad 61t^3 + 19t^2 - 277t - 112 \rangle \\
i &= 30, D_i = 4, \rho_{ij}: 4:1, 17:2, 20:2, 38:2 \\
w_i &\langle 18t^3 + 9t^2 + 6t + 24, \\
&\quad 30t + 36 \rangle \\
i &= 31, D_i = 256, \rho_{ij}: 3:2, 5:1, 6:2, 8:1, 17:1, 19:1, 22:2, 23:1, 25:2, 26:1, 28:2, 29:1, 32:1, 34:2, 37:2, 38:1 \\
T &\langle -6020y^2 + 1414y + 574, \\
&\quad -2072y^2 + 3346y - 707 \rangle \\
U &\langle -67108864x^{27} - 503316480x^{26} + 33554432x^{25}y - 1140850688x^{25} - 67108864x^{24}y^3 \\
&\quad -167772160x^{24}y^2 + 1476395008x^{24}y - 671088640x^{24} - 33554432x^{23}y^3 \\
&\quad + 67108864x^{23}y^2 + 7163871232x^{23}y + 301989888x^{23} + 704643072x^{22}y^4 \\
&\quad + 1862270976x^{22}y^3 + 2701131776x^{22}y^2 + 8858370048x^{22}y - 2164260864x^{22} \\
&\quad + 134217728x^{21}y^6 + 67108864x^{21}y^5 + 3707764736x^{21}y^4 + 3607101440x^{21}y^3 \\
&\quad - 10032775168x^{21}y^2 - 184549376x^{21}y - 4731174912x^{21} + 771751936x^{20}y^6 \\
&\quad + 486539264x^{20}y^5 - 318767104x^{20}y^4 - 7482638336x^{20}y^3 - 38663094272x^{20}y^2 \\
&\quad + 10015997952x^{20}y - 167772160x^{20} + 536870912x^{19}y^7 + 2197815296x^{19}y^6 \\
&\quad - 8740929536x^{19}y^5 - 17431527424x^{19}y^4 - 22322085888x^{19}y^3 - 2734686208x^{19}y^2 \\
&\quad + 39292239872x^{19}y + 3774873600x^{19} + 134217728x^{18}y^9 + 402653184x^{18}y^8 \\
&\quad - 2600468480x^{18}y^7 - 3447717888x^{18}y^6 - 38503710720x^{18}y^5 + 7021264896x^{18}y^4 \\
&\quad + 65968013312x^{18}y^3 + 36540776448x^{18}y^2 + 10603200512x^{18}y - 3196059648x^{18} \\
&\quad - 201326592x^{17}y^9 - 2684354560x^{17}y^8 - 19117637632x^{17}y^7 + 5972688896x^{17}y^6 \\
&\quad + 34653339648x^{17}y^5 + 105360916480x^{17}y^4 + 22930259968x^{17}y^3 \\
&\quad - 86478159872x^{17}y^2 - 30006050816x^{17}y - 5184159744x^{17} - 805306368x^{16}y^{10} \\
&\quad - 2902458368x^{16}y^9 - 2290089984x^{16}y^8 + 1677721600x^{16}y^7 + 114009571328x^{16}y^6 \\
&\quad - 18568183808x^{16}y^5 + 11496587264x^{16}y^4 - 249577865216x^{16}y^3 \\
&\quad - 44027609088x^{16}y^2 + 5200936960x^{16}y + 6257901568x^{16} - 67108864x^{15}y^{12} \\
&\quad - 67108864x^{15}y^{11} - 2181038080x^{15}y^{10} - 2843738112x^{15}y^9 + 49895440384x^{15}y^8 \\
&\quad - 26319257600x^{15}y^7 + 83739279360x^{15}y^6 - 224567230464x^{15}y^5 \\
&\quad - 55197040640x^{15}y^4 - 93310681088x^{15}y^3 + 165062639616x^{15}y^2 + 15355346944x^{15}y \\
&\quad + 14772338688x^{15} - 301989888x^{14}y^{12} - 1711276032x^{14}y^{11} + 5595201536x^{14}y^{10} \\
&\quad + 4810866688x^{14}y^9 + 7187787648x^{14}y^8 - 82598428672x^{14}y^7 - 50004492288x^{14}y^6 \\
&\quad - 188257140736x^{14}y^5 + 390168838144x^{14}y^4 + 77754007552x^{14}y^3 \\
&\quad + 114072485888x^{14}y^2 - 37731958784x^{14}y + 918552576x^{14} - 570425344x^{13}y^{13} \\
&\quad - 1342177280x^{13}y^{12} + 9932111872x^{13}y^{11} + 32350666752x^{13}y^{10} + 6345981952x^{13}y^9 \\
&\quad - 21294481408x^{13}y^8 - 290371665920x^{13}y^7 + 222340055040x^{13}y^6 \rangle
\end{aligned}$$

$$\begin{aligned}
& +93136617472x^{13}y^5 + 555149688832x^{13}y^4 - 397510967296x^{13}y^3 \\
& + 18157142016x^{13}y^2 - 89753911296x^{13}y - 6618611712x^{13} - 67108864x^{12}y^{15} \\
& - 234881024x^{12}y^{14} + 2785017856x^{12}y^{13} + 8900313088x^{12}y^{12} + 19012780032x^{12}y^{11} \\
& - 16747855872x^{12}y^{10} - 245664579584x^{12}y^9 - 43226497024x^{12}y^8 \\
& + 122310098944x^{12}y^7 + 308048560128x^{12}y^6 - 34317795328x^{12}y^5 \\
& - 136323268608x^{12}y^4 - 455023263744x^{12}y^3 + 74998349824x^{12}y^2 - 30270291968x^{12}y \\
& + 13497270272x^{12} + 234881024x^{11}y^{15} + 1409286144x^{11}y^{14} + 6375342080x^{11}y^{13} \\
& - 11681136640x^{11}y^{12} - 101397299200x^{11}y^{11} - 104496889856x^{11}y^{10} \\
& + 48913973248x^{11}y^9 + 79121350656x^{11}y^8 + 454756925440x^{11}y^7 \\
& - 395602558976x^{11}y^6 - 259137732608x^{11}y^5 - 18356371456x^{11}y^4 \\
& - 72934752256x^{11}y^3 + 196570251264x^{11}y^2 + 33013366784x^{11}y + 13828620288x^{11} \\
& + 100663296x^{10}y^{16} + 654311424x^{10}y^{15} - 2214592512x^{10}y^{14} - 19004391424x^{10}y^{13} \\
& - 57787023360x^{10}y^{12} - 14608760832x^{10}y^{11} + 85572190208x^{10}y^{10} \\
& + 461541212160x^{10}y^9 - 170999676928x^{10}y^8 + 247911677952x^{10}y^7 \\
& - 768298975232x^{10}y^6 + 284763881472x^{10}y^5 + 385015087104x^{10}y^4 \\
& - 142606336x^{10}y^3 + 161925300224x^{10}y^2 - 40735080448x^{10}y - 7090470912x^{10} \\
& + 83886080x^9y^{16} - 1224736768x^9y^{15} - 17163091968x^9y^{14} - 13461618688x^9y^{13} \\
& + 43759173632x^9y^{12} + 220142239744x^9y^{11} + 230468616192x^9y^{10} \\
& + 372359823360x^9y^9 - 1249498890240x^9y^8 + 635964489728x^9y^7 \\
& - 873851781120x^9y^6 + 1112728928256x^9y^5 - 102242451456x^9y^4 - 33267122176x^9y^3 \\
& - 101216944128x^9y^2 - 34236006400x^9y - 16328425472x^9 + 33554432x^8y^{18} \\
& + 16777216x^8y^{17} - 2483027968x^8y^{16} - 3510632448x^8y^{15} + 3059744768x^8y^{14} \\
& + 61400416256x^8y^{13} + 262584401920x^8y^{12} + 206919696384x^8y^{11} \\
& - 775363231744x^8y^{10} + 210194399232x^8y^9 - 1457613963264x^8y^8 \\
& + 1665048510464x^8y^7 - 826785923072x^8y^6 + 936717058048x^8y^5 \\
& - 513307312128x^8y^4 - 113774690304x^8y^3 - 66613936128x^8y^2 + 35823550464x^8y \\
& + 3225419776x^8 - 117440512x^7y^{18} - 587202560x^7y^{17} - 2172649472x^7y^{16} \\
& + 10823401472x^7y^{15} + 103620280320x^7y^{14} + 97453604864x^7y^{13} \\
& - 194326298624x^7y^{12} - 255175163904x^7y^{11} - 1009142202368x^7y^{10} \\
& + 1127123779584x^7y^9 - 820859895808x^7y^8 + 1618275205120x^7y^7 \\
& - 1097113534464x^7y^6 + 508964110336x^7y^5 - 560014032896x^7y^4 + 85932900352x^7y^3 \\
& + 33462157312x^7y^2 + 54910779392x^7y + 8659140608x^7 - 50331648x^6y^{19} \\
& - 360710144x^6y^{18} + 1031798784x^6y^{17} + 17597202432x^6y^{16} + 45351960576x^6y^{15} \\
& + 3109027840x^6y^{14} - 253308698624x^6y^{13} - 482297774080x^6y^{12} \\
& + 184680448000x^6y^{11} - 265101508608x^6y^{10} + 1498279837696x^6y^9 \\
& - 867259908096x^6y^8 + 961082818560x^6y^7 - 1445793366016x^6y^6 \\
& + 524275417088x^6y^5 - 418653929472x^6y^4 + 301521174528x^6y^3 - 14964228096x^6y^2 \\
& + 22636658688x^6y - 15255732224x^6 - 58720256x^5y^{19} + 1040187392x^5y^{18} \\
& + 12423528448x^5y^{17} + 12370051072x^5y^{16} - 90073726976x^5y^{15} \\
& - 175351267328x^5y^{14} - 277554921472x^5y^{13} - 23341301760x^5y^{12} \\
& + 1020077277184x^5y^{11} + 42725277696x^5y^{10} + 721323294720x^5y^9 \\
& - 947621199872x^5y^8 - 62980882432x^5y^7 - 841470181376x^5y^6 + 759073079296x^5y^5 \\
& - 109003669504x^5y^4 + 168366702592x^5y^3 - 70388809728x^5y^2 - 6730809344x^5y \\
& - 13935575040x^5 - 16777216x^4y^{21} - 8388608x^4y^{20} + 1535115264x^4y^{19} \\
& + 2686451712x^4y^{18} - 10702815232x^4y^{17} - 48069869568x^4y^{16} - 210197544960x^4y^{15} \\
& - 47177161216x^4y^{14} + 464085057536x^4y^{13} + 516986241024x^4y^{12} \\
& + 702042603520x^4y^{11} - 424990474240x^4y^{10} - 631217324032x^4y^9 \\
& - 400398483456x^4y^8 - 140343508992x^4y^7 + 185079693312x^4y^6 + 729017221120x^4y^5 \\
& - 267141513216x^4y^4 + 68121788416x^4y^3 - 114671747072x^4y^2 + 22000173056x^4y \\
& + 5337251840x^4 + 58720256x^3y^{21} + 318767104x^3y^{20} + 780140544x^3y^{19} \\
& - 8586788864x^3y^{18} - 66447212544x^3y^{17} - 54019489792x^3y^{16} + 105445326848x^3y^{15} \\
& + 433010507776x^3y^{14} + 664570167296x^3y^{13} - 207744925696x^3y^{12} \\
& - 465887821824x^3y^{11} - 249563971584x^3y^{10} - 1088189366272x^3y^9 \\
& + 930949890048x^3y^8 + 96655900672x^3y^7 + 363180064768x^3y^6 + 402239782912x^3y^5 \\
& - 538787184640x^3y^4 + 46345486336x^3y^3 - 27922530304x^3y^2 + 17934843904x^3y \\
& + 8691646464x^3 + 25165824x^2y^{22} + 180355072x^2y^{21} - 671088640x^2y^{20} \\
& - 9922674688x^2y^{19} - 26845118464x^2y^{18} - 1030750208x^2y^{17} + 193623228416x^2y^{16} \\
& + 390602948608x^2y^{15} + 110409285632x^2y^{14} - 105289351168x^2y^{13} \\
& - 624822059008x^2y^{12} - 963444473856x^2y^{11} + 791134863360x^2y^{10} \\
& - 1003239768064x^2y^9 + 1971809288192x^2y^8 - 176141893632x^2y^7 \\
& - 162527969280x^2y^6 + 58768490496x^2y^5 - 480252264448x^2y^4 + 170899013632x^2y^3 \\
& + 34147926016x^2y^2 + 2235564032x^2y - 6285688832x^2 + 37748736xy^{22} - 515899392xy^{21} \\
& - 6752829440xy^{20} - 6349651968xy^{19} + 47007662080xy^{18} + 140046237696xy^{17} \\
& + 237726859264xy^{16} + 58414333952xy^{15} - 588306448384xy^{14} - 662418751488xy^{13} \\
& - 345922338816xy^{12} - 715060674560xy^{11} + 2115301801984xy^{10} - 936801075200xy^9 \\
& + 1659083161600xy^8 - 911920332800xy^7 - 689982668800xy^6 + 308305723392xy^5 \\
& - 229673271296xy^4 + 244108754944xy^3 - 30101995520xy^2 + 1056440320xy - 7169114112x \\
& - 25165824y^{22} + 206569472y^{21} + 3236429824y^{20} + 4776787968y^{19} - 15162408960y^{18} \\
& - 69865308160y^{17} - 142692581376y^{16} - 149743992832y^{15} + 250909425664y^{14} \\
& + 250014990336y^{13} + 308812611584y^{12} + 672729284608y^{11} - 1371107852288y^{10} \\
& + 1068213407744y^9 - 1585608640512y^8 + 641113147392y^7 + 238981341696y^6 - 168701920000y^5 \\
& + 394640181760y^4 - 300761968512y^3 + 36884662976y^2 + 5304988864y + 2336042152,
\end{aligned}$$

$$\begin{aligned}
& -100663296x^{27} - 402653184x^{26} + 167772160x^{25}y - 301989888x^{25} - 100663296x^{24}y^3 \\
& -134217728x^{24}y^2 + 1862270976x^{24}y + 637534208x^{24} - 50331648x^{23}y^3 \\
& + 1509949440x^{23}y^2 + 3758096384x^{23}y - 16777216x^{23} + 1409286144x^{22}y^4 \\
& + 33554432x^{22}y^3 + 2231369728x^{22}y^2 - 2332033024x^{22}y - 2952790016x^{22} \\
& + 201326592x^{21}y^6 - 134217728x^{21}y^5 + 3154116608x^{21}y^4 - 4747952128x^{21}y^3 \\
& - 8883535872x^{21}y^2 - 5268045824x^{21}y - 1694498816x^{21} + 452984832x^{20}y^6 \\
& - 973078528x^{20}y^5 + 2046820352x^{20}y^4 - 23672651776x^{20}y^3 - 419430400x^{20}y^2 \\
& + 15200157696x^{20}y + 3036676096x^{20} + 805306368x^{19}y^7 + 1769996288x^{19}y^6 \\
& - 17397972992x^{19}y^5 + 3623878656x^{19}y^4 - 16131293184x^{19}y^3 + 56849596416x^{19}y^2 \\
& + 15485370368x^{19}y - 209715200x^{19} + 201326592x^{18}y^9 + 134217728x^{18}y^8 \\
& - 4194304000x^{18}y^7 - 9076473856x^{18}y^6 - 19704840192x^{18}y^5 + 66991423488x^{18}y^4 \\
& + 12192841728x^{18}y^3 + 16584278016x^{18}y^2 - 29309796352x^{18}y - 6408896512x^{18} \\
& - 301989888x^{17}y^9 - 7549747200x^{17}y^8 - 3279945728x^{17}y^7 + 29100081152x^{17}y^6 \\
& + 5532286976x^{17}y^5 + 83642810368x^{17}y^4 - 173459636224x^{17}y^3 - 56434360320x^{17}y^2 \\
& - 15443427328x^{17}y + 562036736x^{17} - 1677721600x^{16}y^{10} + 1048576000x^{16}y^9 \\
& + 2936012800x^{16}y^8 + 8006926336x^{16}y^7 + 65288536064x^{16}y^6 - 143722020864x^{16}y^5 \\
& - 25841106944x^{16}y^4 - 223426379776x^{16}y^3 + 108548587520x^{16}y^2 + 27619491840x^{16}y \\
& + 10678697984x^{16} - 100663296x^{15}y^{12} + 134217728x^{15}y^{11} - 805306368x^{15}y^{10} \\
& + 2604662784x^{15}y^9 + 16064184320x^{15}y^8 - 14243856384x^{15}y^7 + 1136656384x^{15}y^6 \\
& - 173564493824x^{15}y^5 + 124765863936x^{15}y^4 + 101917392896x^{15}y^3 \\
& + 137317318656x^{15}y^2 - 2231369728x^{15}y + 5129633792x^{15} - 100663296x^{14}y^{12} \\
& - 1979711488x^{14}y^{11} - 738197504x^{14}y^{10} + 40760246272x^{14}y^9 + 18136170496x^{14}y^8 \\
& - 64722305024x^{14}y^7 + 51535413248x^{14}y^6 + 77422657536x^{14}y^5 \\
& + 364648595456x^{14}y^4 - 35571892224x^{14}y^3 - 50340036608x^{14}y^2 - 47584378880x^{14}y \\
& - 8942256128x^{14} - 973078528x^{13}y^{13} - 486539264x^{13}y^{12} + 23236444160x^{13}y^{11} \\
& + 19738394624x^{13}y^{10} - 17712545792x^{13}y^9 - 32161923072x^{13}y^8 \\
& - 6614836884x^{13}y^7 + 259816161280x^{13}y^6 + 229568937984x^{13}y^5 \\
& - 201148334080x^{13}y^4 - 301109084160x^{13}y^3 - 66379055104x^{13}y^2 - 28420603904x^{13}y \\
& + 2696937472x^{13} - 100663296x^{12}y^{15} + 4882169856x^{12}y^{13} + 10884218880x^{12}y^{12} \\
& - 5729419264x^{12}y^{11} - 58900611072x^{12}y^{10} - 116102529024x^{12}y^9 \\
& + 67779952640x^{12}y^8 + 299731255296x^{12}y^7 - 285590159360x^{12}y^6 \\
& + 129917517824x^{12}y^5 - 736475742208x^{12}y^4 + 176072687616x^{12}y^3 \\
& + 21026045952x^{12}y^2 + 57340329984x^{12}y + 16311648256x^{12} + 352321536x^{11}y^{15} \\
& + 2818572288x^{11}y^{14} - 1828716544x^{11}y^{13} - 28414312448x^{11}y^{12} \\
& - 56249810944x^{11}y^{11} - 41800433664x^{11}y^{10} + 132972019712x^{11}y^9 \\
& + 3913285632x^{11}y^8 + 246651289600x^{11}y^7 - 890998095872x^{11}y^6 \\
& + 601144426496x^{11}y^5 - 294168559616x^{11}y^4 + 430134263808x^{11}y^3 \\
& + 50021269504x^{11}y^2 + 22083010560x^{11}y - 1864368128x^{11} + 268435456x^{10}y^{16} \\
& - 251658240x^{10}y^{15} - 4496293888x^{10}y^{14} - 14722007040x^{10}y^{13} \\
& - 47081062400x^{10}y^{12} + 16680747008x^{10}y^{11} + 175510650880x^{10}y^{10} \\
& + 100837359616x^{10}y^9 - 378087145472x^{10}y^8 + 415015895040x^{10}y^7 \\
& - 962382004224x^{10}y^6 + 1495245258752x^{10}y^5 - 502955769856x^{10}y^4 \\
& + 315363426304x^{10}y^3 - 126315659264x^{10}y^2 - 45189431296x^{10}y - 15443427328x^{10} \\
& + 67108864x^9y^{16} - 2776629248x^9y^{15} - 17347641344x^9y^{14} - 8029995008x^9y^{13} \\
& + 93266640896x^9y^{12} + 90326433792x^9y^{11} + 74180460544x^9y^{10} - 116031225856x^9y^9 \\
& - 966201966592x^9y^8 + 1396238712832x^9y^7 - 1104651223040x^9y^6 \\
& + 1661984571392x^9y^5 - 1073700929536x^9y^4 + 188202614784x^9y^3 \\
& - 200842149888x^9y^2 + 18926796800x^9y - 4087349248x^9 + 50331648x^8y^{18} \\
& - 268435456x^8y^{17} - 2785017856x^8y^{16} - 5324668928x^8y^{15} + 16781410304x^8y^{14} \\
& + 75915853824x^8y^{13} + 151068344320x^8y^{12} - 193344831488x^8y^{11} \\
& - 481618296832x^8y^{10} + 343136010240x^8y^9 - 537743327232x^8y^8 \\
& + 1739758501888x^8y^7 - 1276877209600x^8y^6 + 889016287232x^8y^5 \\
& - 905656139776x^8y^4 + 154706903040x^8y^3 + 25446842368x^8y^2 + 46710915072x^8y \\
& + 13088325632x^8 - 176160768x^7y^{18} - 1409286144x^7y^{17} + 587202560x^7y^{16} \\
& + 26261585920x^7y^{15} + 72077017088x^7y^{14} - 25836912640x^7y^{13} \\
& - 163780231168x^7y^{12} - 277866348544x^7y^{11} + 40509112320x^7y^{10} \\
& + 928338935808x^7y^9 - 685363953664x^7y^8 + 974899380224x^7y^7 \\
& - 1398340583424x^7y^6 + 577669955584x^7y^5 - 515657695232x^7y^4 \\
& + 295540097024x^7y^3 + 51661242368x^7y^2 + 13212057600x^7y - 3952082944x^7 \\
& - 134217728x^6y^{19} - 41943040x^6y^{18} + 3221225472x^6y^{17} + 17624465408x^6y^{16} \\
& + 33195819008x^6y^{15} - 54464086016x^6y^{14} - 255931187200x^6y^{13} \\
& + 28110225408x^6y^{12} + 233857613824x^6y^{11} + 86131081216x^6y^{10} \\
& + 537126240256x^6y^9 - 1073233264640x^6y^8 + 179940622336x^6y^7 \\
& - 815412019200x^6y^6 + 444991537152x^6y^5 - 14436794368x^6y^4 + 136891596800x^6y^3 \\
& + 2498756608x^6y^2 - 30072111104x^6y - 17418944512x^6 - 58720256x^5y^{19} \\
& + 2294284288x^5y^{18} + 14495514624x^5y^{17} - 10309599232x^5y^{16} - 92718235648x^5y^{15} \\
& - 67088416768x^5y^{14} - 100854136832x^5y^{13} + 363559124992x^5y^{12} \\
& - 411324383232x^5y^{11} - 482322415616x^5y^{10} - 267843272704x^5y^9 \\
& - 604335767552x^5y^8 - 460128780288x^5y^7 + 653895401472x^5y^6 + 138693574656x^5y^5 \\
& + 210634276864x^5y^4 - 138589241344x^5y^3 - 19759890432x^5y^2 - 15575547904x^5y \\
& + 2452619264x^5 - 25165824x^4y^{21} + 134217728x^4y^{20} + 2067791872x^4y^{19} \\
& + 1445986304x^4y^{18} - 16234053632x^4y^{17} - 55108960256x^4y^{16} - 111118647296x^4y^{15}
\end{aligned}$$

$$\begin{aligned}
& +159439126528x^4y^{14} + 341189328896x^4y^{13} + 15086911488x^4y^{12} \\
& -193117552640x^4y^{11} - 765365583872x^4y^{10} - 528958881792x^4y^9 \\
& +769357774848x^4y^8 - 483936960512x^4y^7 + 1420158828544x^4y^6 \\
& -529900240896x^4y^5 + 58858143744x^4y^4 - 177164255232x^4y^3 + 15308161024x^4y^2 \\
& +12914262016x^4y + 12013535232x^4 + 88080384x^3y^{21} + 771751936x^3y^{20} \\
& -1002438656x^3y^{19} - 17144741888x^3y^{18} - 43156242432x^3y^{17} - 6494879744x^3y^{16} \\
& +164797874176x^3y^{15} + 233446047744x^3y^{14} - 35751460864x^3y^{13} \\
& -631646453760x^3y^{12} - 161011990528x^3y^{11} + 183611162624x^3y^{10} \\
& -36949521600x^3y^9 + 2087546650624x^3y^8 - 1223615578112x^3y^7 \\
& +1365209513984x^3y^6 - 1071726460928x^3y^5 + 103535607808x^3y^4 \\
& -107007705088x^3y^3 + 100065083392x^3y^2 - 7206862848x^3y - 130547712x^3 \\
& +67108864x^2y^{22} + 20971520x^2y^{21} - 1916796928x^2y^{20} - 9514778624x^2y^{19} \\
& -24906825728x^2y^{18} + 42075684864x^2y^{17} + 169201369088x^2y^{16} \\
& +54513631232x^2y^{15} - 143047262208x^2y^{14} - 175903342592x^2y^{13} \\
& -333820985344x^2y^{12} + 342504505344x^2y^{11} + 1100172492800x^2y^{10} \\
& -679526268928x^2y^9 + 2445060866048x^2y^8 - 2379149475840x^2y^7 \\
& +1184239714304x^2y^6 - 1170807717888x^2y^5 + 424142700544x^2y^4 \\
& +11857559552x^2y^3 + 48370286592x^2y^2 - 8104443904x^2y - 10219421696x^2 \\
& +41943040xy^{22} - 1214251008xy^{21} - 8088715264xy^{20} + 3483893760xy^{19} + 54616653824xy^{18} \\
& +72616247296xy^{17} + 98657894400xy^{16} - 170003529728xy^{15} - 397866172416xy^{14} \\
& +287179014144xy^{13} + 78043676672xy^{12} + 668359983104xy^{11} + 1523161235456xy^{10} \\
& -1937053450240xy^9 + 2160674865152xy^8 - 3268596465664xy^7 + 1650531762176xy^6 \\
& -764078194688xy^5 + 495431974912xy^4 - 7412383744xy^3 - 45791576064xy^2 - 6315048960xy \\
& +2333605888x - 37748736y^{22} + 592445440y^{21} + 4016046080y^{20} + 1660157952y^{19} \\
& -21657288704y^{18} - 36624662528y^{17} - 85356183552y^{16} + 55327326208y^{15} + 221898211328y^{14} \\
& -142863564800y^{13} + 147958923264y^{12} - 436308099072y^{11} - 1024769744896y^{10} \\
& +1299351330816y^9 - 2080060680192y^8 + 2531541916672y^7 - 1486327928832y^6 \\
& +992287815424y^5 - 473649777536y^4 - 42054634752y^3 + 54714338560y^2 + 772362640y - 517770000) \\
V \quad & (8388608x^3y^{24} + 4194304x^3y^{23} - 754974720x^3y^{22} - 1675624448x^3y^{21} \\
& +688914432x^3y^{20} + 27169128448x^3y^{19} + 152914362368x^3y^{18} + 146317770752x^3y^{17} \\
& -9844922776x^3y^{16} - 333857423360x^3y^{15} - 1213194436608x^3y^{14} \\
& -367908225024x^3y^{13} + 591497625600x^3y^{12} - 1007563980800x^3y^{11} \\
& +3922602196992x^3y^{10} - 3214999228416x^3y^9 + 2483648448512x^3y^8 \\
& -1791599724544x^3y^7 - 633931686400x^3y^6 + 974375289600x^3y^5 \\
& -665637046784x^3y^4 + 508568760192x^3y^3 - 125087468224x^3y^2 - 15597810880x^3y \\
& +4459778904x^3 - 25165824x^2y^{24} - 169869312x^2y^{23} - 916455424x^2y^{22} \\
& +3736600576x^2y^{21} + 38814613504x^2y^{20} + 60946907136x^2y^{19} + 37347131392x^2y^{18} \\
& -200705835008x^2y^{17} - 765033709568x^2y^{16} - 324971659264x^2y^{15} \\
& -647774273536x^2y^{14} + 474132709376x^2y^{13} + 2112037634048x^2y^{12} \\
& -2130223898624x^2y^{11} + 4201090629632x^2y^{10} - 5014643705856x^2y^9 \\
& +2165059085312x^2y^8 - 1249192256512x^2y^7 - 413911150848x^2y^6 \\
& +1452727959936x^2y^5 - 791561348352x^2y^4 + 317279418304x^2y^3 - 93410141536x^2y^2 \\
& -12735339104x^2y + 8387652012x^2 - 12582912xy^{25} - 102760448xy^{24} + 206569472xy^{23} \\
& +4761059328xy^{22} + 19462094848xy^{21} + 25201999872xy^{20} - 71112327168xy^{19} \\
& -28207622400xy^{18} - 407154982912xy^{17} - 661259288576xy^{16} + 558699839488xy^{15} \\
& +643666444288xy^{14} + 447487246336xy^{13} + 2748973473792xy^{12} - 4612848848896xy^{11} \\
& +4303422152704xy^{10} - 5934952170496xy^9 + 2829864464896xy^8 - 14078218752xy^7 \\
& -603837397120xy^6 + 1370203275456xy^5 - 980913607296xy^4 + 262360652768xy^3 \\
& -44273160880xy^2 + 7539593936xy + 447001814x - 25165824y^{25} + 206569472y^{24} + 3454533632y^{23} \\
& +5617745920y^{22} - 10640949248y^{21} - 59588739072y^{20} - 187323645952y^{19} - 263373455360y^{18} \\
& +26933854208y^{17} + 101071323136y^{16} + 957020012544y^{15} + 915742932992y^{14} \\
& -984438702080y^{13} + 2082981875712y^{12} - 4984447502336y^{11} + 4131736381440y^{10} \\
& -3988503969280y^9 + 2699468559616y^8 + 221960259072y^7 - 1015736187584y^6 \\
& +915146390368y^5 - 728070697344y^4 + 288913574032y^3 - 49050239928y^2 + 11809059604y \\
& -3361056149, \\
& 12582912x^3y^{24} - 67108864x^3y^{23} - 1015021568x^3y^{22} - 1915224064x^3y^{21} \\
& +4178575360x^3y^{20} + 36134977536x^3y^{19} + 105987440640x^3y^{18} - 9196535808x^3y^{17} \\
& -130881683456x^3y^{16} - 167529414656x^3y^{15} - 402957664256x^3y^{14} \\
& +1144013520896x^3y^{13} + 221291872256x^3y^{12} + 175310061568x^3y^{11} \\
& +2272576757760x^3y^{10} - 5005593862144x^3y^9 + 4916993665024x^3y^8 \\
& -5736946998272x^3y^7 + 3898696816640x^3y^6 - 1643919865600x^3y^5 \\
& +757623032704x^3y^4 - 53249127168x^3y^3 - 89925520640x^3y^2 - 7941214608x^3y \\
& +9544436496x^3 - 37748736x^2y^{24} - 452984832x^2y^{23} - 104857600x^2y^{22} \\
& +8315207680x^2y^{21} + 31524388864x^2y^{20} + 37828952064x^2y^{19} - 23442489344x^2y^{18} \\
& -192316178432x^2y^{17} - 221056860160x^2y^{16} + 311314612224x^2y^{15} \\
& -69592350720x^2y^{14} + 1229978763264x^2y^{13} - 57746718720x^2y^{12} \\
& -1919504703488x^2y^{11} + 2886788100096x^2y^{10} - 7002115698688x^2y^9 \\
& +6835488396288x^2y^8 - 5779691408896x^2y^7 + 4485663435776x^2y^6 \\
& -1769667115904x^2y^5 + 501371700672x^2y^4 - 148579457408x^2y^3 - 13803013760x^2y^2 \\
& +5920610104x^2y + 3196994952x^2 - 33554432xy^{25} - 29360128xy^{24} + 643825664xy^{23} \\
& +5693767680xy^{22} + 20871380992xy^{21} - 877133824xy^{20} - 74287939584xy^{19} \\
& -115789529088xy^{18} - 181693841408xy^{17} + 55501520896xy^{16} + 715259510784xy^{15}
\end{aligned}$$

	$ \begin{aligned} & -358001115136xy^{14} + 335839576064xy^{13} - 551093043200xy^{12} - 3863455510528xy^{11} \\ & + 4880365934592xy^{10} - 8007563491328xy^9 + 9308785199616xy^8 - 6345926772480xy^7 \\ & + 4369622242816xy^6 - 2143564897728xy^5 + 420070458080xy^4 - 43729243328xy^3 \\ & + 1567841472xy^2 + 16215092124xy - 4417183036x - 37748736y^{25} + 592445440y^{24} + 4343201792y^{23} \\ & + 1607729152y^{22} - 12607029248y^{21} - 40541618176y^{20} - 128183435264y^{19} + 4406378496y^{18} \\ & + 89689554944y^{17} + 52476182528y^{16} + 532804141056y^{15} - 936887009280y^{14} \\ & - 590283300864y^{13} + 277434032128y^{12} - 3698258126848y^{11} + 6160181705728y^{10} \\ & - 6943384700928y^9 + 8131003143936y^8 - 5975700068224y^7 + 3281784590208y^6 \\ & - 1681211345184y^5 + 476396431344y^4 + 5789651744y^3 - 23128767080y^2 + 4083260246y \\ & - 1041788574) \end{aligned} $				
$i = 32$	$D_i = 2$	$\rho_{ij}: 3:1, 8:2, 14:1, 22:2, 23:2, 31:2$	w_i	$ \begin{aligned} & \langle -2t^3 + 4t^2 + 2t + 2, \\ & \quad -4t^3 + 2t^2 - 2t + 4 \rangle \end{aligned} $	
$i = 33$	$D_i = 18$	$\rho_{ij}: 18:1, 20:1, 26:2$	T	$ \begin{aligned} & \langle -4y^2 - 8y + 23, \\ & \quad 0 \rangle \end{aligned} $	
	U	$ \begin{aligned} & \langle 64y^3 + 384y^2 + 5400y + 10156, \\ & \quad 0 \rangle \end{aligned} $			
	V	$ \begin{aligned} & \langle -64x^3y^3 - 384x^3y^2 - 5400x^3y - 7240x^3 + 32x^2y^3 + 192x^2y^2 - 216x^2y \\ & \quad - 2212x^2 - 16xy^3 - 96xy^2 + 108xy + 4022x - 32y^4 - 3100y^3 - 2652y^2 - 2216y - 2011, \\ & \quad 0 \rangle \end{aligned} $			
$i = 34$	$D_i = 8$	$\rho_{ij}: 0:2, 2:1, 8:2, 9:1, 12:2, 16:1, 20:1, 28:1, 31:1, 39:2$	w_i	$ \begin{aligned} & \langle 192t^3 - 282t^2 + 208t - 124, \\ & \quad 166t^3 - 158t^2 + 142t - 68 \rangle \end{aligned} $	
$i = 35$	$D_i = 4$	$\rho_{ij}: 7:1, 9:1, 14:1, 15:1, 19:1, 20:1, 21:1, 29:2$	w_i	$ \begin{aligned} & \langle -4t^5 + 4t^4 + 4t^3 - 4t^2, \\ & \quad 4t^4 - 8t^3 - 8t^2 \rangle \end{aligned} $	
$i = 36$	$D_i = 7$	$\rho_{ij}: 11:1$	w_i	$ \begin{aligned} & \langle 4t^3 + t + 8, \\ & \quad 0 \rangle \end{aligned} $	
$i = 37$	$D_i = 4$	$\rho_{ij}: 2:1, 3:1, 6:1, 16:2, 17:2, 18:1, 19:2, 20:1, 23:2, 26:1, 31:1$	w_i	$ \begin{aligned} & \langle -56t^5 + 56t^3 - 24t - 16, \\ & \quad -28t^5 + 28t^3 - 12t - 8 \rangle \end{aligned} $	
$i = 38$	$D_i = 4$	$\rho_{ij}: 1:1, 2:1, 3:2, 9:2, 10:1, 13:2, 15:2, 23:1, 28:2, 30:1, 31:2$	w_i	$ \begin{aligned} & \langle 46t^3 + 110t^2 + 4t - 32, \\ & \quad 50t^3 - 26t^2 - 106t - 28 \rangle \end{aligned} $	
$i = 39$	$D_i = 1944$	$\rho_{ij}: 1:1, 2:2, 3:1, 7:1, 9:2, 10:2, 15:2, 17:2, 19:1, 21:1, 27:2, 29:1, 34:1$	w_i	$ \begin{aligned} & \langle -83592t^5 + 101088t^4 - 147744t^3 + 276048t^2 - 104976t + 186624, \\ & \quad -44712t^5 + 48600t^4 - 56376t^3 + 128304t^2 - 34992t + 93312 \rangle \end{aligned} $	

Table 5: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
$(0, 4)$	0	1	$(9, 34)$	9	2^2
$(0, 5)$	0	7^2	$(9, 35)$	9	1
$(0, 14)$	0	1	$(9, 38)$	9	2^2
$(0, 21)$	0	7	$(9, 39)$	9	2^4
$(0, 24)$	0	1	$(10, 18)$	10	1
$(0, 34)$	0	7	$(10, 38)$	10	7^2
$(1, 3)$	1	$2^2 \cdot 7$	$(10, 39)$	10	$3 \cdot 7$
$(1, 5)$	1	$7^5 \cdot 13$	$(11, 21)$	11	1
$(1, 11)$	1	$3 \cdot 13$	$(11, 36)$	11	1
$(1, 12)$	1	$3^4 \cdot 7$	$(12, 26)$	12	3^2
$(1, 13)$	1	7	$(12, 34)$	12	1
$(1, 15)$	1	$3 \cdot 7$	$(13, 16)$	13	$3 \cdot 7$
$(1, 22)$	1	$2^2 \cdot 3^2 \cdot 7$	$(13, 19)$	13	$3 \cdot 7$
$(1, 38)$	1	$2^4 \cdot 7^2$	$(13, 25)$	13	7
$(1, 39)$	1	$2^4 \cdot 7^3$	$(13, 38)$	13	13
$(2, 6)$	2	$2^4 \cdot 13$	$(14, 24)$	14	$3^2 \cdot 7$
$(2, 7)$	2	$2^4 \cdot 3^3 \cdot 7$	$(14, 26)$	14	1
$(2, 11)$	2	3	$(14, 32)$	14	1
$(2, 13)$	2	13	$(14, 35)$	14	3^2

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(2, 14)	2	7	(15, 19)	15	3^2
(2, 16)	2	3	(15, 26)	15	3^2
(2, 24)	2	$2^4 \cdot 7$	(15, 27)	15	7
(2, 34)	2	2^4	(15, 29)	15	$3^2 \cdot 7$
(2, 37)	2	13	(15, 35)	15	7
(2, 38)	2	13^2	(15, 38)	15	7
(2, 39)	2	$2^4 \cdot 7$	(15, 39)	15	7
(3, 4)	3	$2^2 \cdot 7 \cdot 13^2$	(16, 17)	16	1
(3, 5)	3	$3^6 \cdot 5^2$	(16, 24)	16	$2^4 \cdot 3^3$
(3, 7)	3	$2^2 \cdot 7^3 \cdot 13$	(16, 34)	16	$7 \cdot 13$
(3, 10)	3	7^2	(16, 37)	16	7^2
(3, 11)	3	13	(17, 30)	17	3^4
(3, 12)	3	7	(17, 31)	17	2^6
(3, 16)	3	13	(17, 37)	17	3^4
(3, 17)	3	3^6	(17, 39)	17	1
(3, 18)	3	$3^2 \cdot 13$	(18, 33)	18	$2^4 \cdot 13$
(3, 28)	3	5^2	(18, 37)	18	3^2
(3, 29)	3	$5^2 \cdot 7^3 \cdot 13$	(19, 21)	19	$7 \cdot 13$
(3, 31)	3	$5^2 \cdot 7^2 \cdot 13$	(19, 27)	19	3^2
(3, 32)	3	7	(19, 31)	19	$2^4 \cdot 3 \cdot 7$
(3, 37)	3	$3^6 \cdot 13$	(19, 35)	19	$3 \cdot 7$
(3, 38)	3	$3^2 \cdot 7^2$	(19, 37)	19	7^2
(3, 39)	3	$7^4 \cdot 13$	(19, 39)	19	$3 \cdot 13^2$
(4, 7)	4	$2^4 \cdot 3^3 \cdot 7$	(20, 21)	20	$2^2 \cdot 7$
(4, 10)	4	7	(20, 30)	20	3^4
(4, 11)	4	$3 \cdot 13$	(20, 33)	20	$2^4 \cdot 13$
(4, 15)	4	3	(20, 34)	20	7
(4, 29)	4	$3 \cdot 7 \cdot 13$	(20, 35)	20	1
(4, 30)	4	2^{12}	(20, 37)	20	3^2
(5, 6)	5	$7^2 \cdot 13^2$	(21, 27)	21	2^{10}
(5, 8)	5	$3^2 \cdot 7 \cdot 13^2$	(21, 35)	21	$2^4 \cdot 7$
(5, 11)	5	13	(21, 39)	21	$2^4 \cdot 13$
(5, 16)	5	7	(22, 23)	22	7^3
(5, 17)	5	3^6	(22, 25)	22	1
(5, 20)	5	$3^2 \cdot 13$	(22, 26)	22	3^6
(5, 21)	5	$7^2 \cdot 13$	(22, 31)	22	$2^4 \cdot 3^2 \cdot 7$
(5, 27)	5	7^2	(22, 32)	22	3^6
(5, 29)	5	5^2	(23, 31)	23	2^{14}
(5, 31)	5	$5^2 \cdot 7^2 \cdot 13$	(23, 32)	23	7^3
(6, 13)	6	7^2	(23, 37)	23	2^6
(6, 15)	6	7	(23, 38)	23	2^{10}
(6, 24)	6	$2^4 \cdot 7$	(25, 31)	25	$2^6 \cdot 7$
(6, 28)	6	$2^2 \cdot 7^2 \cdot 13$	(26, 28)	26	$2^4 \cdot 3^2 \cdot 13$
(6, 31)	6	$7^2 \cdot 13^2$	(26, 29)	26	$2^4 \cdot 3^2 \cdot 13$
(6, 37)	6	$2^2 \cdot 7^2 \cdot 13^2$	(26, 31)	26	$2^4 \cdot 3^4$
(7, 8)	7	3^9	(26, 33)	26	$2^6 \cdot 13^2$
(7, 11)	7	3^2	(26, 37)	26	2^4
(7, 24)	7	$2^2 \cdot 3 \cdot 7$	(27, 28)	27	$2^4 \cdot 3 \cdot 7^2$
(7, 28)	7	$2^4 \cdot 3^5$	(27, 39)	27	$2^4 \cdot 3 \cdot 7^3$
(7, 35)	7	$3 \cdot 7$	(28, 29)	28	$2^2 \cdot 3^6 \cdot 5^2$
(7, 39)	7	$3^2 \cdot 7^3 \cdot 13$	(28, 31)	28	$2^4 \cdot 3 \cdot 5^2 \cdot 7$
(8, 9)	9	1	(28, 34)	28	7^3
(8, 10)	10	$3 \cdot 7^2$	(28, 38)	28	2^2
(8, 16)	16	$3^3 \cdot 7^2$	(29, 31)	29	$2^4 \cdot 3 \cdot 5^2 \cdot 7$
(8, 20)	20	$3^2 \cdot 7$	(29, 35)	29	$2^4 \cdot 3 \cdot 7$
(8, 21)	21	$2^4 \cdot 7^2 \cdot 13$	(29, 39)	29	$2^4 \cdot 3^2 \cdot 7^3$
(8, 27)	27	$2^4 \cdot 3$	(30, 38)	30	$3^4 \cdot 7^3$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(8, 32)	32	3^2	(31, 32)	32	$2^4 \cdot 3^2 \cdot 7$
(8, 34)	34	7^4	(31, 34)	34	$2^4 \cdot 7^2$
(9, 22)	9	2^2	(31, 37)	37	$7^2 \cdot 13^2$
(9, 23)	9	2^4	(31, 38)	38	$2^6 \cdot 7 \cdot 13$
(9, 29)	9	2^2	(34, 39)	34	$7^2 \cdot 13$

Table 6: The data of the identities (72) for the edges without a linear endpoint.

$(i, j) = (8, 31), D_{ij} = \langle 1029, 0 \rangle$	
λ	$\langle 512, -2048 \rangle$
μ	$\langle 256y^2 + 256y + 812, -1024y^2 - 352y - 2660 \rangle$
ν	$\langle -512x^3 - 256x^2 + 512xy^2 + 176xy + 1202x - 256y^2 - 344y - 729, 2048x^3 + 1024x^2 + 640xy^2 + 304xy + 2248x - 320y^2 + 872y - 612 \rangle$

Table 7: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 6, grouped by p, k and the joint value $(\mathcal{B}_p; B_p)$.

p	k	value	classes (a, b)
2	5	(0; -1)	(0, 3), (2, 21), (4, 7), (6, 9), (8, 11), (10, 29), (12, 15), (14, 17), (16, 19), (18, 5), (20, 23), (22, 25), (24, 27), (26, 13), (28, 31)
2	6	(0; -1)	(30, 1), (62, 33)
3	3	(1; 1)	(2, 19), (5, 13), (11, 1), (14, 22), (20, 10), (23, 4)
3	3	(2; -1)	(13, 5), (13, 14), (13, 23)
3	4	(1; 1)	(44, 52)
3	4	(2; -1)	(13, 8), (13, 17), (13, 26), (13, 35), (13, 44), (13, 53), (13, 62), (13, 71), (13, 80), (40, 2), (40, 11), (40, 20), (40, 29), (40, 38), (40, 47), (40, 56), (40, 65), (40, 74)
3	5	(1; 1)	(8, 124), (26, 169), (35, 70), (53, 115), (62, 16), (80, 61), (89, 205), (107, 7), (116, 151), (134, 196), (143, 97), (161, 142), (170, 43), (188, 88), (197, 232), (215, 34), (224, 178), (242, 223)
3	6	(1; 1)	(17, 349), (71, 484), (98, 187), (152, 322), (179, 25), (233, 160), (260, 592), (314, 727), (341, 430), (395, 565), (422, 268), (476, 403), (503, 106), (557, 241), (584, 673), (638, 79), (665, 511), (719, 646)
5	1	(0; 1)	(1, 2), (2, 3)
5	2	(0; 1)	(16, 11), (18, 14)
5	3	(0; 1)	(1, 71), (3, 64), (6, 26), (8, 114), (11, 56), (13, 114), (21, 91), (23, 89), (26, 96), (28, 64), (31, 51), (33, 114), (36, 81), (38, 114), (46, 116), (48, 89), (51, 121), (53, 64), (56, 76), (58, 114), (61, 106), (63, 114), (71, 16), (73, 89), (76, 21), (78, 64), (81, 101), (83, 114), (86, 6), (88, 114), (96, 41), (98, 89), (101, 46), (103, 64), (106, 1), (108, 114), (111, 31), (113, 114), (121, 66), (123, 89)
7	2	(2; 1)	(1, 39), (2, 45), (3, 4), (3, 38), (4, 21), (8, 46), (9, 3), (10, 10), (10, 11), (11, 28), (11, 29), (15, 4), (16, 10), (18, 22), (18, 35), (22, 11), (23, 17), (24, 3), (24, 21), (24, 25), (25, 42), (30, 24), (31, 24), (32, 0), (32, 8), (36, 25), (37, 31), (38, 39), (38, 45), (39, 1), (43, 32), (44, 38), (45, 17), (45, 46), (46, 14), (46, 43)

p	k	value	classes (a, b)
7	3	(2; 1)	(3, 105), (4, 183), (4, 188), (10, 77), (11, 237), (18, 139), (25, 64), (29, 67), (31, 42), (31, 326), (32, 188), (38, 161), (39, 105), (39, 335), (45, 231), (46, 335), (52, 252), (53, 134), (53, 188), (59, 224), (60, 237), (66, 67), (66, 129), (66, 147), (67, 139), (74, 15), (74, 237), (78, 116), (80, 32), (80, 189), (81, 188), (87, 308), (88, 154), (88, 335), (94, 35), (95, 335), (101, 56), (102, 85), (102, 188), (108, 28), (109, 237), (115, 116), (115, 276), (115, 294), (116, 139), (123, 237), (123, 309), (127, 165), (129, 81), (129, 336), (130, 188), (136, 112), (137, 203), (137, 335), (143, 182), (144, 335), (150, 203), (151, 36), (151, 188), (157, 175), (158, 237), (164, 80), (164, 98), (164, 165), (165, 139), (172, 237), (176, 214), (178, 130), (178, 140), (179, 188), (186, 252), (186, 335), (192, 329), (193, 335), (199, 7), (200, 188), (200, 330), (206, 322), (207, 237), (213, 214), (213, 227), (213, 245), (214, 139), (221, 211), (221, 237), (225, 263), (227, 179), (227, 287), (228, 188), (234, 63), (235, 301), (235, 335), (241, 133), (242, 335), (248, 154), (249, 188), (249, 281), (255, 126), (256, 237), (262, 31), (262, 49), (262, 263), (263, 139), (270, 162), (270, 237), (274, 312), (276, 91), (276, 228), (277, 188), (283, 210), (284, 7), (284, 335), (290, 280), (291, 335), (297, 301), (298, 188), (304, 273), (305, 237), (311, 178), (311, 196), (311, 312), (312, 139), (319, 113), (319, 237), (325, 238), (325, 277), (326, 188), (332, 14), (333, 56), (333, 335), (339, 84), (340, 335)
7	4	(2; 1)	(17, 686), (17, 1733), (17, 2383), (25, 923), (172, 1975), (185, 602), (298, 575), (323, 2076), (360, 1011), (360, 1715), (360, 2076), (368, 923), (515, 1632), (528, 1631), (641, 232), (666, 18), (703, 18), (703, 343), (703, 2040), (711, 923), (858, 1289), (871, 259), (984, 2290), (1009, 361), (1046, 361), (1046, 668), (1046, 1372), (1054, 923), (1201, 946), (1214, 1288), (1327, 1947), (1352, 704), (1389, 0), (1389, 704), (1389, 1697), (1397, 923), (1544, 603), (1557, 2317), (1670, 1604), (1695, 1047), (1732, 325), (1732, 1029), (1732, 1047), (1740, 923), (1887, 260), (1900, 945), (2013, 1261), (2038, 1390), (2075, 1354), (2075, 1390), (2075, 2058), (2083, 923), (2230, 2318), (2243, 1974), (2356, 918), (2381, 1733)
13	1	(0; 1)	(0, 11), (2, 12), (3, 11), (5, 9), (6, 7), (9, 6), (9, 11), (10, 4), (11, 2), (11, 5)
13	2	(0; 1)	(6, 96), (6, 118), (7, 4), (9, 165), (11, 6), (12, 7), (19, 18), (19, 118), (20, 160), (22, 61), (24, 110), (25, 111), (32, 109), (33, 147), (35, 126), (37, 45), (38, 46), (40, 141), (45, 31), (45, 118), (46, 134), (48, 22), (50, 149), (51, 150), (53, 102), (58, 118), (58, 122), (59, 121), (61, 87), (63, 84), (64, 85), (71, 44), (71, 118), (72, 108), (74, 152), (76, 19), (77, 20), (84, 118), (84, 135), (85, 95), (87, 48), (90, 124), (97, 57), (97, 118), (98, 82), (100, 113), (102, 58), (103, 59), (105, 115), (110, 118), (110, 148), (111, 69), (113, 9), (115, 162), (116, 163), (118, 76), (123, 70), (123, 118), (124, 56), (126, 74), (128, 97), (129, 98), (136, 118), (136, 161), (137, 43), (139, 139), (142, 33), (149, 83), (149, 118), (150, 30), (152, 35), (154, 136), (155, 137), (162, 5), (162, 118), (163, 17), (165, 100), (167, 71), (168, 72)

p	k	value	classes (a, b)
13	3	(0; 1)	(1, 89), (14, 895), (27, 518), (32, 1470), (66, 1077), (79, 362), (89, 461), (92, 661), (131, 1051), (141, 1891), (144, 1012), (157, 1987), (170, 1779), (183, 388), (196, 11), (201, 1470), (235, 570), (248, 2052), (258, 1813), (261, 154), (300, 544), (310, 1046), (313, 505), (326, 1480), (339, 1272), (352, 2078), (365, 1701), (370, 1470), (404, 63), (417, 1545), (427, 968), (430, 1844), (469, 37), (479, 201), (482, 2195), (495, 973), (508, 765), (521, 1571), (534, 1194), (539, 1470), (573, 1753), (586, 1038), (596, 123), (599, 1337), (638, 1727), (648, 1553), (651, 1688), (664, 466), (677, 258), (690, 1064), (703, 687), (708, 1470), (742, 1246), (755, 531), (765, 1475), (768, 830), (807, 1220), (817, 708), (820, 1181), (833, 2156), (846, 1948), (859, 557), (872, 180), (877, 1470), (911, 739), (924, 24), (934, 630), (937, 323), (976, 713), (986, 2060), (989, 674), (1002, 1649), (1015, 1441), (1028, 50), (1041, 1870), (1046, 1470), (1080, 232), (1093, 1714), (1103, 1982), (1106, 2013), (1145, 206), (1155, 1215), (1158, 167), (1171, 1142), (1184, 934), (1197, 1740), (1210, 1363), (1215, 1470), (1249, 1922), (1262, 1207), (1272, 1137), (1275, 1506), (1314, 1896), (1324, 370), (1327, 1857), (1340, 635), (1353, 427), (1366, 1233), (1379, 856), (1384, 1470), (1418, 1415), (1431, 700), (1441, 292), (1444, 999), (1483, 1389), (1493, 1722), (1496, 1350), (1509, 128), (1522, 2117), (1535, 726), (1548, 349), (1553, 1470), (1587, 908), (1600, 193), (1610, 1644), (1613, 492), (1652, 882), (1662, 877), (1665, 843), (1678, 1818), (1691, 1610), (1704, 219), (1717, 2039), (1722, 1470), (1756, 401), (1769, 1883), (1779, 799), (1782, 2182), (1821, 375), (1831, 32), (1834, 336), (1847, 1311), (1860, 1103), (1873, 1909), (1886, 1532), (1891, 1470), (1925, 2091), (1938, 1376), (1948, 2151), (1951, 1675), (1990, 2065), (2000, 1384), (2003, 2026), (2016, 804), (2029, 596), (2042, 1402), (2055, 1025), (2060, 1470), (2094, 1584), (2107, 869), (2117, 1306), (2120, 1168), (2159, 1558), (2169, 539), (2172, 1519), (2185, 297)